MATHEMATICS

COMPLEX NUMBER

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COMPLEX NUMBER

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BASICS OF COMPLEX NUMBERS

1. The argument of the complex number $\sin\frac{6\pi}{5}+i\left(1+\cos\frac{6\pi}{5}\right)$ is :

सम्मिश्र संख्या $\sin\frac{6\pi}{5}+i\left(1+\cos\frac{6\pi}{5}\right)$ का कोणांक है :

(A) $\frac{6\pi}{5}$ (B) $\frac{5\pi}{6}$ (C) $\frac{9\pi}{10}$ (D) $\frac{2\pi}{5}$

Ans. C

Sol. $z=\sin\frac{6\pi}{5}+i\left(1+\cos\frac{6\pi}{5}\right)$

$\Rightarrow z=\sin\frac{\pi}{5}+i\left(1+\cos\left(\pi+\frac{\pi}{5}\right)\right)$

$z=\sin\frac{\pi}{5}+i\left(1-\cos\frac{\pi}{5}\right)$

arg (z) = lies in IInd quadrant

$\arg(z)=17-\tan^{-1}\left(\frac{1-\cos\frac{\pi}{5}}{\sin\frac{\pi}{5}}\right)$

$=\pi-\tan^{-1}\left(\frac{2\sin^2\frac{\pi}{5}}{2\sin\frac{\pi}{10}\cos\frac{\pi}{10}}\right)$

$=\pi-\tan^{-1}\left(\tan\frac{\pi}{10}\right)$

$=\pi-\frac{\pi}{10}$

$=\frac{9\pi}{10}$

2. Number of complex numbers z satisfying $z^3 = \bar{z}$ is :

सम्मिश्र संख्या z के मानों की संख्या जो $z^3 = \bar{z}$ को सन्तुष्ट करते है :

(A) 1 (B) 2 (C) 4 (D) 5

Ans. D

Sol. $z = 0$; $z = \pm 1$; $z = \pm i$;

$z^3 = \bar{z} \Rightarrow |z|^3 = |\bar{z}| = |z|$

note that $z^n = |\bar{z}|$ has $n + 2$ solutions

hence $|z| = 0$ or $|z|^2 = 1$

again $z^4 = z\,\bar{z} = |z|^2 = 1 \Rightarrow z^4 = 1 \Rightarrow$ total number of roots are 5

Note that (a) the equation $z^n = \bar{z}$ will have $(n + 2)$ solutions.

(b) all non-zero solutions of $z^3 = \bar{z}$ are $(1, 0), (-1, 0), (0, 1), (0, -1)$ i.e. a square.

3. For $Z_1 = \sqrt[6]{\frac{1-i}{1+i\sqrt{3}}}$; $Z_2 = \sqrt[6]{\frac{1-i}{\sqrt{3}+i}}$; $Z_3 = \sqrt[6]{\frac{1+i}{\sqrt{3}-i}}$ which of the following holds good?

$Z_1 = \sqrt[6]{\frac{1-i}{1+i\sqrt{3}}}$; $Z_2 = \sqrt[6]{\frac{1-i}{\sqrt{3}+i}}$; $Z_3 = \sqrt[6]{\frac{1+i}{\sqrt{3}-i}}$ के लिए सत्य कथन है?

(A) $\sum |Z_1|^2 = \frac{3}{2}$ (B) $|Z_1|^4 + |Z_2|^4 = |Z_3|^{-8}$

(C) $\sum |Z_1|^3 + |Z_2|^3 = |Z_3|^{-6}$ (D) $|Z_1|^4 + |Z_2|^4 = |Z_3|^8$

Ans. B

Sol. $|z_1| = \left|\frac{1-i}{1+i\sqrt{3}}\right|^{\frac{1}{6}} = \left|\frac{\sqrt{2}}{2}\right|^{\frac{1}{6}} = 2^{-\frac{1}{12}}$

|||ly $|z_2| = 2^{-\frac{1}{12}}$; $|z_3| = 2^{-\frac{1}{12}}$ hence the result

4. Let z and w are two non zero complex number such that $|z| = |w|$, and $\text{Arg}(z) + \text{Arg}(w) = \pi$ then :

माना दो सम्मिश्र संख्याऐं z एवं w इस प्रकार से हैं कि $|z| = |w|$, तथा $\text{Arg}(z) + \text{Arg}(w) = \pi$ तो :

(A) $z = w$ (B) $z = \bar{w}$ (C) $\bar{z} = \bar{w}$ (D) $z = -\bar{w}$

Ans. D

Sol. Let arg $w = \theta_1$

$w = |w| (\cos\theta_1 + \sin\theta_1)$

$\text{Arg}(z) + \text{Arg}(w) = \pi$

$\arg z = \pi - \theta_1$

$Z = |z|\left(\cos(\pi - \theta_1) + \sin(\pi - \theta_1)\right) \because |z| = |w|$

$z = -|z|\left(\cos\theta_1 - i\sin\theta_1\right)$

$z = -\overline{w}$

5. If z and ω are two non- zero comlex numbers such that $|z\omega| = 1$, and $\text{Arg}(z) - \text{Arg}(\omega) = \frac{\pi}{2}$, then $\overline{z}\,\omega$ is equal to:

यदि z तथा ω दो अशून्य सम्मिश्र संख्याऐं इस प्रकार हैं कि $|z\omega| = 1$, एवं $\text{Arg}(z) - \text{Arg}(\omega) = \frac{\pi}{2}$, तो $\overline{z}\,\omega$ बराबर है :

(A) $-i$ (B) 1 (C) -1 (D) i

Ans. A

Sol. $|z\omega| = 1$

$\overline{z}\,\omega = \frac{1}{z\overline{w}}$

let arg cos = θ_1

$w = |w|\left(\cos\theta_1 + i\sin\theta_1\right)$

$\therefore \arg(x)$

$z = |z|\left(\cos\left(\frac{\pi}{2} + \theta_1\right) + \sin\left(\frac{\pi}{2} + \theta_1\right)\right) - \arg|w| = \frac{\pi}{2}$

$\overline{z}w = \frac{1}{|z||w|\left(-\sin\theta_1 + i\cos\theta_1\right)\left(\cos\theta_1 - \sin\theta_1\right)}$

$\overline{z}w = \frac{1}{|z||w|i}$

$\overline{z}w = -i$

6. If $\left(\frac{1+i}{1-i}\right)^x = 1$, then :

(A) $x = 2n + 1$, where n is any positive integer (B) $x = 4n$, where n is any positive integer

(C) $x = 2n$, where n is any positive integer (D) $x = 4n + 1$, where n is any positive integer

यदि $\left(\frac{1+i}{1-i}\right)^x = 1$, तो :

(A) $x = 2n + 1$, जहाँ n एक धनात्मक पूर्णांक है (B) $x = 4n$, जहाँ n एक धनात्मक पूर्णांक है

(C) $x = 2n$, जहाँ n एक धनात्मक पूर्णांक है (D) $x = 4n + 1$, जहाँ n एक धनात्मक पूर्णांक है

Ans. B

Sol. $\left(\frac{1+i}{1-i}\right)^x = 1$

$$\left(\frac{(1+i)^2}{(1-i)(1+i)}\right)^x = 1$$

(i) $x = 4$

$x = 4n$, where x is positive integer.

7. If $z = x - iy$ and $z^{\frac{1}{3}} = p + iq$, then $\frac{\left(\frac{x}{p}+\frac{y}{q}\right)}{\left(p^2+q^2\right)}$ is equal to :

यदि $z = x - iy$ तथा $z^{\frac{1}{3}} = p + iq$, तो $\frac{\left(\frac{x}{p}+\frac{y}{q}\right)}{\left(p^2+q^2\right)}$ बराबर है :

(A) 1 (B) – 1 (C) 2 (D) – 2

Ans. D

Sol. $z = x - iy$

$z^{\frac{1}{3}} = p + iq$

$z = p^3 - iq^3 + 3ip^2q - 3pq$

By comparsion

$p^3 - pq^2 = x$

$\frac{x}{p} = p^2 - 3q^2$

$3p^2q - q^3 = - y$

$\frac{y}{q} = q^2 - 3p^2$

So that $\frac{\frac{x}{p}+\frac{y}{q}}{p^2+q^2}=\frac{p^2-3q^2+q^2-3p^2}{p^2+q^2}=-2$

8. If $|z+4|\le 3$, then the maximum and minimum value of $|z+1|$ are :

यदि $|z+4|\le 3$, तब $|z+1|$ का अधिकतम तथा न्यूनतम मान है :

(A) 4, 1 (B) 4, 0 (C) 6, 0 (D) 6, 1

Ans. C

Sol. $|z+4|\le 3$

$-3\le z+4\le 3$

$-6\le z+1\le 0$

$0\le |z+1|\le 6$

9. Let α, β be real and z be a complex number. If $z^2+\alpha z+\beta=0$ has two distinct roots on the line Re (z) = 1, then it is necessary that : [AIEEE-2010]

माना α, β वास्तविक हैं तथा z एक सम्मिश्र संख्या है। यदि $z^2+\alpha z+\beta=0$ के रेखा Re (z) = 1 पद दो विभिन्न मूल हैं, तो यह अनिवार्य है कि : [AIEEE-2010]

(A) $|\beta|=1$ (B) $\beta\in(1,\infty)$ (C) $\beta\in(0,1)$ (D) $\beta\in(-1,0)$

Ans. B

Sol. Let $z^2+\alpha z+\beta=0$ has $(1+iy_1)$ and $(1+iy_2)$

so $z_1z_2=\beta$

$(1+iy_1)(1+iy_2)=\beta$

$\beta=1-y_1y_2+i(y_1+y_2)$ ($\because$ β is purely real)

here $y_1+y_2=0$

$y_1=-y_2$

$\beta=1-y_1y_2$

$\beta=1+y_1^2$

$\beta>1$

$\Rightarrow \beta\in(1,\infty)$

10. If z is a complex number of unit modulus and argument θ, then $\arg\left(\frac{1+z}{1+\bar{z}}\right)$ equals : [JEE-(Main)2013]

यदि z एक ऐसी सम्मिश्र संख्या है जिसका मापांक एक इकाई है तथा कोणांक θ है, तो कोणांक $\arg\left(\frac{1+z}{1+\bar{z}}\right)$ बराबर है :

[JEE-(Main)2013]

(A) $-\theta$ (B) $\frac{\pi}{2}-\theta$ (C) θ (D) $\pi-\theta$

Ans. C

Sol. $\overline{z}=\frac{1}{z}$

$$\arg\left(\frac{1+z}{1+\frac{1}{z}}\right)\Rightarrow \arg z\Rightarrow\theta$$

11. A complex number z is said to be unimodular if $|z| = 1$. Suppose z_1 and z_2 are complex numbers such that $\frac{z_1-2z_2}{2-\overline{z}_1z_2}$ is unimodular and z_2 is not unimodular. Then the point z_1 lies on a : [JEE-(Main)2015]

(A) straight line parallel to x-axis (B) straight line parallel to y-axis

(C) circle of radius 2 (D) circle of radius $\sqrt{2}$

एक सम्मिश्र संख्या z एकमापांकी कहलाती है यदि $|z| = 1$ है। माना z_1 तथा z_2 ऐसी सम्मिश्र संख्याएँ हैं कि $\frac{z_1-2z_2}{2-\overline{z}_1z_2}$ एकमापांकी है तथा z_2 एकमापांकी नहीं है, तो बिन्दु z_1 स्थित है : [JEE-(Main)2015]

(A) x-अक्ष के समांतर एक रेखा पर (B) y-अक्ष के समांतर एक रेखा पर

(C) 2 त्रिज्या वाले वृत पर (D) $\sqrt{2}$ त्रिज्या वाले वृत पर

Ans. C

Sol. $\frac{1}{3}=1$

$(z_1-2z_2)(\overline{z}_1-2\overline{z}_2)=(2-z_1\overline{z}_2)(2-\overline{z}_1z_2)$

$z_1\overline{z}_1-2z_2\overline{z}_1-2z_1\overline{z}_2+4z_2\overline{z}_2=4-2z_1\overline{z}_2+z_1\overline{z}_1z_2\overline{z}_2$

$|z_1|^2|z_2|^2-4|z_2|^2-|z_1|^2+4=0$

$(|z_1|^2-4)(|z_2|^2-1)=0$

$|z_1|^2-4=0$

$|z_1|=2$

Circle of radius 2

12. A value of θ for which $\frac{2+3i\sin\theta}{1-2i\sin\theta}$ is purely imaginary, is : [JEE-(Main)2016]

θ का वह एक मान जिसके लिए $\frac{2+3i\sin\theta}{1-2i\sin\theta}$ पूर्णतः काल्पनिक है, है : [JEE-(Main)2016]

(A) $\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (B) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Ans. B

Sol. $z = \frac{2+3i\sin\theta}{1-2i\sin\theta}$

$z + \bar{z} = 0$

$\Rightarrow \sin^2\theta = 1/3$

$\sin\theta = \frac{1}{\sqrt{3}}$

13. Let $z = \cos\theta + i\sin\theta$. Then the value of $\sum_{m=1}^{15} \text{Im}(z^{2m-1})$ at $\theta = 2°$ is : [JEE-2009]

माना $z = \cos\theta + i\sin\theta$. तब $\sum_{m=1}^{15} \text{Im}(z^{2m-1})$ का मान $\theta = 2°$ पर निम्न है : [JEE-2009]

(A) $\frac{1}{\sin 2°}$ (B) $\frac{1}{3\sin 2°}$ (C) $\frac{1}{2\sin 2°}$ (D) $\frac{1}{4\sin 2°}$

Ans. D

Sol. Given that $z = \cos\theta + i\sin\theta = e^{i\theta}$

$$\therefore \sum_{m=1}^{15} \text{Im}(z^{2m-1}) = \sum_{m=1}^{15} \text{Im}\ (e^{i\theta})^{2m-1} = \sum_{m=1}^{15} \text{Im}\ (e^{i(2m-1)\theta}) = \sin\theta + \sin3\theta + \sin5\theta + \ldots\ldots\ldots + \sin29\theta$$

$$= \frac{\sin\left(\frac{\theta+29\theta}{2}\right)\sin\left(\frac{15\times 2\theta}{2}\right)}{\sin\left(\frac{2\theta}{2}\right)} = \frac{\sin(15\theta)\sin(15\theta)}{\sin\theta} = \frac{1}{4\sin 2°}$$

Hindi. दिया गया है $z = \cos\theta + i\sin\theta = e^{i\theta}$

$$\therefore \sum_{m=1}^{15} \text{Im}(z^{2m-1}) = \sum_{m=1}^{15} \text{Im}\ (e^{i\theta})^{2m-1} = \sum_{m=1}^{15} \text{Im}\ (e^{i(2m-1)\theta}) = \sin\theta + \sin3\theta + \sin5\theta + \ldots\ldots\ldots + \sin29\theta$$

$$= \frac{\sin\left(\frac{\theta+29\theta}{2}\right)\sin\left(\frac{15\times 2\theta}{2}\right)}{\sin\left(\frac{2\theta}{2}\right)} = \frac{\sin(15\theta)\sin(15\theta)}{\sin\theta} = \frac{1}{4\sin 2°}$$

14. Let $z = x + iy$ be a complex number where x and y are integers. Then the area of the rectangle whose vertices are the roots of the equation $z\bar{z}^3 + \bar{z}z^3 = 350$ is : [JEE-2009]

माना $z = x + iy$ एक ऐसी सम्मिश्र संख्या है जिसमें x व y पूर्णांक है। उस आयत का क्षेत्रफल जिसके शीर्ष बिन्दु समीकरण $z\bar{z}^3 + \bar{z}z^3 = 350$ के मूल है, निम्न है : [JEE-2009]

(A) 48 (B) 32 (C) 40 (D) 80

Ans. A

Sol. $z\bar{z}(z^2 + \bar{z}^2) = 350 \quad \Rightarrow 2(x^2 + y^2)(x^2 - y^2) = 350 \quad \Rightarrow (x^2 + y^2)(x^2 - y^2) = 175$

Since $x, y \in I$, the only possible case which gives integral solution, is

$x^2 + y^2 = 25$ ..... (1)

$x^2 - y^2 = 7$ .....(2)

From (1) and (2) $x^2 = 16$; $y^2 = 9$

$\Rightarrow x = \pm 4$; $y = \pm 3 \quad \Rightarrow$ Area = 48

GEOMETRICAL INTERPRETATION AND LOCUS

15. The points z_1, z_2, z_3, z_4 in the complex plane are the vertices of a parallelogram taken in order if and only if :

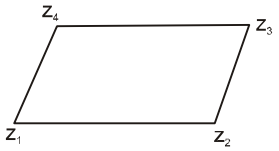
(A) $z_1 + z_4 = z_2 + z_3$ (B) $z_1 + z_3 = z_2 + z_4$ (C) $z_1 + z_2 = z_3 + z_4$ (D) None

सम्मिश्र समतल में बिन्दु z_1, z_2, z_3, z_4 एक क्रम में लेने पर किसी समान्तर चतुर्भुज के शीर्ष बिन्दु होगें यदि और केवल यदि :

(A) $z_1 + z_4 = z_2 + z_3$ (B) $z_1 + z_3 = z_2 + z_4$ (C) $z_1 + z_2 = z_3 + z_4$ (D) कोई नहीं

Ans. B

Sol.



Mid points of diagonals are equal (विकर्णो के मध्य बिन्दु समान होगें)

$$\frac{z_1 + z_3}{2} = \frac{z_2 + z_4}{2} \quad \Rightarrow z_1 + z_3 = z_2 + z_4$$

16. The number of solutions of the system of equations $\text{Re}(z^2) = 0, |z| = 2$ is :

समीकरण निकाय $\text{Re}(z^2) = 0, |z| = 2$ के हलों की संख्या हैं :

(A) 4 (B) 3 (C) 2 (D) 1

Ans. A

Sol. Let $z = x + iy$

$z^2 = x^2 - y^2 + 2ixy$

$\text{Re}(z^2) = 0$

$x^2 - y^2 = 0 \Rightarrow x = y$ or $x = -y$

$|z| = -2$

$x^2 + y^2 = 4$

$x - y \Rightarrow 2x^2 = 4$

$x = +\sqrt{2}$, $y = +\sqrt{2}$

$x = -\sqrt{2}$, $y = -\sqrt{2}$

$x = -y \Rightarrow 2x^2 = 4$

$x = \sqrt{2}$, $y = -\sqrt{2}$

$x = -\sqrt{2}$, $y = \sqrt{2}$

17. If $z\ (\neq -1)$ is a complex number such that $\frac{z-1}{z+1}$ is purely imaginary, then $|z|$ is equal to :

यदि $z\ (\neq -1)$ एक सम्मिश्र संख्या इस प्रकार है कि $\frac{z-1}{z+1}$ विशुद्ध काल्पनिक है, तो $|z|$ का मान है :

(A) 1 (B) 2 (C) 3 (D) 5

Ans. A

Sol. $\frac{z-1}{z+1} = \lambda i, \lambda \text{ real} \quad \Rightarrow \quad z = \frac{1+i\lambda}{1-\lambda i} \quad \Rightarrow \quad |z| = 1$

18. The curve represented by $|z| = \text{Re}(z) + 2$ is :

(A) a straight line (B) a circle (C) an ellipse (D) a parabola

$|z| = \text{Re}(z) + 2$ द्वारा प्रदर्शित वक्र है :

(A) सरल रेखा (B) वृत्त (C) दीर्घवृत्त (D) परवलय

Ans. D

Sol. $|z| = \text{Re}(z) + 2$

Let $z = x + iy$

$\sqrt{x^2 + y^2} = x + 2$

$x^2 + y^2 = (x + 2)^2$

$y^2 = 4x + 4$

Hence the curve is parabola

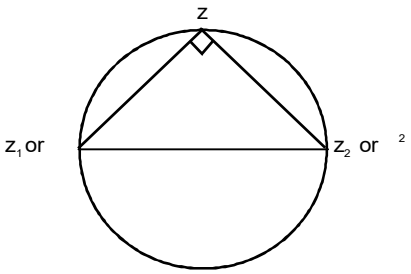
19. Let z_1 and z_2 be two nonreal complex cube roots of unity and $|z-z_1|^2+|z-z_2|^2=\lambda$ be the equation of a circle with z_1, z_2 as ends of a diameter then the value of λ is :

माना इकाई के दो अवास्तविक घनमूल z_1 व z_2 है तथा $|z-z_1|^2+|z-z_2|^2=\lambda$ उस वृत्त को प्रदर्शित करता है जिसके व्यास के सिरे z_1 व z_2 हो, तो λ का मान है :

(A) 4 (B) 3 (C) 2 (D) $\sqrt{2}$

Ans. B

Sol. ∵ Circle



so by pythogorous theorem $\lambda=|w-w^2|^2=\left|\sqrt{3}\right|^2=3$

20. If z is a complex number satisfying the equation $|z-(1+i)|^2=2$ and $\omega=\frac{2}{z}$, then the locus traced by 'ω' in the complex plane is :

यदि सम्मिश्र संख्या z समीकरण $|z-(1+i)|^2=2$ और $\omega=\frac{2}{z}$ को सन्तुष्ट करता है तब सम्मिश्र तल में 'ω' का बिन्दुपथ होगा :

(A) $x-y-1=0$ (B) $x+y-1=0$ (C) $x-y+1=0$ (D) $x+y+1=0$

Ans. A

Sol. We have $|z-(1+i)|^2=2$

$\Rightarrow \quad (x-1)^2+(y-1)^2=2 \qquad (\text{Put } z=x+iy)$

$\Rightarrow \quad x^2+y^2=2(x+y) \qquad .....(1)$

Let $\omega=h+ik=\frac{2}{z}=\frac{2}{x+iy}=\frac{2(x-iy)}{x^2+y^2}$, so

$$h=\frac{2x}{x^2+y^2}\ ,\ k=\frac{-2y}{x^2+y^2}$$

$\Rightarrow \quad h-k=\frac{2(x+y)}{x^2+y^2}=1 \quad (\text{from equation (1)})$

$\therefore \quad$ Locus of the point $\omega(h,k)$ will be $x-y=1$ Ans. (A)

21. If $|z-2| \geq |z-4|$ then correct statement is :

यदि $|z-2| \geq |z-4|$ तो सही कथन है :

(A) $\text{Re}(z) \geq 3$ (B) $\text{Re}(z) \leq 3$ (C) $\text{Re}(z) \geq 2$ (D) $\text{Re}(z) \leq 2$

Ans. A

Sol. $|z-2| \geq |z-4|$

Let $z = x + iy$

$|x + iy - 2| \geq |x + iy - 4|$

$|(x-2) + iy| \geq |x - 4 + iy|$

$(x-2)^2 + y^2 \geq (x-4)^2 + y^2$

$x + 4 - 4x \geq x^2 + 16 - 8x$

$x \geq 3$

22. If $w = \dfrac{z}{z - \frac{1}{3}i}$ and $|w| = 1$, then z lies on :

(A) an ellipse (B) a circle (C) a straight line (D) a parabola

यदि $w = \dfrac{z}{z - \frac{1}{3}i}$ एवं $|w| = 1$, तो z स्थित है :

(A) एक दीर्घ वृत्त पर (B) एक वृत्त पर (C) एक रेखा पर (D) एक परवलय पर

Ans. C

Sol. $w = \dfrac{z}{z - \frac{1}{3}i}, |w| = 1$

$$\left|\frac{z}{z - \frac{i}{3}}\right| = 1$$

$$\frac{|z|}{|z - \frac{i}{3}|} = 1 \Rightarrow |z| = \left|z - \frac{i}{3}\right|$$

$$x^2 + y^2 = (x)^2 + \left(y - \frac{1}{3}\right)^2$$

$6y = 1$

It's equaiton of straight line.

23. For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$, the minimum value of $|z_1 - z_2|$ is :

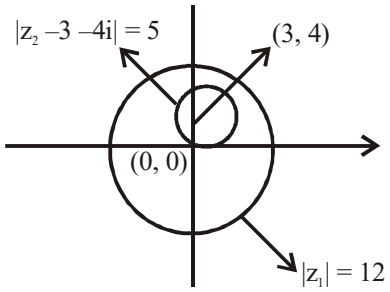
$|z_1| = 12$ और $|z_2 - 3 - 4i| = 5$ को संतुष्ट करने वाली सभी सम्मिश्र संख्याओं z_1, z_2 के लिये $|z_1 - z_2|$ का न्यूनतम मान है :

(A) 0 (B) 2 (C) 7 (D) 17

Ans. B

Sol. $|z_1| = 12 \Rightarrow$ It is a circle whose centre (0, 0) and radius = 12

$|z_2 - 3 - 4i| = 5 \Rightarrow$ It is a circle whose centre (3, 4) and radius 5



minimum of $|z_1 - z_2| = 12 - 10 = 2$

24. If $Z \neq 1$ and $\frac{z^2}{z-1}$ is real, then the point represented by the complex number z lies : [AIEEE-2012]

(A) On the imaginary axis

(B) Either on the real axis or on a circle passing through the origin

(C) On a circle with centre at the origin

(D) Either on the real axis or on a circle not passing through the origin

यदि $Z \neq 1$ तथा $\frac{z^2}{z-1}$ वास्तविक है, तो z द्वारा निरूपित सम्मिश्र संख्या वाला बिन्दु : [AIEEE-2012]

(A) काल्पनिक अक्ष पर है

(B) वास्तविक अक्ष पर है या मूल बिन्दु से होकर जाने वाले वृत्त पर है

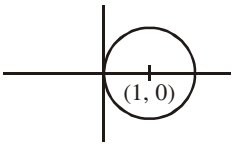
(C) एक वृत्त पर है जिसका केन्द्र मूल बिन्दु पर है

(D) वास्तविक अक्ष पर है या एक वृत्त पर है जो मूल बिन्दु से होकर नहीं जाता

Ans. B

Sol. $\frac{z^2}{z-1}$ is purely real where ($Z \neq 1$)

so $\dfrac{\overline{z}^2}{\overline{z}-1}=\dfrac{z^2}{z-1}$



$z\overline{z}^2-\overline{z}^2=\overline{z}z^2-z^2$

$z\overline{z}(z-\overline{z})=z^2-\overline{z}^2$

$z\overline{z}(z-\overline{z})=(z+\overline{z})(z-\overline{z})$

$\Rightarrow \overline{z}-z=0$ or $z+\overline{z}=z\overline{z}$

$\Rightarrow \overline{z}=z$ or $x^2+y^2-2x=0$

$(x-1)^2+y^2=1$

so either lie on z real axis or on a circle passing through the origin.

25. If z is a complex number such that $|z|\geq 2$, then the minimum value of $\left|z+\dfrac{1}{2}\right|$: [JEE-(Main)-2014]

(A) is strictly greater than $\dfrac{5}{2}$

(B) is strictly greater than $\dfrac{3}{2}$ but less than $\dfrac{5}{2}$

(C) is equal to $\dfrac{5}{2}$

(D) lies in the interval (1, 2)

यदि z एक ऐसी सम्मिश्र संख्या है कि $|z|\geq 2$ है, तो $\left|z+\dfrac{1}{2}\right|$ का न्यूनतम मान : [JEE-(Main)2014]

(A) $\dfrac{5}{2}$ से निरंतर बड़ा है

(B) $\dfrac{3}{2}$ से निरंतर बड़ा है परन्तु $\dfrac{5}{2}$ से कम है

(C) $\dfrac{5}{2}$ के बराबर है

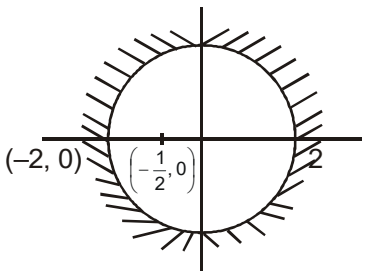
(D) अन्तराल (1, 2) में स्थित है

Ans. D

Sol. $\left|z+\dfrac{1}{2}\right|$ is distance of 'z' from $-\dfrac{1}{2}$.

Clearly its minimum value is when $z=-2$ which is 3/2.

(D) Ans.



26 If $|z|=1$ and $z\neq\pm 1$, then all the values of $\dfrac{z}{1-z^2}$ lie on : [JEE-2007]

(A) a line not passing through the origin (B) $|z| = \sqrt{2}$

(C) the x-axis (D) the y-axis

यदि $|z| = 1$ तथा $z \neq \pm 1$ तो $\frac{z}{1-z^2}$ के सभी मान स्थित है : [JEE-2007]

(A) एक रेखा जो मूल बिन्दु से नहीं गुजरती है (B) $|z| = \sqrt{2}$

(C) x-अक्ष (D) y-अक्ष

Ans. D

Sol. $|z| = 1,\ z \neq \pm 1$

Let $z = e^{i\theta}$

$$\Rightarrow \quad \frac{z}{1-z^2} = \frac{e^{i\theta}}{1-e^{i2\theta}} = \frac{1}{-2i\sin\theta} \text{ which is purely imaginary}$$

Hence $\frac{z}{1-z^2}$ lies on the y-axis

27. A particle P starts from the point $z_0 = 1 + 2i$, where $i = \sqrt{-1}$. It moves first horizontally away from origin by 5 units and then vertically away from origin by 3 units to reach a point z_1. From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $\hat{i} + \hat{j}$ and then it moves through an angle $\frac{\pi}{2}$ in anticlockwise direction on a circle with centre at origin, to reach a point z_2. The point z_2 is given by : [JEE-2007]

एक कण P बिन्दु $z_0 = 1 + 2i$ से शुरू होता है, जहाँ $i = \sqrt{-1}$ है। यदि वह पहले मूल बिन्दु से विपरित क्षैतिज दिशा में 5 इकाई तथा इसके बाद मूल बिन्दु से विपरीत उर्ध्वाधर दिशा में 3 इकाई जाता है तो बिन्दु z_1 पर पहुँचता है। z_1 से सदिश $\hat{i} + \hat{j}$ की दिशा में कण $\sqrt{2}$ इकाई दूरी तय करता है। तत्पश्चात् कण मूल बिन्दु को केन्द्र मानते हुये वामावृत दिशा में $\frac{\pi}{2}$ से घूर्णन करके z_2 पर पहुचता है। बिन्दु z_2 है : [JEE-2007]

(A) $6 + 7i$ (B) $-7 + 6i$ (C) $7 + 6i$ (D) $-6 + 7i$

Ans. D

Sol. $z_0 \equiv (1 + 2\,i)$

$z_1 \equiv (6 + 5\,i)$

$z_2 \equiv (-6 + 7i)$

ANGLES AND ROTATION OF COMPLEX NUMBER

28. If z_1, z_2, z_3 are vertices of an equilateral triangle inscribed in the circle $|z| = 2$ and if $z_1 = 1 + i\sqrt{3}$, then :

यदि एक वृत्त $|z| = 2$ के अन्तर्गत बनाए गए समबाहु त्रिभुज के शीर्ष z_1, z_2, z_3 हो तथा $z_1 = 1 + i\sqrt{3}$ हो, तो :

(A) $z_2 = -2, z_3 = 1 + i\sqrt{3}$ (B) $z_2 = 2, z_3 = 1 - i\sqrt{3}$

(C) $z_2 = -2, z_3 = 1 - i\sqrt{3}$ (D) $z_2 = 1 - i\sqrt{3}, z_3 = -1 - i\sqrt{3}$

Ans. C

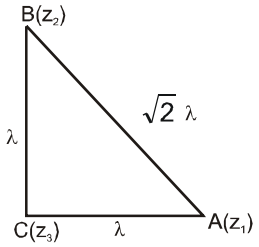
Sol. Apply Rotation (घूर्णन प्रमेय से)

$$\sqrt{2}(z_3 - z_1) = (z_2 - z_1)\, e^{i\pi/4}$$

$$(z_1 - z_2) = \sqrt{2}\,(z_3 - z_2)e^{i\pi/4}$$

$$\frac{\sqrt{2}(z_3 - z_1)}{(z_1 - z_2)} = \frac{(z_2 - z_1)}{\sqrt{2}\,(z_3 - z_2)}$$

$$2(z_3 - z_2)(z_1 - z_3) = (z_1 - z_2)^2$$



29. If z_1 and $\bar{z}_1$ represent adjacent vertices of a regular polygon of n sides with centre at the origin and if $\frac{\text{Im}\, z_1}{\text{Re}\, z_1} = \sqrt{2} - 1$ then the value of n is equal to :

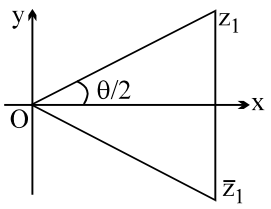
यदि z_1 और $\bar{z}_1$ किसी n भुजा वाले समबहुभुज के क्रमागत शीर्ष है जिसका केन्द्र मूल बिन्दु है। यदि $\frac{\text{Im}\, z_1}{\text{Re}\, z_1} = \sqrt{2} - 1$ तब n बराबर :

(A) 8 (B) 12 (C) 16 (D) 24

Ans. A

Sol. $\frac{y}{x} = \tan\frac{\theta}{2} = \sqrt{2} - 1 = \tan\frac{\pi}{8}$

$$= \frac{\theta}{2} = \frac{\pi}{8} \Rightarrow \theta = 45^\circ \Rightarrow n = \frac{360^\circ}{45^\circ} = 8$$



if $\frac{y}{x} = 2 - \sqrt{3} \quad \Rightarrow \quad n = 12$

30. Let P denotes a complex number z on the Argand's plane, and Q denotes a complex number $\sqrt{2|z|^2}\ \text{CiS}\left(\frac{\pi}{4} + \theta\right)$ where $\theta = \text{amp}\, z$. If 'O' is the origin, then the Δ OPQ is :

(A) isosceles but not right angled (B) right angled but not isosceles

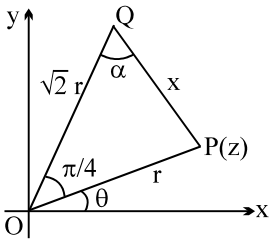
(C) right isosceles (D) equilateral

यदि सम्मिश्र तल में P सम्मिश्र संख्या z को निरूपित करता है तथा Q सम्मिश्र संख्या $\sqrt{2|z|^2}\ \text{CiS}\left(\frac{\pi}{4}+\theta\right)$ है जहां $\theta = \text{amp } z$. यदि 'O' मूल बिन्दु है तब Δ OPQ है :

(A) समद्विबाहु लेकिन समकोण नहीं
(B) समकोण लेकिन समद्विबाहु नहीं
(C) समकोण समद्विबाहु
(D) समबाहु त्रिभुज

Ans. C

Sol. $Z_P = r\,\text{Cis}\theta$; $Z_Q = \sqrt{2}\,|z|\,\text{Cis}\left(\theta+\frac{\pi}{4}\right) = \sqrt{2}\,r\left[\cos\left(\theta+\frac{\pi}{4}\right)+i\sin\left(\theta+\frac{\pi}{4}\right)\right]$



$$\cos\frac{\pi}{4} = \frac{2r^2+r^2-x^2}{2\cdot\sqrt{2}\,r\cdot r} = \frac{3r^2-x^2}{2\sqrt{2}\,r^2}$$

$$\therefore \quad 1 = \frac{3r^2-x^2}{2r^2} \quad \Rightarrow \quad r^2 = x^2 \quad \Rightarrow \quad x = r$$

31. z_1 and z_2 are two distinct points in an argand plane. If $a|z_1| = b|z_2|$, (where $a, b \in R$) then the point $\frac{a z_1}{b z_2}+\frac{b z_2}{a z_1}$ is a point on the :

(A) line segment $[-2, 2]$ of the real axis
(B) line segment $[-2, 2]$ of the imaginary axis
(C) unit circle $|z| = 1$
(D) the line with $\arg z = \tan^{-1}2$

यदि z_1 और z_2 सम्मिश्र तल में कोई दो बिन्दु है। यदि $a|z_1| = b|z_2|$, (जहां $a, b \in R$) तब बिन्दु $\frac{a z_1}{b z_2}+\frac{b z_2}{a z_1}$ कहाँ स्थित होगा :

(A) वास्तविक अक्ष पर रेखा खण्ड़ $[-2, 2]$ पर
(B) काल्पनिक अक्ष पर रेखा खण्ड़ $[-2, 2]$ पर
(C) इकाई वृत्त $|z| = 1$ पर
(D) रेखा जिसके लिए $\arg z = \tan^{-1}2$ पर

Ans. A

Sol. Assuming $\arg z_1 = \theta$ and $\arg z_2 = \theta+\alpha$.

$$\frac{a z_1}{b z_2}+\frac{b z_2}{a z_1} = \frac{a|z_1|e^{i\theta}}{b|z_2|e^{i(\theta+\alpha)}}+\frac{b|z_2|e^{i(\theta+\alpha)}}{a|z_1|e^{i\theta}} = e^{-i\alpha}+e^{i\alpha} = 2\cos\alpha$$

Alternatively: Let $\alpha = \frac{az_1}{bz_2}$; $\frac{1}{\alpha} = \frac{bz_2}{az_1}$; Also $|\alpha| = \frac{|az_1|}{|bz_2|} = \frac{a|z_1|}{b|z_2|} = 1 \Rightarrow \alpha = \frac{1}{\overline{\alpha}}$

$$\Rightarrow \alpha+\frac{1}{\alpha} = \alpha+\overline{\alpha} = 2\,\text{Re.}(\alpha) = 2\cos\alpha$$

32. The complex numbers z_1, z_2 and z_3 satisfying $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$ are the vertices of a triangle which is :

(A) of area zero
(B) right-angled isosceles
(C) equilateral
(D) obtuse – angled isosceles

$\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$ को संतुष्ट करने वाली सम्मिश्र संख्याएँ z_1, z_2 और z_3 जिस त्रिभुज के शीर्ष है, वह है :

(A) शून्य क्षेत्रफल वाला त्रिभुज
(B) समकोण समद्विबाहु त्रिभुज
(C) समबाहु त्रिभुज
(D) अधिक कोण समद्विबाहु

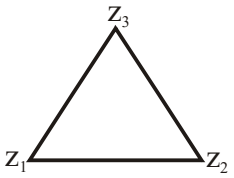
Ans. C

Sol. $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$

$\arg\left(\frac{z_1 - z_3}{z_2 - z_3}\right) = \left(\frac{-\pi}{3}\right)$

(-ve sign due to clockwise direction)

$\left|\frac{z_1 - z_3}{z_2 - z_3}\right| = \left|\frac{i - 1\sqrt{3}}{2}\right|$



$\left|\frac{z_1 - z_3}{z_2 - z_3}\right| = 1 \Rightarrow |z1 - z_3| = |z_2 - z_3|$

Two sides are equal then corresponding angles are also equal.

Every angle of triangle is $\frac{\pi}{3}$, so that it's an equilatent triangle.

33. The vector $z = -4 + 5i$ is turned counter clockwise through an angle of 180° and stretched 1.5 times. The complex number corresponding to the newly obtained vector is :

(A) $6 - \frac{15}{2}i$ (B) $-6 + \frac{15}{2}i$ (C) $6 + \frac{15}{2}i$ (D) None of these

सदिश $z = -4 + 5i$ को वामावर्त दिशा में 180° कोण पर घुमाया जाता है तथा 1.5 गुणा विस्तारित कर दिया जाता है। इस प्रकार प्राप्त सदिश के संगत सम्मिश्र संख्या है :

(A) $6 - \frac{15}{2}i$ (B) $-6 + \frac{15}{2}i$ (C) $6 + \frac{15}{2}i$ (D) इनमें से कोई नहीं

Ans. A

Sol. New $z = \frac{3}{2}(-4+5i)e^{i\pi}s = 6 - \frac{15}{2}i$

34. Points z_1 and z_2 are adjacent vertices of a regular octagon. The vertex z_3 adjacent to z_2 $(z_3 \neq z_1)$ is represented by:

(A) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 + z_2)$ (B) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 - z_2)$

(C) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_2 - z_1)$ (D) None of these

समअष्टभुज के आसन्न शीर्ष z_1 एवं z_2 है, तो z_2 के आसन्न शीर्ष z_3 $(z_3 \neq z_1)$ को प्रदर्शित किया जा सकता है :

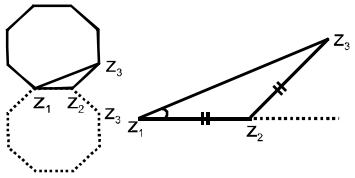
(A) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 + z_2)$ (B) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 - z_2)$

(C) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_2 - z_1)$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\frac{z_3 - z_2}{z_2 - z_1} = e^{\pm i\frac{\pi}{4}}$

$\because \quad |z_1 - z_2| = |z_2 - z_3|$



$\Rightarrow \quad z_3 = z_2 + \left(\frac{1 \pm i}{\sqrt{2}}\right)(z_2 - z_1)$

DE MOIVRE'S THEOREM

35. If $(\cos\theta + i\sin\theta)(\cos 2\theta + i\sin 2\theta) \ldots (\cos n\theta + i\sin n\theta) = 1$, then the value of θ is :

यदि $(\cos\theta + i\sin\theta)(\cos 2\theta + i\sin 2\theta) \ldots (\cos n\theta + i\sin n\theta) = 1$ हो, तो θ का मान है :

(A) $4m\pi, m \in Z$ (B) $\frac{2m\pi}{n(n+1)}, m \in Z$ (C) $\frac{4m\pi}{n(n+1)}, m \in Z$ (D) $\frac{m\pi}{n(n+1)}, m \in Z$

Ans. C

Sol. $e^{i\theta} \cdot e^{i2\theta} \ldots\ldots e^{in\theta} = 1 \quad \Rightarrow e^{i\theta\frac{(n)(n+1)}{2}} = e^{i2m\pi} \quad \Rightarrow \frac{\theta n(n+1)}{2} = 2m\pi$

$\theta = \frac{4m\pi}{n(n+1)} \quad m \in z.$

36. The complex number ω satisfying the equation $\omega^3 = 8i$ and lying in the second quadrant on the complex plane is :

द्वितीय चतुर्थांश में सम्मिश्र संख्या ω जो समीकरण $\omega^3 = 8i$ को सन्तुष्ट करता है, है :

(A) $-\sqrt{3}+i$ (B) $-\frac{\sqrt{3}}{2}+\frac{1}{2}i$ (C) $-2\sqrt{3}+i$ (D) $-\sqrt{3}+2i$

Ans. A

Sol. $\omega = 2\cdot i^{1/3} = 2\left(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}\right)^{1/3} = 2\left[\cos\frac{2n\pi+\frac{\pi}{2}}{3}+i\sin\frac{2n\pi+\frac{\pi}{2}}{3}\right]$

put $n=1$ $\quad = 2\left[\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}\right] = -\sqrt{3}+i$ Ans.

37. If m and n are the smallest positive integers satisfying the relation $\left(2\text{Cis}\frac{\pi}{6}\right)^m = \left(4\text{Cis}\frac{\pi}{4}\right)^n$, then $(m+n)$ has the value equal to :

यदि m और n न्यूनतम धनात्मक पूर्णांक है, जो सम्बन्ध $\left(2\text{Cis}\frac{\pi}{6}\right)^m = \left(4\text{Cis}\frac{\pi}{4}\right)^n$ को सन्तुष्ट करते हैं तब :

(A) 120 (B) 96 (C) 72 (D) 60

Ans. C

Sol. $2^{m-2n}\cdot\left[\cos\frac{m\pi}{6}+i\sin\frac{m\pi}{6}\right] = \left[\cos\frac{n\pi}{4}+i\sin\frac{n\pi}{4}\right]$

for equality $m=2n$

$\frac{n\pi}{4} = \frac{m\pi}{6}+2k\pi \qquad k\in I$

put $m=2n$

$\frac{n\pi}{4} = \frac{n\pi}{3}+2k\pi$; $\quad -\left(\frac{n\pi}{12}\right) = 2k\pi$ (ignore (–)ve sign)

$n=24k;$ $\quad m=48k;$ for m, n to be smallest $m+n=72$

38. Least positive argument of the 4th root of the complex number $2-i\sqrt{12}$ is :

$2-i\sqrt{12}$ के चतुर्थ मूल का न्यूनतम धनात्मक कोणांक है :

(A) $\frac{\pi}{6}$ (B) $\frac{5\pi}{12}$ (C) $\frac{7\pi}{12}$ (D) $\frac{11\pi}{12}$

Ans. B

Sol. $z^4 = 2\left(1-\sqrt{3}\,i\right) = 4\left(\frac{1}{2}-\frac{\sqrt{3}}{2}i\right) = 4\left[\cos\left(-\frac{\pi}{3}\right)+i\sin\left(-\frac{\pi}{3}\right)\right]$

$$z = \sqrt{2}\left[\cos\frac{2m\pi-(\pi/3)}{4}+i\sin\frac{2m\pi-(\pi/3)}{4}\right]$$

$$m = 1,\ z = \sqrt{2}\left[\cos\left(\frac{5\pi}{12}\right)+i\sin\left(\frac{5\pi}{12}\right)\right]$$

CUBE ROOT OF UNITY

39. If ω is an imaginary cube root of unity then $(1+\omega-\omega^2)(1+\omega^2-\omega)$ equals :

यदि ω इकाई का काल्पनिक घनमूल हो तो $(1+\omega-\omega^2)(1+\omega^2-\omega)$ बराबर है :

(A) 0 (B) 1 (C) 2 (D) 4

Ans. D

Sol. $(1+\omega-\omega^2)(1+\omega^2-\omega)$ $\therefore\ 1+\omega+\omega^2=0$

$= (-\omega^2-\omega^2)\ \ (-\omega-\omega)$

$= (-2\omega^2)(-2\omega)$

$= 4\omega^3$

$= 4$

40. If the cube roots of unity are 1, ω, ω^2 then the roots of the equation $(x-1)^3+8=0$, are :

यदि 1, ω तथा ω^2 घनमूल हैं 1 के, तो समीकरण $(x-1)^3+8=0$, के मूल हैं :

(A) $-1, -1+2\omega, -1-2\omega^2$ (B) $-1, -1, -1$

(C) $-1, 1-2\omega, 1-2\omega^2$ (D) $-1, 1+2\omega, 1+2\omega^2$

Ans. C

Sol. Since, $(x-1)^3+8=0$

$\Rightarrow (x-1)^3 = -8 = (-2)^3$ $\Rightarrow \left(\frac{x-1}{-2}\right)^3 = 1$ $\Rightarrow \left(\frac{x-1}{-2}\right) = (1)^{1/3}$

$\therefore$ Cube roots of $\left(\frac{x-1}{-2}\right)$ are 1, ω and ω^2. $\Rightarrow$ Cube roots of $(x-1)$ are $-2, -2\omega$ and $-2\omega^2$

$\Rightarrow$ Cube roots of x are $-1, 1-2\omega$ and $1-2\omega^2$.

41. If $\omega(\neq 1)$ is a cube root of unity and $(1+\omega)^7 = A + B\omega$. Then (A, B) equals : [AIEEE-2011]

यदि $\omega(\neq 1)$ इकाई का एक धन मूल है तथा $(1+\omega)^7 = A + B\omega$ है, तो (A, B) बराबर है : [AIEEE-2011]

(A) (1, 0) (B) (–1, 1) (C) (0, 1) (D) (1, 1)

Ans. D

Sol. $(1+\omega)^7 = A + B\omega$

$(-\omega^2)^7 = A + B\omega$

$-\omega^2 = A + B\omega$

$1 + \omega = A + B\omega$

$A = 1$

$B = 1 \qquad (1, 1)$

42. Let ω be a complex number such that $2\omega + 1 = z$ where $z = \sqrt{-3}$. If $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$, then k is equal to :

[JEE-(Main)2017]

माना ω एक सम्मिश्र संख्या ऐसी है कि $2\omega + 1 = z$ जहाँ $z = \sqrt{-3}$ है। यदि $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$ है, तो k बराबर है :

[JEE-(Main)2017]

(A) –1 (B) 1 (C) –z (D) z

Ans. C

Sol. $x + ky + 3z = 0$..... (i)

$3x + ky - 2z = 0$ .... (ii) system will have ∞ solutions

$2x + 4y - 3z = 0$ .... (iii)

(i) – (ii)

$x = \dfrac{5z}{2}$ then $y = \dfrac{-z}{2}$

$\dfrac{xy}{z^2} = 10$

43. If $\alpha, \beta \in C$ are the distinct roots, of the equation $x^2 - x + 1 = 0$, then $\alpha^{101} + \beta^{107}$ is equal to :

[JEE-(Main)2018]

यदि $\alpha, \beta \in C$ समीकरण $x^2 - x + 1 = 0$ के विभिन्न मूल है, तो $\alpha^{101} + \beta^{107}$ बराबर है : [JEE-(Main)2018]

(A) 1 (B) 2 (C) –1 (D) 0

Ans. A

Sol. Roots of $x^2 - x + 1 = 0$ are $\alpha = -\omega, \beta = -\omega^2$

$\alpha^{101} + \beta^{107} = -\omega^{101} - \omega^{214}$

$-\omega^2 - \omega = 1$

n^{th} ROOT OF UNITY

44. The value of $\sum_{k=1}^{10}\left(\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right)$ is :

$\sum_{k=1}^{10}\left(\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right)$ का मान है :

(A) 1 (B) – 1 (C) – i (D) i

Ans. C

Sol. $\sum_{k=1}^{10}\left(\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right) = i\sum_{k=1}^{10}\left(\cos\frac{2k\pi}{11} - i\sin\frac{2k\pi}{11}\right) = i\left\{\sum_{k=0}^{10}\left(e^{-\frac{2k\pi}{11}}\right) - 1\right\}$

$= -i$

Exercise J-2

BASICS OF COMPLEX NUMBERS

1. Solve the following equations over C and express the result in the form $a + ib,\ a, b \in R$.

निम्नलिखित समीकरणों को C में हल कीजिये तथा अपने परिणाम को $a + ib,\ a, b \in R$ के रूप में व्यक्त कीजिये।

(a) $ix^2 - 3x - 2i = 0$ (b) $2(1+i)x^2 - 4(2-i)x - 5 - 3i = 0$

Ans. (a) $-i, -2i$ (b) $\frac{3-5i}{2}$ or $-\frac{1+i}{2}$

Sol. (a) $a + ib,\ a, b \in R$

$$x = \frac{3 \pm \sqrt{9-8}}{2i}$$

$$x = \frac{3 \pm 1}{2i}$$

$$x = \frac{4}{2i} \times \frac{i}{i} = -2i$$

(b) $(1+i)x^2 - 4(2-i)x - 5 - 3i = 0$

$$x = \frac{4(2-i) \pm \sqrt{16(2i)^2 + 4 \times 2(1+i)(5+3i)}}{4(1+i)}$$

$$x = \frac{4(2-i) \pm \sqrt{16(3i-4i)^2 + 16(1+4i)}}{4(1+i)}$$

$$x = \frac{(2-i) \pm 2}{1+i}$$

$$x = \frac{4-i}{1+i} \times \frac{1-i}{1-i} = \frac{3-5i}{2}$$

$$x = \frac{-i}{1+i} \times \frac{1-i}{1-i} = -\frac{(1+i)}{2}$$

2. Let z be a complex number such that $z \in C\backslash R$ and $\frac{1+z+z^2}{1-z+z^2} \in R$, then prove that $|z| = 1$.

माना z एक सम्मिश्र संख्या इस प्रकार है कि $z \in C\backslash R$ तथा $\frac{1+z+z^2}{1-z+z^2} \in R$, तो सिद्ध कीजिये $|z| = 1$.

Ans.

Sol. $\frac{1+z+z^2}{1-z+z^2} \in R$

$$\text{Im}\left(\frac{1+z+z^2}{1-z+z^2}\right)=0$$

So that

$$\frac{1+z+z^2}{1-z+z^2}=\left(\frac{1+z+z^2}{1-z+z^2}\right)$$

$(1+z+z^2)(1-z+z^2)=(1-z+z^2)(1+z+z^2)$

$$\Rightarrow 2z-2\bar{z}=22\left(\bar{z}z^2-z\bar{z}^2\right)$$

$$=2\left(z-\bar{z}\right)=2\left(z|z|-\bar{z}|z|\right)$$

$|z| = 1$

3. Prove the identity, $|1-z_1\bar{z}_2|^2-|z_1-z_2|^2=\left(1-|z_1|^2\right)\left(1-|z_2|^2\right)$

सर्वसमिका को सिद्ध कीजिये, $|1-z_1\bar{z}_2|^2-|z_1-z_2|^2=\left(1-|z_1|^2\right)\left(1-|z_2|^2\right)$

Ans.

Sol. $|1-z_1\bar{z}_2|^2-|z_1-z_2|^2=\left(1-|z_1\bar{z}_2|\right)\left(1-\bar{z}_1z_2\right)-\left(z_1-z_2\right)\left(\bar{z}_1-\bar{z}_2\right)$

$$=-\bar{z}_1z_2+z_1\bar{z}_2+z_1\bar{z}_1\bar{z}_2\bar{z}_2-z_1\bar{z}_1+z_1\bar{z}_2+z_2\bar{z}_1-\bar{z}_2\bar{z}_2+z_2\bar{z}_1-z_2\bar{z}_2$$

$$=1-|z_1|^2|z_2|^2-|z_1|^2-|z_2|^2$$

$$=\left(1-|z_1|^2\right)-|z_2|^2-\left(1-|z_1|^2\right)$$

$= (1 - |z_1|^2)(1 - |z_2|^2)$

4. Prove the identity, $|1+z_1\bar{z}_2|^2+|z_1-z_2|^2=\left(1+|z_1|^2\right)\left(1+|z_2|^2\right)$

सर्वसमिका को सिद्ध कीजिये, $|1+z_1\bar{z}_2|^2+|z_1-z_2|^2=\left(1+|z_1|^2\right)\left(1+|z_2|^2\right)$

Ans.

Sol. $|1+z_1\bar{z}_2|^2+|z_1-z_2|^2$

$|1+z_1\bar{z}_2|^2+|z_1-z_2|^2=\left(1+|z_1|^2\right)\left(1+|z_2|^2\right)$

$= 1 + \overline{z}_1 z_2 + z_1 \overline{z}_2 + z_1 \overline{z}_1 z_2 \overline{z}_2 + z_1 \overline{z}_1 - z_1 \overline{z}_2 - z_2 \overline{z}_1 + z_2 \overline{z}_2$

$= 1 + |z_1|^2 |z_2|^2 + |z_1|^2 + |z_2|^2$

$= (1 + |z_1|^2) + |z^2|^2 (1 + |z_1|^2)$

$= (1 + |z_1|^2)(1 + |z_2|^2)$

5. (a) Find all non–zero complex numbers Z satisfying $\overline{Z} = i Z^2$.

(b) If the complex numbers $z_1, z_2, \ldots\ldots\ldots\ldots z_n$ lie on the unit circle $|z| = 1$ then show that

$$|z_1 + z_2 + \ldots\ldots\ldots\ldots + z_n| = |z_1^{-1} + z_2^{-1} + \ldots\ldots\ldots\ldots + z_n^{-1}|.$$

(a) $\overline{Z} = i z^2$ को संतुष्ट करने वाले सभी अशुन्य सम्मिश्र संख्याओं Z को ज्ञात करें।

(b) यदि सम्मिश्र संख्यायें $z_1, z_2, \ldots\ldots\ldots\ldots z_n$ एक ईकाई वृत $|z| = 1$ पर स्थित है तो सिद्ध कीजिये।

$$|z_1 + z_2 + \ldots\ldots\ldots\ldots + z_n| = |z_1^{-1} + z_2^{-1} + \ldots\ldots\ldots\ldots + z_n^{-1}|.$$

Ans. (a) $\frac{\sqrt{3}}{2} - \frac{i}{2}, -\frac{\sqrt{3}}{2} - \frac{i}{2}, i$

Sol. (i) $\overline{Z} = i z^2$

let $z = x + iy$

$x - iy = i (x + iy)^2$

$x - iy = i (x^2 - y^2 + 2ixy)$

$x - iy = i (x^2 - y^2) - 2xy$

By comperision

$\Rightarrow x = -2xy$

$x (2y + 1) = 0$

$x = 0, y = \frac{1}{2}$

$\Rightarrow x^2 - y^2 = -y$

when $x = 0, y = 0, -1$

and $y = \frac{1}{2}, x = \pm\frac{\sqrt{3}}{2}$

Non zero, complex number

$Z = -i, \frac{\sqrt{3}}{2} - \frac{i}{2}, \frac{-\sqrt{3}}{2} - \frac{i}{2}$

(B) $|z| = 1$

$|z|^2 = 1 \Rightarrow z\overline{z} = 1$

So that $Z = \frac{1}{\overline{z}}$

$|z_1 + z_2 + z_3 + ... + z_n| = |z_1 + z_2 + ... + z_n|$

$= |\overline{z}_1 + \overline{z}_1 + ... + \overline{z}_n|$

$= \left|\frac{1}{z_1} + \frac{1}{z_2} + ... + \frac{1}{z_n}\right|$

6. Let $z = (0, 1) \in C$. Express $\sum_{k=0}^{n} z^k$ in terms of the positive integer n.

माना $z = (0, 1) \in C$. $\sum_{k=0}^{n} z^k$ को धनात्मक पूर्णांक n के रूप में व्यक्त कीजिये।

Ans. $\begin{cases} (1,0) & \text{for } n = 4k \\ (1,1) & \text{for } n = 4k+1 \\ (0,1) & \text{for } n = 4k+2 \\ (0,0) & \text{for } n = 4k+3 \end{cases}$

Sol. $z = (0, 1) \in C$

$\sum_{k=0}^{n} z^k = z^0 + z^1 + z^2 + ... + z^n$

7. Let Z is complex satisfying the equation, $z^2 - (3+i)z + m + 2i = 0$, where $m \in R$. Suppose the equation has a real root, then find the value of *m*.

माना Z एक सम्मिश्र संख्या है जो समीकरण $z^2 - (3 + i)z + m + 2i = 0$ को संतुष्ट करती है, जहाँ $m \in R$ है। माना समीकरण वास्तविक मूल है तो *m* का मान ज्ञात कीजियें।

Ans. 2

Sol. $z^2 - (3 + i)z + m + 2i = 0$

Let t be the real root of equation

$i^2 - (3 + i)t + m + 2i = 0$

$i^2 - 3 - it + m + 2i = 0$

$(t^2 - 3t + m) + i(-t + 2) = 0$

Real and Imaginary parts are equal to zero.

$-t + 2 = 0 \Rightarrow t = +2$

So that $t^2 - 3t + m = 0$

$4 - 6 + m = 0$

$m = 2$

8. a, b, c are real numbers in the polynomial, $P(Z) = 2Z^4 + aZ^3 + bZ^2 + cZ + 3$
If two roots of the equation $P(Z) = 0$ are 2 and i, then find the value of 'a'.

a, b, c वास्तविक संख्याये है, बहुपद $P(Z) = 2Z^4 + aZ^3 + bZ^2 + cZ + 3$ है। यदि समीकरण $P(Z) = 0$ के दो मूल 2 तथा i है, तो 'a' का मान ज्ञात करें।

Ans. $-11/2$

Sol. $P(Z) = 2Z^4 + aZ^3 + bZ^2 + cZ + 3$

Roots of of the equation are $\alpha = 2, \beta = i, \gamma = -i$ and δ.

$\therefore$ Complex roots appears in conjugate form product of roots $= -\frac{3}{2}$.

$-t + 2 = 0$ $t = +2$

Sum of ro

9. (a) Find the real values of x & y for which $z_1 = 9y^2 - 4 - 10ix$ and $z_2 = 8y^2 - 20i$ are conjugate complex of each other.

वास्तविक संख्यायें x तथा y ज्ञात कीजिये जिनके लिये $z_1 = 9y^2 - 4 - 10ix$ तथा $z_2 = 8y^2 - 20i$ एक दूसरे के सम्मिश्र संयुग्मी है।

(b) Find the value of $x^4 - x^3 + x^2 + 3x - 5$ if $x = 2 + 3i$
यदि $x = 2 + 3i$ तो $x^4 - x^3 + x^2 + 3x - 5$ का मान ज्ञात कीजिये।

Ans. (a) $[(-2, 2) ; (-2, -2)]$ (b) $-(77 + 108i)$

Sol. (a) $\bar{z}_1 = z_2 \Rightarrow 9y^2 - 4 + 10ix = 8y^2 - 20i \Rightarrow y^2 = 4$

$\Rightarrow y = \pm 2 \quad x = -2$

(b) $\Rightarrow x^4 - x^3 + x^2 + 3x - 5 = (2 + 3i)^4 - (2 + 3i)^3 + (2 + 3i)^2 + 3(2 + 3i) - 5 = -(77 + 108i)$

10. Solve the following equation $z^2 - (3 - 2i)z = (5i - 5)$ expressing your answer in the form of $(a + ib)$.

समीकरण $z^2 - (3 - 2i)z = (5i - 5)$ को हल करे एवं उत्तर $(a + ib)$ के रूप में व्यक्त करें।

Ans. $z = (2 + i)$ or $(1 - 3i)$

Sol. $z^2 - (3 - 2i)z + (5 - 5i) = 0$

$$z = \frac{(3-2i) \pm \sqrt{(3-2i)^2 - 4(5-5i)}}{2} = \frac{(3-2i) \pm \sqrt{-15+8i}}{2} = \frac{(3-2i) \pm (1+4i)}{2}$$

$= 2 + i$ or $1 - 3i$

11. If $(1 - i)$ is a root of the equation, $z^3 - 2(2 - i)z^2 + (4 - 5i)z - 1 + 3i = 0$, then find the other two roots.

यदि समीकरण $z^3 - 2(2 - i)z^2 + (4 - 5i)z - 1 + 3i = 0$ का एक मूल $(1 - i)$ है, तो अन्य दो मूल ज्ञात करें।

Ans. $z = 1$ or $2 - i$

Sol. $z^3 - 2(2 - i)z^2 + (4 - 5i)z - 1 + 3i = 0$
Let roots of equation are α,β,γ.
$\alpha = 1 - i$
Sum of roots
$\alpha + \beta + \gamma = 2\ (2 - i)$
$1 - i + \beta + \gamma = 4 - 2i$
$\beta + \gamma = 3 - i$ ...(1)
Product of roots
$\alpha\beta\gamma = 3i - 1$
$(1 - i)\beta\gamma = 3i - 1$
$\beta\gamma = i - 2$ ...(2)
So that other roots are
$z = 1$ or $2 - i$

12. If $iZ^3 + Z^2 - Z + i = 0$, then find $|Z|$.

यदि $iZ^3 + Z^2 - Z + i = 0$ है, तो $|Z|$ ज्ञात करें।

Ans. 1

Sol. $iz^3 + z^2 - z + i = 0$
$\Rightarrow z^2\ (iz + 1) + (-z + i) = 0$
multiply by i
$iz^2\ (iz + 1) + (-iz - 1) = 0$
$(iz + 1)\ (iz^2 - 1) = 0$

Case-I : $z = -\frac{1}{i} = i \Rightarrow |z| = 1$

Case II : $z^2 = \frac{1}{i} \Rightarrow z^2 = -i$

$z = \frac{1}{\sqrt{2}}i + \frac{1}{\sqrt{2}}i$ or $-\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}} \Rightarrow |z| = 1$

13. Show that the product, $\left[1+\left(\frac{1+i}{2}\right)\right]\left[1+\left(\frac{1+i}{2}\right)^2\right]\left[1+\left(\frac{1+i}{2}\right)^{2^2}\right].....\left[1+\left(\frac{1+i}{2}\right)^{2^n}\right]$ is equal to $\left(1 - \frac{1}{2^{2^n}}\right)(1+i)$ where $n \geq 2$.

दर्शाइये की गुणन, $\left[1+\left(\frac{1+i}{2}\right)\right]\left[1+\left(\frac{1+i}{2}\right)^2\right]\left[1+\left(\frac{1+i}{2}\right)^{2^2}\right].....\left[1+\left(\frac{1+i}{2}\right)^{2^n}\right]$ का मान $\left(1 - \frac{1}{2^{2^n}}\right)(1+i)$ के बराबर है, जहाँ $n \geq 2$ है।

Ans.

Sol. $\left[1+\left(\frac{1+i}{2}\right)\right]\left[1+\left(\frac{1+i}{2}\right)^2\right]\left[1+\left(\frac{1+i}{2}\right)^{2^2}\right].....\left[1+\left(\frac{1+i}{2}\right)^{2^n}\right]$

Let $z=\frac{1+i}{2}$

$=(1+z)(1+z)^2(1+z)^{2^2}.....\left(1+z^{2^n}\right)$

multiply and divide by $1-z$

$=\frac{\left(1-z^2\right)}{1-z}\left(1+z^2\right)\left(1+z^{2^2}\right).....\left(1+z^{2^n}\right)$

$\Rightarrow\frac{\left(1-\left(z^{2^n}\right)^2\right)}{1-z}$

$\Rightarrow\frac{\left(1-\left(z^2\right)^{2^n}\right)}{1-z}$

$=\frac{\left(1-\left(\frac{1-i}{2}\right)^{2^n}\right)}{1-\left(\frac{1-i}{2}\right)}$

$=\left(1-\frac{1}{2^{2^n}}\right)(1+i)$

14. Find the minimum value of the expression $E = |z|^2 + |z-3|^2 + |z-6i|^2$ (where $z = x + iy$, $x, y \in R$).

व्यंजक $E = |z|^2 + |z-3|^2 + |z-6i|^2$ का न्यूनतम मान ज्ञात कीजिये। (जहाँ $z = x + iy$, $x, y \in R$)

Ans. 30

Sol. $E = |z|^2 + |z-3|^2 + |z-6i|^2$
$z = x + iy$
$E = x^2 + y^2 + x^2 + 9 - 6x + y^2 + x^2 + y^2 + 36 - 12y$
$E = 3x^2 - 6x + 3y^2 - 12y + 45$
$E = 3(x^2 - 2x + y^2 - 4y + 15)$
$E = 3((x-1)^2 + (y-2)^2 + 10)$
$E = 3(x-1)^2 + 3(y-2)^2 + 30$
So that minimum value of E is 30.

15. Let $A = \{a \in R \mid \text{the equation } (1+2i)x^3 - 2(3+i)x^2 + (5-4i)x + 2a^2 = 0\}$ has at least one real root. Find the value of $\sum_{a \in A} a^2$.

माना $A = \{a \in R \mid$ समीकरण $(1+2i)x^3 - 2(3+i)x^2 + (5-4i)x + 2a^2 = 0$ का कम से कम एक वास्तविक मूल है$\}$ $\sum_{a \in A} a^2$ का मान ज्ञात कीजिये।

Ans. 18

Sol. $(1+2i)x^3 - 2(3+i)x^2 + (5-4i)x + 2a^2 = 0$
Let t is real root of the equation
$(t^3 - 2t^2 + 5t + 2a^2) + i(2t^3 - 2t^2 - 4t) = 0$
By comparison
$\Rightarrow 2t^3 - 2t^2 - 4t = 0 \quad \Rightarrow t = 0, 2, -1$
and $t^3 - 2t^2 + 5t + 2a^2 = 0$
At $t = 0$, $a^2 = 0 \Rightarrow a = 0$
$t = 2$, $a = \pm\sqrt{3}$

16. One root of the cubic $2z^3 - (5+6i)z^2 + 9iz + 1 - 3i = 0$ is real. If all the three roots of this cubic are plotted on the complex plane, the find the area of the triangle formed by them.

घन समीकरण $2z^3 - (5+6i)z^2 + 9iz + 1 - 3i = 0$ का एक मूल वास्तविक है। इस घन समीकरणों के मूलों को सम्मिश्र तल में निरूपित करने से बने त्रिभुज का क्षेत्रफल ज्ञात करें।

Ans. $\frac{1}{4}$

Sol. $2z^3 - (5+6i)z^2 + 9iz + 1 - 3i = 0$

Put $z = \frac{1}{2}$

$$\Rightarrow 2\left(\frac{1}{8}\right) - (5+6i)\frac{1}{4} + \frac{9i}{2} + 1 - 3i$$

$$\Rightarrow \frac{1}{4} - \frac{5}{4} - \frac{6i}{4} + \frac{9i}{2} + 1 - 3i$$

$$\Rightarrow 0$$

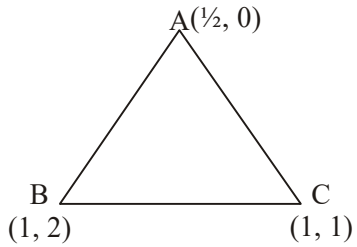
$$\Rightarrow \left(2 - \frac{1}{2}\right)(2z^2 - 14 + 6i)z - 2 + 6i = 0$$

$$\Rightarrow z^2 - (2+3i)z - 1 + 3i = 0$$

$$z=\frac{2+3i\pm\sqrt{4-9+12i+4-12i}}{2\times1}$$

$$z=\frac{2+3i\pm i}{2}$$

$$z=1+2i,\ 1+i$$



$$\text{Ar}(\Delta ABC)=\frac{1}{2}\begin{vmatrix}\frac{1}{2} & 0 & 1\\ 1 & 2 & 1\\ 1 & 1 & 1\end{vmatrix}=\left|-\frac{1}{4}\right|$$

$$=\frac{1}{4}$$

17. Prove that $\left(\frac{1+\sin\theta+i\cos\theta}{1+\sin\theta-i\cos\theta}\right)^n=\cos\left(\frac{n\pi}{2}-n\theta\right)+i\sin\left(\frac{n\pi}{2}-n\theta\right)$. Hence deduce that

$$\left(1+\sin\frac{\pi}{5}+i\cos\frac{\pi}{5}\right)^5+i\left(1+\sin\frac{\pi}{5}-i\cos\frac{\pi}{5}\right)^5=0$$

सिद्ध करो कि $\left(\frac{1+\sin\theta+i\cos\theta}{1+\sin\theta-i\cos\theta}\right)^n=\cos\left(\frac{n\pi}{2}-n\theta\right)+i\sin\left(\frac{n\pi}{2}-n\theta\right)$ तदानुसार

$\left(1+\sin\frac{\pi}{5}+i\cos\frac{\pi}{5}\right)^5+i\left(1+\sin\frac{\pi}{5}-i\cos\frac{\pi}{5}\right)^5=0$ बताइयें।

Ans.

Sol. $$\Rightarrow\left(\frac{1+\sin\theta+i\cos\theta}{1+\sin\theta-i\cos\theta}\times\frac{1+\sin\theta+i\cos\theta}{1+\sin\theta+i\cos\theta}\right)^n$$

$$\Rightarrow\left(\frac{1+\sin^2\theta+2\sin\theta-\cos^2\theta+2i\cos\theta(1+\sin\theta)}{1+\sin^2\theta+2\sin\theta+\cos^2\theta}\right)^n$$

$$\Rightarrow\left(\frac{1+\sin^2\theta+2\sin\theta-1+\sin^2\theta+2i\cos\theta(1+\sin\theta)}{2(1+\sin\theta)}\right)^n$$

$$\Rightarrow \left(\frac{2\sin\theta(1+\sin\theta)+2i\cos\theta(1+\sin\theta)}{2(1+\sin\theta)} \right)^n$$

$$\Rightarrow (\sin\theta + i\cos\theta)^n$$

$$\Rightarrow \left(\cos\left(\frac{\pi}{2}-\theta\right)+i\sin\left(\frac{\pi}{2}-\theta\right) \right)^n$$

$$\Rightarrow \left(\cos\left(\frac{n\pi}{2}-n\theta\right)+i\sin\left(\frac{n\pi}{2}-n\theta\right) \right)$$

18. Dividing f(z) by z – i, we get the remainder i and dividing it by z + i, we get the remainder 1 + i. Find the remainder upon the division of f(z) by $z^2 + 1$.

f(z) को z – i से भाग देने पर, हमें शेषफल i प्राप्त होता है तथा इसे z + i से भाग देने पर हमे शेषफल 1 + i प्राप्त होता है। f(z) का z^2 + 1 से भाग देने पर क्या शेषफल प्राप्त होगा?

Ans. $\frac{iz}{2} + \frac{1}{2} + i$

Sol. Remainder is of form az + b az + b के रूप का शेषफल

ai + b = i

– ai + b = 1 + i

$b = \frac{1}{2} + i \qquad a = \frac{i}{2}$

Remainder $= \frac{i}{2}z + \frac{1}{2} + i$ शेषफल $= \frac{i}{2}z + \frac{1}{2} + i$

19. If z_1 and z_2 are two complex numbers such that $|z_1| < 1 < |z_2|$ then prove that $\left|\frac{1-z_1\bar{z}_2}{z_1-z_2}\right| < 1$.

यदि z_1 और z_2 दो सम्मिश्र संख्याएँ इस प्रकार है कि $|z_1| < 1 < |z_2|$ हो, तब सिद्ध कीजिए कि $\left|\frac{1-z_1\bar{z}_2}{z_1-z_2}\right| < 1$.

Ans.

Sol. Let $\left|\frac{1-z_1\bar{z}_2}{z_1-z_2}\right| < 1 \quad \Rightarrow \quad |1-z_1\bar{z}_2| < |z_2-z_1|$

$\Rightarrow \quad (1-z_1\bar{z}_2)(1-\bar{z}_1z_2) < (z_2-z_1)(\bar{z}_2-\bar{z}_1) \quad \Rightarrow \quad 1+|z_1|^2|z_2|^2-|z_1|^2-|z_2|^2 < 0$

$\Rightarrow \quad (1 - |z_1|^2) + (|z_1|^2 - 1)\,|z_2|^2 < 0 \qquad \Rightarrow \quad (1 - |z_1|^2)\,(1 - |z_2|^2) < 0$

which is true because of $|z_1| < 1 < |z_2|$.

GEOMETRICAL INTERPRETATION AND LOCUS

20. Among the complex numbers z satisfying the condition $\left|z + 3 - \sqrt{3}\,i\right| = \sqrt{3}$, find the number having the least positive argument.

सम्मिश्र संख्याओं, जो $\left|z + 3 - \sqrt{3}\,i\right| = \sqrt{3}$ को संतुष्ट करती है, में से वह संख्या ज्ञात कीजिये जिसका न्यूनतम धनात्मक कोणांक है।

Ans. $-\frac{3}{2} + \frac{3\sqrt{3}}{2}\,i$

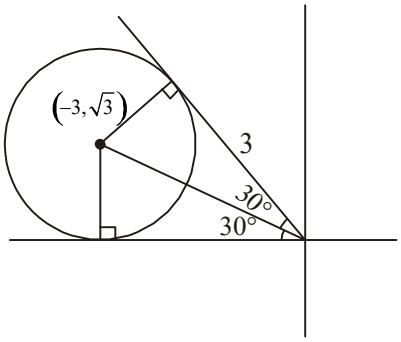
Sol. $12 - \left(-3 + \sqrt{3}i\right) = \sqrt{3}$

$$z = 3e^{\frac{i2\pi}{3}}$$

$$z = 3\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

$$z = 3\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)$$

$$z = \frac{-3}{2} + \frac{i3\sqrt{3}}{2}$$



21. Let $z_1 = 1 + i$ and $z_2 = -1 - i$. Find $z_3 \in C$ such that triangle z_1, z_2, z_3 is equilaterial.

माना $z_1 = 1 + i$ तथा $z_2 = -1 - i$ है। $z_3 \in C$ का मान निकालिये जब z_1, z_2, z_3 समबाहु त्रिभुज है।

Ans. $z_3 = \sqrt{3}(1 - i)$ and $z'_3 = \sqrt{3}(-1 + i)$

Sol. Let us have point z_3 as $x + y$

Now, for $\Delta z_1 z_2 z_3$ to be equilateral,

$z_1 z_2 = z_2 z_3 = z_1 z_3$

$$\therefore z_1 z_2 = \sqrt{\left(1-(-1)\right)^2 + \left(1-(-1)\right)^2} = 2\sqrt{2} \qquad \text{_____(1)}$$

$$\therefore\ z_2 z_3 = \sqrt{\left(x-(-1)\right)^2 + \left(y-(-1)\right)^2} \qquad \text{_____(2)}$$

$\therefore\ z_1z_3 = \sqrt{(x+1)^2+(y+1)^2}$ _____(3)

equating (1) and (2),

$2\sqrt{2} = \sqrt{(x-(-1))^2+(y-(-1))^2}$

squaring both sides,

$8 = x^2+2x+y^2+2y+2$

$\therefore x^2+2x+y^2+2y-6=0$ _____(4)

equating (3) and (1),

$2\sqrt{2} = \sqrt{(x-1)^2+(y-1)^2}$

squarinig both sides,

$8 = x^2-2x+y^2-2y+2$

$\therefore x^2-2x+y^2-2y-6=0$ _____(5)

subtracting (5) from (4), we have

$x^2+2x+x^2-2x-6=0$

$\therefore x^2=3$

$\therefore x=\pm\sqrt{3}$

Hence, z_3 can be $(\sqrt{3},-\sqrt{3})$ or $(-\sqrt{3},\sqrt{3})$.

22. Find the Cartesian equation of the locus of 'z' in the complex plane satisfying, $|z-4|+|z+4|=16$.

z के बिन्दु पथ का कार्तीय समीकरण ज्ञात करे जो सम्मिश्र तल में $|z-4|+|z+4|=16$ को संतुष्ट करता है।

Ans. $\frac{x^2}{64}+\frac{y^2}{48}=1$

Sol. $|z-4|+|z-4|=16$

$|z-4|=16-|z+4|$

$(|z-4|)^2=(16-12+41)^2$

$z = x + y$

$$(x-y)^2 + y^2 = 16^2 + (x+4)^2 + y^2 - 32\sqrt{(x+4)^2 + y^2}$$

$$x^2 + 16 - 8x = 16^2 + x^2 + 16 + 8x - 32\sqrt{(x+4)^2 + y^2}$$

$$x^2 + 16 - 8x = 16^2 + x^2 + 16 + 8x - 32\sqrt{(x+4)^2 + y^2}$$

$$32\sqrt{(x+4) + y^2} = 16x + 16$$

$$2\sqrt{(x+y)^2 + y^2} = x + 16$$

$$(x+4)^2 + y^2 = \left(\frac{y}{2} + 8\right)^2$$

$$x^2 + 16 + 8x + y^2 = \frac{x^2}{4} + 64 + 8x$$

$$\frac{3x^2}{4} + y^2 = 48$$

23. Interpret the following locii in $z \in C$:

$z \in C$ में निम्नलिखित बिन्दु पथों का विवेचना कीजिये :

(a) $1 < |z - 2i| < 3$

(b) $\text{Re}\left(\frac{z+2i}{iz+2}\right) \le 4 \ (z \neq 2i)$

(c) $\text{Arg}(z+i) - \text{Arg}(z-i) = \pi/2$

(d) $\text{Arg}(z-a) = \pi/3$ where $a = 3 + 4i$

$z \in C$ में निम्नलिखित बिन्दु पथों का विवेचना कीजिये :

(a) $1 < |z - 2i| < 3$

(b) $\text{Re}\left(\frac{z+2i}{iz+2}\right) \le 4 \ (z \neq 2i)$

(c) कोणांक $(z+i)$ – कोणांक $(z-i) = \pi/2$

(d) कोणांक $(z-a) = \pi/3$ जहाँ $a = 3 + 4i$

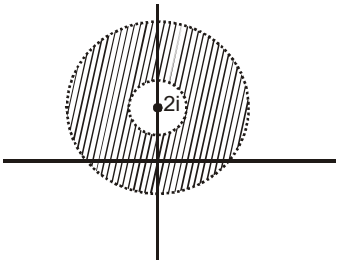
Ans. (a) The region between the co encentric circles with centre at $(0, 2)$ and radii 1 and 3 units

(b) region outside or on the circle with centre $\frac{1}{2} + 2i$ and radius $\frac{1}{2}$.

(c) semi circle (in the 1st & 4th quadrant) $x^2 + y^2 = 1$

(d) a ray emanating from the point $(3+4i)$ directed away from the origin and having equation $\sqrt{3}x - y + 4 - 3\sqrt{3} = 0$

Sol.



24. Let $A \equiv z_1$; $B \equiv z_2$; $C \equiv z_3$ are three complex numbers denoting the vertices of an acute angled triangle. If the origin 'O' is the orthocentre of the triangle, then prove that :

$z_1\bar{z}_2 + \bar{z}_1 z_2 = z_2\bar{z}_3 + \bar{z}_2 z_3 = z_3\bar{z}_1 + \bar{z}_3 z_1$

hence show that the ΔABC is a right angled triangle $\Leftrightarrow z_1\bar{z}_2 + \bar{z}_1 z_2 = z_2\bar{z}_3 + \bar{z}_2 z_3 = z_3\bar{z}_1 + \bar{z}_3 z_1 = 0$

माना $A \equiv z_1$; $B \equiv z_2$; $C \equiv z_3$ तीन सम्मिश्र संख्यायें एक न्यूनकोण त्रिभुज के शीर्ष है। यदि 'O' मूल बिन्दु त्रिभुज का लम्बकेन्द्र है तो सिद्ध कीजिये :

$z_1\bar{z}_2 + \bar{z}_1 z_2 = z_2\bar{z}_3 + \bar{z}_2 z_3 = z_3\bar{z}_1 + \bar{z}_3 z_1$

तदानुसार दर्शाइये कि त्रिभुज ABC समकोणीय होगा $\Leftrightarrow z_1\bar{z}_2 + \bar{z}_1 z_2 = z_2\bar{z}_3 + \bar{z}_2 z_3 = z_3\bar{z}_1 + \bar{z}_3 z_1 = 0$

Ans.

Sol. Here O is othocenter, then $OA \perp BC$

$$\therefore \arg\left(\frac{z_1 - 0}{z_2 - z_3}\right) = \frac{\pi}{2}$$

$$\therefore \frac{z_1 - 0}{z_2 - z_3} \text{ is purely imaginary.}$$

$$\Rightarrow \frac{z_1 - 0}{z_2 - z_3} + \left(\frac{z_1 - 0}{z_2 - z_3}\right) = 0$$

$$\Rightarrow \frac{z_1}{z_2 - z_3} + \frac{\bar{z}}{\bar{z}_2 - \bar{z}_3} = 0$$

$$\Rightarrow z_1\left(\bar{z}_2 - \bar{z}_3\right) + \bar{z}_1\left(z_2 - z_3\right) = 0$$

$$\Rightarrow z_1\bar{z}_2 + \bar{z}_1 z_2 = z_1\bar{z}_3 + \bar{z}_1 z_3 \qquad \text{_____(1)}$$

Similarly, $OB \perp AC$

$$\Rightarrow z_1\bar{z}_2 + \bar{z}_1\bar{z}_2 = z_2\bar{z}_3 + \bar{z}_2 z_3 \qquad \text{_____(2)}$$

From (1) and (2),

$z_1\overline{z}_2 + \overline{z}_1 z_2 = z_2\overline{z}_3 + \overline{z}_2 z_3 = z_1\overline{z}_3 + \overline{z}_1 z_3$

25. Let $z = x + iy$ be a complex number, where x and y are real numbers. Let A and B be the sets defined by $A = \{z : |z| \le 2\}$ and $B = \{z : (1-i)z + (1+i)\overline{z} \ge 4\}$. Find the area of the region $A \cap B$.

माना $z = x + iy$ एक सम्मिश्र संख्या है जहाँ x तथा y वास्तविक है। माना A तथा B समुच्चय द्वारा परिभाषित है $A = \{z : |z| \le 2\}$ तथा $B = \{z : (1-i)z + (1+i)\overline{z} \ge 4\}$. क्षेत्र $A \cap B$ द्वारा दर्शाया गये क्षेत्र का क्षेत्रफल ज्ञात कीजिये।

Ans. $\pi - 2$

Sol. $z = x + iy$

$A = \{z'' : |z| \le 2\}$

$\Rightarrow \sqrt{x^2 + y^2} \le 2$

or $x^2 + y^2 \le 4$

$\Rightarrow$ z lies on or inside the circle

$x^2 + y^2 = 4$

$B = \{z : (1-i)z + (1+i)\overline{z} \ge 4\}$

$\Rightarrow (1-i)(x+iy) + (1+i)(x-iy) \ge 4$

$\Rightarrow x + iy - ix + y + x - iy + ix + y \ge 4$

$\Rightarrow x + y \ge 2$

Area of region $A \cap B$

$\text{Area} = \frac{\pi(2)}{4} - \frac{1}{2} \times 2 \times 2 = \pi - 2$

26. For all real numbers x, let the mapping $f(x) = \frac{1}{x - i}$, where $i = \sqrt{-1}$. If there exist real number a, b, c and d for which $f(a), f(b), f(c)$ and $f(d)$ form a square on the complex plane. Find the area of the square.

सभी वास्तविक संख्याओं x के लिये माना प्रतिचित्रण $f(x) = \frac{1}{x - i}$, जहाँ $i = \sqrt{-1}$. यदि वास्तविक संख्या a, b, c तथा d जिनके लिये $f(a), f(b), f(c)$ तथा $f(d)$ सम्मिश्र तल में एक वर्ग का निर्माण करते है। वर्ग का क्षेत्रफल ज्ञात करो ।

Ans. 1/2

Sol. $f(t)=\frac{1}{t-i}\times\frac{t+i}{t+i}$

also let $f(t)=x+iy$

$x+iy=\frac{t+i}{t^2+1}$

$x=\frac{t}{t^2+1}, y=\frac{1}{t^2+1} \Rightarrow \frac{x}{y}=t$

from $y=\frac{1}{\left(\frac{x}{y}\right)^2+1}$

$y=\frac{y^2}{x^2+y^2}\Rightarrow 1=\frac{y}{x^2+y^2}$

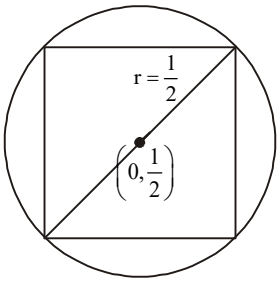
$x^2+y^2-y=0$

centre $=\left(0,\frac{1}{2}\right)$

radius $=\frac{1}{2}$

So the side of square $=\frac{1}{\sqrt{2}}$

area of square $=\frac{1}{2}$



27. Find centre and radius of the circle determined by all complex numbers $z=x+iy$ satisfying $\left|\frac{(z-\alpha)}{(z-\beta)}\right|=k$, where $\alpha=\alpha_1+i\alpha_2, \beta=\beta_1+i\beta_2$ are fixed complex and $k\neq 1$.

$\left|\frac{(z-\alpha)}{(z-\beta)}\right|=k$, $k\neq 1$ जहाँ $z=x+iy$, $\alpha=\alpha_1+i\alpha_2$, $\beta=\beta_1+i\beta_2$ द्वारा दिए जाने वाले वृत का केन्द्र ओर त्रिज्या का मान ज्ञात कीजिये।

Ans. Centre $\equiv \frac{k^2\beta-\alpha}{k^2-1}$, Radius $=\frac{1}{(k^2-1)}\sqrt{|\alpha-k^2\beta|^2-\left(k^2.|\beta|^2-|\alpha|^2\right)\left(k^2-1\right)}$

Sol. As we know $|z|^2 = z \cdot \bar{z}$

Given, $\dfrac{|z-\alpha|^2}{|z-\beta|^2} = k^2$

$\Rightarrow (z-\alpha)(\bar{z}-\bar{\alpha}) = k^2 (z-\beta)(\bar{z}-\bar{\beta})$

$\Rightarrow |z|^2 - \alpha\bar{z} - \bar{\alpha}z + |\alpha|^2 = k^2\left(|z|^2 - \beta\bar{z} - \beta\bar{z} + |\beta|^2\right)$

$\Rightarrow |z|^2 (1-k^2) - (\alpha - k^2\beta)\bar{z} - (\bar{\alpha} - \bar{\beta}k^2)z + \left(|\alpha|^2 - k^2|\beta|^2\right) = 0$

$\Rightarrow |z|^2 - \dfrac{(\alpha - k^2\beta)}{(1-k^2)}\bar{z} - \dfrac{(\bar{\alpha} - \bar{\beta}k^2)}{(1-k^2)}z$

$\Rightarrow |z|^2 - \dfrac{(\alpha - k^2\beta)}{(1-k^2)}\bar{z} - \dfrac{(\bar{\alpha} - \bar{\beta}k^2)}{(1-k^2)}z + \dfrac{|\alpha|^2 - k^2|\beta|^2}{(1-k^2)} = 0$ _____(1)

On comparing with equation of circle, $|z|^2 + a\bar{z} + \bar{a}z + b = 0$

whose centre is (–a) and radius = $\sqrt{|a|^2 - b}$

$\therefore$ Centre for eq. (1)

$= \dfrac{\alpha - k^2\beta}{1-k^2}$ and radius

$= \sqrt{\left(\dfrac{\alpha - k^2\beta}{1-k^2}\right)\left(\dfrac{\bar{\alpha} - k^2\bar{\beta}}{1-k^2}\right) - \dfrac{\alpha\bar{\alpha} - k^2\beta\bar{\beta}}{1-k^2}}$

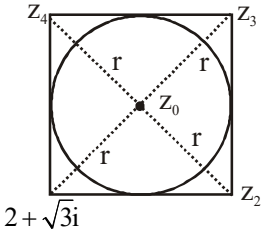
radius = $\left|\dfrac{k(\alpha-\beta)}{1-k^2}\right|$

28. If one of the vertices of the square circumscribing the circle $|z-1| = \sqrt{2}$ is $2+\sqrt{3}\,i$. Find the other vertices of square.

एक वृत $|z-1| = \sqrt{2}$ के परिगत वर्ग का एक शीर्ष $2+\sqrt{3}\,i$ है। वर्ग के अन्य शीर्ष ज्ञात कीजिये।

Ans. $z_2 = -\sqrt{3}\,i\ ;\ z_3 = \left(1-\sqrt{3}\right)+i\ ;\ z_4 = \left(1+\sqrt{3}\right)-i$

Sol. $z_0 = 1$



$$z_1 = 2+\sqrt{3}\,i$$

$$z_2 - z_0 = \left(2+\sqrt{3}\,i - z_0\right)e^{i\pi/2}$$

$$\Rightarrow \quad z_2 = 1+\left(1+\sqrt{3}\,i\right)e^{i\pi/2} = \left(1-\sqrt{3}\right)+i$$

Similary

$$z_3 = 1+\left(1+\sqrt{3}\,i\right)e^{i\pi} = -\sqrt{3}\,i$$

$$z_4 = 1+\left(1+\sqrt{3}\,i\right)e^{-i\pi/2} = \left(1+\sqrt{3}\right)-i$$

29. Let $|z| = 2$ and $w = \dfrac{z+1}{z-1}$ where $z, w \in C$ (where C is the set of complex numbers). If M and m respectively be the greatest and least modulus of w, then find the value of $(2010\,m + M)$.

माना $|z| = 2$ और $w = \dfrac{z+1}{z-1}$ जहां $z, w \in C$ (जहां C सम्मिश्र संख्याओं का समुच्चय है) यदि M और m क्रमशः w के अधिकतम और न्यूनतम मांपाक है। तब $(2010\,m + M)$ का मान ज्ञात कीजिए।

Ans. 673

Sol. Let $z = a + ib$ hence $a, b \in [-2, 2]$

$$w = \frac{(a+1)+ib}{(a-1)+ib}; \qquad a^2 + b^2 = 4$$

$$|w| = \sqrt{\frac{(a+1)^2+b^2}{(a-1)^2+b^2}} = \sqrt{\frac{a^2+b^2+2a+1}{a^2+b^2-2a+1}} = \sqrt{\frac{5+2a}{5-2a}}$$

$$|w|_{max} = \sqrt{\frac{5+4}{1}} = 3 = M, \qquad \text{when } a = 2 \quad (z = 2)$$

$$|w|_{min} = \sqrt{\frac{5-4}{9}} = \frac{1}{3} = m, \qquad \text{when } a = -2 \quad (z = -2)$$

Hence $(2010\,m + M) = 673$ Ans.

30. If z is any complex number satisfying $|z-3-2i| \le 2$, then the minimum value of $|2z-6+5i|$ is_____.
[JEE-2011]

यदि z एक सम्मिश्र संख्या है जो $|z-3-2i| \le 2$, को संन्तुष्ट करती है तो $|2z-6+5i|$ का न्यूनतम मान होगा_____
[JEE-2011]

Ans. 5

Sol. Given, $|z-3-2i| \le 2$ _____(1)

To find minimum of $|2z-6+5i|$

Or $2\left|z-3+\frac{5}{2}i\right|$,

using triangle inequality

i.e., $\left||z_1|-|z_2|\right| \le |z_1+\bar{z}_2|$

$\therefore \left|z-3+\frac{5}{2}i\right|$

$$= \left|z-3-2i+2i+\frac{5}{2}i\right| = \left|(z-3-2i)+\frac{9}{2}i\right| \ge \left||z-3-2i|-\frac{9}{2}\right| \ge \left|2-\frac{9}{2}\right| \ge \frac{5}{2}$$

$$\Rightarrow \left|z-3+\frac{5}{2}i\right| \ge \frac{5}{2} \text{ or } |2z-6+5i| \ge 5$$

DE MOIVRE'S THEOREM

31. If Z is a non-real complex number, then find the minimum value of $\frac{\text{Im}\,Z^5}{\text{Im}^5\,Z}$.

यदि Z एक अवास्तविक सम्मिश्र संख्या है, तो $\frac{\text{Im}\,Z^5}{\text{Im}^5\,Z}$ का न्यूनतम मान ज्ञात करें।

Ans. –4

Sol. Let $z = re^{i\theta}$

$$\frac{\text{Im}\,z^5}{(\text{Im}\,z)^5} = \frac{r^5 \sin 5\theta}{r^5 \sin 5\theta}$$

say, $f(\theta) = \frac{\sin 5\theta}{\sin^5 \theta}$

equatinig first derivative to 0

$f'(\theta) = 0$

$$\Rightarrow \sin^5\theta.5\cos 5\theta - \sin 5\theta \cdot 5\sin^4\theta \cdot \cos\theta$$

$$\Rightarrow \sin\theta \cdot \cos 5\theta = \cos\theta \cdot \sin 5\theta$$

$\Rightarrow \tan\theta = \tan 5\theta$

$\Rightarrow \theta = 0, \frac{\pi}{4} \cdots$

but f(0) is indeterminate

$f\left(\frac{\pi}{4}\right) = -4$

$\therefore$ Minimum value of $\frac{\text{Im}\, z^5}{(\text{Im}\, z)^5}$ is -4

CUBE ROOT OF UNITY

32. Find the sum of the series $1(2-\omega)(2-\omega^2) + 2(3-\omega)(3-\omega^2) \ldots\ldots (n-1)(n-\omega)(n-\omega^2)$ where ω is one of the imaginary cube root of unity.

श्रेणी $1(2-\omega)(2-\omega^2) + 2(3-\omega)(3-\omega^2) \ldots\ldots (n-1)(n-\omega)(n-\omega^2)$ का योग ज्ञात कीजिये जहाँ ω इकाई का काल्पनिक घनमूल है।

Ans. $\left[\frac{n(n+1)}{2}\right]^2 - n$

Sol. $1(2-w)(2-w^2) + 2(3-w)(3-w^2) + (n-1)(n-w)(n-w^2)$

$T_n = n(n+1-w)(n+1-w^2)$

$= (n^2 + n - nw)(n+1-w^2)$

$= n^3 + n^2 - n\ 2w^2 + n^3 + n - nw^2 - n^2w - nw + nw^3$

$= n^3 + n^2(2 - w^2 - w) + n(1 - w - w^2 + 1)$

$= n^3 + 3n^2 + 3n$

$S_n = \Sigma T_n = \sum (n^3 + 3n^2 + 3n)$

$= \Sigma n^3 + 3\Sigma n^2 + 3\Sigma n$

$= \left[\frac{n(n+1)}{2}\right]^2 + \frac{3n(n+1)(2n+1)}{6} + \frac{3n(n+1)}{2}$

$= \frac{n(n+1)}{2}\left[\frac{n(n+1)}{2} + (2n+1) + 3\right]$

$$= \frac{n(n+1)}{4}\left[n^2+n+6+4n+2\right]$$

$$Sn = \Sigma T_n = \frac{n(n+1)}{4}\left[n^2+5n+8\right]$$

$$S_{n-1} = \frac{(n-1)n}{4}\left[n^2+1-2n+5n-5+8\right]$$

$$= \frac{(n-1)n}{4}\left[n^2+3n+4\right]$$

33. Let ω be the complex number $\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}$. Then the number of distinct complex number z satisfying

$$\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0 \text{ is equal to.}$$ [JEE-2010]

माना ω एक सम्मिश्र संख्या $\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}$ है। तो विभिन्न सम्मिश्र संख्यायें z जो $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$ को संतुष्ट करती है, की संख्या है। [JEE-2010]

Ans. 1

Sol. $\omega = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$

$R_1 \to R_1 + R_2 + R_3$

$$\begin{vmatrix} z & \omega & \omega^2 \\ z & z+\omega^2 & 1 \\ z & 1 & z+\omega \end{vmatrix} = 0 \quad \Rightarrow \quad z\begin{vmatrix} 1 & \omega & \omega^2 \\ 1 & z+\omega^2 & 1 \\ 1 & 1 & z+\omega \end{vmatrix} = 0$$

$z = 0$

$$\begin{vmatrix} 1 & \omega & \omega^2 \\ 0 & z+\omega^2-\omega & 1-\omega^2 \\ 0 & 1-\omega & z+\omega-\omega^2 \end{vmatrix} = 0$$

$(z + \omega^2 - \omega)(z + \omega - \omega^2) - (1 - \omega)(1 - \omega^2) = 0$

$z^2 = 0$

only one solution

n^{th} ROOT OF UNITY

34. If $\cos(\alpha-\beta)+\cos(\beta-\gamma)+\cos(\gamma-\alpha)=-3/2$ then prove that :

यदि $\cos(\alpha-\beta)+\cos(\beta-\gamma)+\cos(\gamma-\alpha)=-3/2$ तो सिद्ध कीजिये :

(a) $\Sigma\cos 2\alpha = 0 = \Sigma\sin 2\alpha$

(b) $\Sigma\sin(\alpha+\beta) = 0 = \Sigma\cos(\alpha+\beta)$

(c) $\Sigma\sin^2\alpha = \Sigma\cos^2\alpha = 3/2$

(d) $\Sigma\sin 3\alpha = 3\sin(\alpha+\beta+\gamma)$

(e) $\Sigma\cos 3\alpha = 3\cos(\alpha+\beta+\gamma)$

Ans.

Sol. Given $\cos(\alpha-\beta)+\cos(\beta-\gamma)+\cos(\gamma-a)=-3/2$

$2\cos(\alpha-\beta)+2\cos(\beta-\alpha)+2\cos(\gamma-\alpha)=-3$

$2\cos(\alpha-\beta)+2\cos(\beta-\gamma)+2\cos(\gamma-\alpha)+3=0$

$[2\cos\alpha\cos\beta+2\sin\alpha\sin\beta]+[2\cos\beta\cos\gamma+2\sin\beta\sin\gamma]+[2\cos\gamma\cos\alpha+\sin\gamma\sin\alpha]+3=0$

$[2\cos\alpha\cos\beta+2\cos\beta\cos\gamma+2\cos\gamma\cos\alpha]+[2\sin\alpha\sin\beta+2\sin\beta\sin\gamma+2\sin\gamma\sin\alpha]+(\sin^2\alpha+\cos^2\alpha)+(\sin^2\beta+\cos^2\beta)+(\sin^2\gamma+\cos^2\gamma)=0$

$\Rightarrow(\cos^2\alpha+\cos^2\beta+\cos^2\gamma+2\cos\alpha\cos\beta+2\cos\beta\cos\gamma+2\cos\gamma\cos\alpha)+(\sin^2\alpha+\sin^2\beta)+(\sin^2\gamma+2\sin\alpha\sin\beta+2\sin\beta\sin\gamma+2\sin\gamma\sin a)=0$

$(\cos\alpha+\cos\beta+\cos\gamma)^2+(\sin\alpha+\sin\beta+\sin\gamma)^2=0$

$=(\cos\alpha+\cos\beta+\cos\gamma)=0$ and $\sin\alpha+\sin\beta+\sin\gamma=0$

35. Resolve Z^5+1 into linear & quadratic factors with real coefficients. Deduce that : $4\cdot\sin\frac{\pi}{10}\cdot\cos\frac{\pi}{5}=1$.

Z^5+1 को रेखीय तथा द्विघातीय गुणनखण्डो के रूप में व्यक्त कीजिये जिनके गुणांक वास्तविक है। दर्शाइये : $4\cdot\sin\frac{\pi}{10}\cdot\cos\frac{\pi}{5}=1$.

Ans. $(Z+1)(Z^2-2Z\cos 36^\circ+1)(Z^2-2Z\cos 108^\circ+1)$

Sol. $Z^5+1=0 \qquad \Rightarrow Z^5=1$

$=\cos(2n\pi+\pi)+\sin(2n\pi+\pi)$

$=e^{i(2n\pi+\pi)} \qquad =e^{i(2n+1)\pi}$

$n=0,1,2,3,4$

$n=0 \quad \Rightarrow Z=e^{i\frac{\pi}{5}}$

$n = 1 \quad \Rightarrow z = e^{i\frac{3\pi}{5}}$

$n = 2 \quad Z = e^{i\pi}$

$n = 3 \quad \Rightarrow Z = e^{i\frac{7\pi}{5}} = e^{-i\frac{\pi}{5}}$

$$\left(Z - e^{i\frac{3\pi}{5}}\right)\left(Z - e^{-i\frac{3\pi}{5}}\right) = Z^2 + 1 - Z\left(e^{i\frac{3\pi}{5}} + e^{-\frac{3\pi}{5}}\right)$$

$$= Z^2 - 2Z\cos\frac{3\pi}{5} + 1$$

$$Z^5 + 1 = (z+1)\left(z^2 - 2z\cos\frac{3\pi}{5} + 1\right)\left(Z^2 - 2Z\cos\frac{\pi}{5} + 1\right)$$

$$4\sin\frac{\pi}{10}\cos\frac{\pi}{5} = 4(0.309016)(0.809016) = 1$$

Paragraph for Question Nos. 36 to 38

For the complex number $w = \dfrac{4z - 5i}{2z + 1}$

सम्मिश्र संख्या $w = \dfrac{4z - 5i}{2z + 1}$ के लिए

36. The locus of z, when w is a real number other than 2, is :

(A) a point circle

(B) a straight line with slope $-\dfrac{5}{2}$ and y-intercept $\dfrac{5}{4}$

(C) a straight line with slope $\dfrac{5}{2}$ and y-intercept $\dfrac{5}{4}$

(D) a straight line passing through the origin

z का बिन्दु पथ, जबकि w, 2 को छोड़ कर कोई भी वास्तविक संख्या है :

(A) एक बिन्दु वृत्त

(B) $-\dfrac{5}{2}$ प्रवणता तथा $\dfrac{5}{4}$, y अन्तःखण्ड वाली सरल रेखा

(C) $\dfrac{5}{2}$ प्रवणता तथा $\dfrac{5}{4}$, y अन्तःखण्ड वाली सरल रेखा

(D) मूल बिन्दु से जाने वाली सरल रेखा

Ans. C

Sol. (i) $w = \dfrac{4z-5i}{2z+1}$ $(w \neq z)$

if w is real then $w = \overline{w}$

$$\overline{w} = \frac{4\overline{z}+5i}{\underbrace{2\overline{z}+1}_{\overline{w}}} = \frac{4z-5i}{\underbrace{2z+1}_{w}} 7$$

$(4\overline{z}+5i)(2z+1) = (4z-5i)(2\overline{z}+1)$

$8z\overline{z} + 4\overline{z} + 10zi + 5i = 8z\overline{z} + 4z - 10\overline{z}i - 5i$

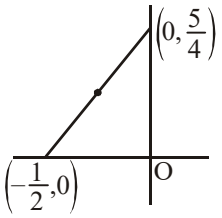
$4(z-\overline{z}) - 10i(z+\overline{z}) - 10i = 0$

$8iy - 20ix - 10i = 0$

$4y - 10x - 5 = 0 \quad \Rightarrow \quad 10x - 4y + 5 = 0$

Alternatively:

$$\omega = \frac{2\left(z-\frac{5}{4}\right)}{\left(z+\frac{1}{2}\right)} \text{ given } \frac{z-\frac{5}{4}}{z+\frac{1}{2}} \text{ is root } \left(-\frac{1}{2}, 0\right)$$



$\Rightarrow$ z moves on line formed $\left(-\frac{1}{2}, 0\right)$ and $\left(0, \frac{5}{4}\right)$ i.e., $m = \frac{5}{2}$ and y-intercept $\frac{5}{4}$.]

37. The locus of z, when w is a purely imaginary number is

(A) a circle with centre $\left(\frac{1}{2}, -\frac{5}{4}\right)$ passing through origin.

(B) a circle with centre $\left(-\frac{1}{4}, \frac{5}{8}\right)$ passing through origin.

(C) a circle with centre $\left(\frac{1}{4}, -\frac{5}{8}\right)$ and radius $\frac{\sqrt{29}}{8}$

(D) any other circle

z का बिन्दु पथ जबकि w पूर्णतः काल्पनिक संख्या है :

(A) $\left(\frac{1}{2}, -\frac{5}{4}\right)$ केन्द तथा (0, 0) से जाने वाला वृत्त

(B) $\left(-\frac{1}{4},\frac{5}{8}\right)$ केन्द्र तथा (0, 0) से जाने वाला वृत्त

(C) केन्द्र $\left(\frac{1}{4},-\frac{5}{8}\right)$ और त्रिज्या $\frac{\sqrt{29}}{8}$ का वृत्त

(D) कोई अन्य वृत्त

Ans. B

Sol. (ii) If w is purely imaginary then

$$w+\overline{w}=0$$

$$\frac{4z-5i}{2z+1}+\frac{4\bar{z}+5i}{2\bar{z}+1}=0$$

$$(4\bar{z}+5i)(2z+1)+(4z-5i)(2\bar{z}+1)=0$$

simplifying $16z\bar{z}+4(z+\bar{z})+10i(z-\bar{z})=0$

$$16(x^2+y^2)+8x-20y=0$$

$$x^2+y^2+\frac{x}{2}-\frac{5}{4}y=0$$

Alternatively: When ω is purely imaginary i.e., $\frac{x-\frac{5}{4}}{z+\frac{1}{2}}$ is purely imaginary, hence the curve wll be the circle

described on $\left(0,\frac{5}{4}\right)$ and $\left(-\frac{1}{2},0\right)$ as denotes is $x\left(x+\frac{1}{2}\right)+t\left(y-\frac{5}{4}\right)=0$

which passes through origin.

38. The locus of z, when $|w|=1$ is :

(A) a circle with centre $\left(-\frac{5}{8},\frac{1}{4}\right)$ and radius $\frac{1}{2}$ (B) a circle with centre $\left(\frac{1}{4},-\frac{5}{8}\right)$ and radius $\frac{1}{2}$

(C) a circle with centre $\left(\frac{5}{8},-\frac{1}{4}\right)$ and radius $\frac{1}{2}$ (D) any other circle

z का बिन्दुपथ, यदि $|w|=1$:

(A) केन्द्र $\left(-\frac{5}{8},\frac{1}{4}\right)$ और त्रिज्या $\frac{1}{2}$ का वृत्त (B) केन्द्र $\left(\frac{1}{4},-\frac{5}{8}\right)$ और त्रिज्या $\frac{1}{2}$ का वृत्त

(C) केन्द्र $\left(\frac{5}{8},-\frac{1}{4}\right)$ और त्रिज्या $\frac{1}{2}$ का वृत्त (D) कोई अन्य वृत्त

Ans. D

Sol. $|w| = 1$

$$w\bar{w} = 1$$

$$\left(\frac{4z - si}{2z + 1}\right)\left(\frac{4\bar{z} + si}{2\bar{z} + 1}\right) = 1$$

$$16(z)^2 + 25 = 4(z)^2 + 2z + 2z + 1$$

$$12(z)^2 - 2(z + \bar{z}) + 24 = 0$$

$$12(x^2 + y^2) - 2(2x) + 24 = 0$$

$$x^2 + y^2 - \frac{1}{3}x + 2 = 0$$

Paragraph for Question Nos. 39 to 41

Let A, B, C be three sets of complex numbers as defined below.

$A = \{z : |z+1| \le 2 + Re(z)\}$, $B = \{z : |z-1| \ge 1\}$ and $C = \left\{z : \left|\frac{z-1}{z+1}\right| \ge 1\right\}$

माना A, B, C तीन सम्मिश्र संख्याऐं इस प्रकार परिभाषित है कि

$A = \{z : |z+1| \le 2 + Re(z)\}$, $B = \{z : |z-1| \ge 1\}$ तथा $C = \left\{z : \left|\frac{z-1}{z+1}\right| \ge 1\right\}$

39. The number of point(s) having integral coordinates in the region $A \cap B \cap C$ is :

क्षेत्र $A \cap B \cap C$ में बिन्दुओं की संख्या जिनके निर्देशांक पूर्णांक है :

(A) 4 (B) 5 (C) 6 (D) 10

Ans. B

Sol. For A, $|z + 1| \le 2 + Re(z)$

$\Rightarrow$ $(x+1)^2 + y^2 \le 4 + 4x + x^2$

$\Rightarrow$ $y^2 \le 3 + 2x$

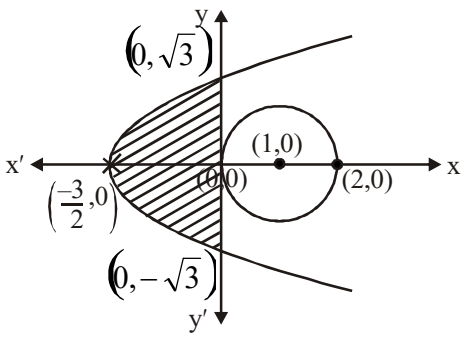
$\Rightarrow$ $y^2 \le 2\left(x + \frac{3}{2}\right)$ .....(1)

For B, $|z - 1| \ge 1$

$\Rightarrow$ $(x-1)^2 + y^2 \ge 1$ .....(2)

For C, $|z - 1|^2 \ge |z + 1|^2$

$\Rightarrow$ $(z - 1)(\bar{z} - 1) \ge (z + 1)(\bar{z} + 1)$



$\Rightarrow \quad (z\,\bar{z} - \bar{z} - z + 1) \geq (z\,\bar{z} + \bar{z} + z + 1)$

$\Rightarrow \quad z + \bar{z} \leq 0$

i.e. $\quad x \leq 0 \qquad\qquad .....(3)$

(i) $(-1,0), (-1, 1) , (-1,-1), (0,0), (0,1), (0,-1)$ but $z = -1$ is not in the domain in set C

$\therefore$ Total number of point(s) having integral coordinates in the region $A \cap B \cap C$ is 5.

40. The area of region bounded by $A \cap B \cap C$ is :

$A \cap B \cap C$ से परिबद्ध क्षेत्र का क्षेत्रफल :

(A) $2\sqrt{3}$ (B) $\sqrt{3}$ (C) $4\sqrt{3}$ (D) 2

Ans. A

Sol. (ii) Required area $= 2\int_{\frac{-3}{2}}^{0}\sqrt{2\left(x+\frac{3}{2}\right)}\,dx = 2\sqrt{3}$(square units)

41. The real part of the complex number in the region $A \cap B \cap C$ and having maximum amplitude is :

क्षेत्र $A \cap B \cap C$ में सम्मिश्र संख्या का वास्तविक भाग, जिसका कोणांक का मुख्य मान अधिकतम है :

(A) –1 (B) $\frac{-3}{2}$ (C) $\frac{1}{2}$ (D) – 2

Ans. B

Sol. (iii) Clearly $z = \frac{-3}{2} + i\,0$ is the complex number in the region $A \cap B \cap C$ and having maximum amplitude.

$\therefore \quad Re(z) = \frac{-3}{2}$

42. If the expression $(1 + ir)^3$ is of the form of $s(1 + i)$ for some real 's' where 'r' is also real and $i = \sqrt{-1}$, then the value of 'r' can be :

यदि $(1 + ir)^3$ को $s(1 + i)$ के रूप में लिखा जाता है कुछ वास्तविक 's' के लिए जहां 'r' भी वास्तविक है और $i = \sqrt{-1}$ तब 'r' का मान हो सकता है :

(A) $\cot\frac{\pi}{8}$ (B) $\sec\pi$ (C) $\tan\frac{\pi}{12}$ (D) $\tan\frac{5\pi}{12}$

Ans. BCD

Sol. $(1+ir)^3 = 1^3 + 3ir + 3(ir)^2 + (ir)^3$

$$= 1 + i(3r) + 3(-1)r^2 - r^3 i = s + si$$

$$= (1 - 3r^2) + i(3r - r^3) = s + si$$

$$= 1 - 3r^2 = s \text{ and } 3r - r^3 = s$$

$$1 - 3r^2 = 3r - r^2$$

$$1 = \frac{3r - r^3}{1 - 3r^2}$$

$$1 = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} = \tan 3\theta$$

$$3\theta = \frac{\pi}{4}, \qquad r = \tan\theta$$

$$\theta = \frac{\pi}{12}, \qquad r = \tan\frac{\pi}{12}$$

ONE OR MORE THAN ONE CORRECT TYPE

43. Let z_1 and z_2 be two distinct complex numbers and let $z = (1-t)z_1 + tz_2$ for some real number t with $0 < t < 1$. If Arg(w) denotes the principal argument of a nonzero complex number w, then : **[JEE-2010]**

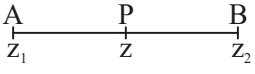
माना z_1 तथा z_2 दो भिन्न सम्मिश्र संख्यायें है तथा $z = (1 - t)z_1 + tz_2$ है जहाँ t कोई वास्तविक संख्या है जिसमें $0 < t < 1$ है। यदि Arg (w) का अशून्य सम्मिश्र संख्या w का मुख्य कोण दर्शाता है, तो : **[JEE-2010]**

(A) $|z - z_1| + |z - z_2| = |z_1 - z_2|$ (B) $\text{Arg}(z - z_1) = \text{Arg}(z - z_2)$

(C) $\begin{vmatrix} z - z_1 & \bar{z} - \bar{z}_1 \\ z_2 - z_1 & \bar{z}_2 - \bar{z}_1 \end{vmatrix} = 0$ (D) $\text{Arg}(z - z_1) = \text{Arg}(z_2 - z_1)$

Ans. ACD

Sol. Given, $z = \frac{(1-t)z_1 + tz_2}{(1-t) + 1}$



t : (1 – t)

Clearly, z divides z_1 and z_2 in the ratio of t : (1 – t), $0 < t < 1$

$\Rightarrow AP + BP = AB$

ie, $\Rightarrow |z - z_1| + |z - z_2| = |z_1 - z_2|$

$\Rightarrow$ Option (a) is true.

And $\arg(z - z_1) = \arg(z_2 - z) = \arg(z_2 - z_1)$

$\Rightarrow$ (b) is false and (d) is true.

Also, $\arg(z - z_1) = \arg(z_2 - z_1)$

$$\Rightarrow \arg\left(\frac{z - z_1}{z_2 - z_1}\right) = 0$$

$$\therefore \frac{z - z_1}{z_2 - z_1} \text{ is purely real.}$$

$$\Rightarrow \frac{z - z_1}{z_2 - z_1} = \frac{\overline{z} - \overline{z}_1}{\overline{z}_2 - \overline{z}_1} \text{ or } \begin{vmatrix} z - z_1 & \overline{z} - \overline{z}_1 \\ z_2 - z_1 & \overline{z}_2 - \overline{z}_1 \end{vmatrix} = 0$$

$\therefore$ option c is correct.

44. POQ is a straight line through the origin O . P and Q represent the complex number a + i b and c + i d respectively and OP = OQ. Then :

(A) $|a + i\,b| = |c + i\,d|$ (B) $a + c = b + d$

(C) $\arg(a + i\,b) = \arg(c + i\,d)$ (D) None of these

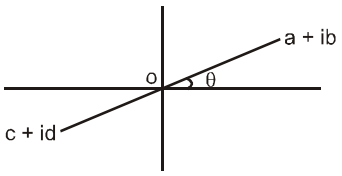
मूलबिन्दु O से गुजरने वाली रेखा POQ है। P तथा Q सम्मिश्र संख्याओं $a + i\,b$, $c + i\,d$ को प्रदर्शित करते हैं तथा OP = OQ है। तब :

(A) $|a + i\,b| = |c + i\,d|$ (B) $a + c = b + d$

(C) कोणांक $(a + i\,b)$ = कोणांक $(c + i\,d)$ (D) इनमें से कोई नहीं

Ans. AB

Sol.



$|a + ib| = |(c + id)|$

$a + c = b + d$

Alter :

It is given that origin is the mid point of the line POQ as OP = OQ.

So, coordinates of orgin will be $\frac{(a+c)}{2} + \frac{(b+d)}{2}i$, but coordinates of the origin are given as $0 + 0_i$.

On comparing we get,

$\frac{a+c}{2}=0$ _____(1) and $\frac{b+d}{2}=0$ _____(2)

Equation (1) and (2)

$\Rightarrow \frac{a+c}{2}=\frac{b+d}{2}$

$\Rightarrow a+c=b+d$

45. If z satisfies the inequality $|z-1-2i| \le 1$, then :

(A) $\min(\arg(z))=\tan^{-1}\left(\frac{3}{4}\right)$ (B) $\max(\arg(z))=\frac{\pi}{2}$

(C) $\min(|z|)=\sqrt{5}-1$ (D) $\max(|z|)=\sqrt{5}+1$

यदि z असमिका $|z-1-2i| \le 1$ को संतुष्ट करता हो, तो :

(A) न्यूनतम (कोणांक (z)) $=\tan^{-1}\left(\frac{3}{4}\right)$ (B) अधिकतम (कोणांक (z)) $=\frac{\pi}{2}$

(C) न्यूनतम $(|z|)=\sqrt{5}-1$ (D) अधिकतम $(|z|)=\sqrt{5}+1$

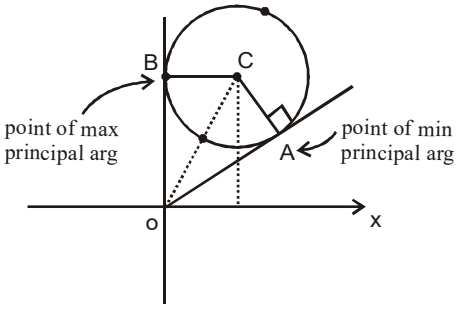
Ans. ABCD

Sol. $\max|z|=d+r=\sqrt{5}+1$

$\min|z|=d-r=\sqrt{5}-1$

$d=OC=\sqrt{5}$

$r=1$



$\theta=\angle OCX=\tan^{-1}\frac{2}{1}$

$\alpha=\angle OCA=\tan^{-1}\frac{1}{2}$ $\left(\because \sin\alpha=\frac{1}{\sqrt{5}}\right)$

So principal Arg of $A=\theta-\alpha=\tan^{-1}2-\tan^{-1}\frac{1}{2}$

$=\tan^{-1}\frac{2-\frac{1}{2}}{1+1}=\tan^{-1}\frac{3}{4}$

46. If $|z_1|=|z_2|=|z_3|=1$ and z_1, z_2, z_3 are represented by the vertices of an equilateral triangle then :

(A) $z_1+z_2+z_3=0$ (B) $z_1z_2z_3=1$ (C) $z_1z_2+z_2z_3+z_3z_1=0$ (D) None of these

यदि $|z_1| = |z_2| = |z_3| = 1$ हो तथा एक समबाहु त्रिभुज के शीर्ष z_1, z_2, z_3 हो, तो :

(A) $z_1 + z_2 + z_3 = 0$ (B) $z_1z_2z_3 = 1$ (C) $z_1z_2 + z_2z_3 + z_3z_1 = 0$ (D) इनमें से कोई नहीं

Ans. AC

Sol. All the three vertices lies on circle $|z| = 1$

take $z_1 = z_1$, $z_2 = z_1\omega$, $z_3 = z_1\omega^2$

so $z_1 + z_2 + z_3 = z_1(1 + \omega + \omega^2) = 0$

$z_1z_2z_3 = z_1^3$ ... (i)

$z_1z_2 + z_2z_3 + z_3z_1 = z_1^2(\omega + \omega^3 + \omega^2) = 0$

if $z_1 + z_2 + z_3 = 0$ then $z_1^3 + z_2^3 + z_3^3 = 3z_1z_2z_3$

from (i) $z_1^3 + z_2^3 + z_3^3 = 3z_1^3$

$z_2^3 + z_3^3 = 2z_1^3$

hence proved.

47. Let A and B be two distinct points denoting the complex numbers α and β respectively. A complex number z lies between A and B where $z \neq \alpha, z \neq \beta$. Which of the following relation(s) hold good?

(A) $|\alpha - z| + |z - \beta| = |\alpha - \beta|$ (B) $\exists$ a positive real number 't' such that $z = (1 - t)\alpha + t\beta$

(C) $\begin{vmatrix} z-\alpha & \bar{z}-\bar{\alpha} \\ \beta-\alpha & \bar{\beta}-\bar{\alpha} \end{vmatrix} = 0$ (D) $\begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0$

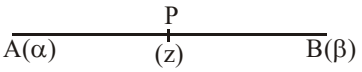
माना A और B दो भिन्न बिन्दु सम्मिश्र संख्या α और β को दर्शाते है। सम्मिश्र संख्या z , A और B के मध्य है जहां $z \neq \alpha, z \neq \beta$. तब सत्य कथन होंगे :

(A) $|\alpha - z| + |z - \beta| = |\alpha - \beta|$ (B) $\exists$ एक धनात्मक वास्तविक संख्या 't' इस प्रकार की $z=(1-t)\alpha + t\beta$

(C) $\begin{vmatrix} z-\alpha & \bar{z}-\bar{\alpha} \\ \beta-\alpha & \bar{\beta}-\bar{\alpha} \end{vmatrix} = 0$ (D) $\begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0$

Ans. ABCD

Sol. $AP + PB = AB$



$|z - \alpha| + |\beta - z| = |\beta - \alpha| \Rightarrow$ A is True

Now $z = \alpha + t(\beta - \alpha)$

$= (1 - t)\alpha + t\beta$ where $t \in (0,1) \Rightarrow$ B is True

again $\dfrac{z-\alpha}{\beta-\alpha}$ is real $\Rightarrow \dfrac{z-\alpha}{\beta-\alpha} = \dfrac{\bar{z}-\bar{\alpha}}{\bar{\beta}-\bar{\alpha}}$

$\Rightarrow \begin{vmatrix} z-\alpha & \bar{z}-\bar{\alpha} \\ \beta-\alpha & \bar{\beta}-\bar{\alpha} \end{vmatrix} = 0$ Ans.

also $\begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0$ if and only if $\begin{vmatrix} z-\alpha & \bar{z}-\bar{\alpha} & 0 \\ \alpha & \bar{\alpha} & 1 \\ \beta-\alpha & \bar{\beta}-\bar{\alpha} & 0 \end{vmatrix} = 0$

$\Rightarrow \quad \begin{vmatrix} (z-\alpha) & \bar{z}-\bar{\alpha} \\ \beta-\bar{\alpha} & \bar{\beta}-\bar{\alpha} \end{vmatrix} = 0$ Ans.

48. Let z_1, z_2, z_3 be non-zero complex numbers satisfying the equation $z^4 = iz$. Which of the following statement(s) is/ are correct?

(A) The complex number having least positive argument is $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

(B) $\sum_{k=1}^{3} \text{Amp}(z_k) = \frac{\pi}{2}$

(C) Centroid of the triangle formed by z_1, z_2 and z_3 is $\left(\frac{1}{\sqrt{3}}, \frac{-1}{3}\right)$

(D) Area of triangle formed by z_1, z_2 and z_3 is $\frac{3\sqrt{3}}{2}$

माना z_1, z_2, z_3 अशून्य सम्मिश्र संख्याऐं हैं जो समीकरण $z^4 = iz$ को सन्तुष्ट करते हैं तब निम्न में से सत्य कथन होंगे?

(A) सम्मिश्र संख्या, जिसका न्यूनतम धनात्मक कोणांक है, $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ है

(B) $\sum_{k=1}^{3} \text{Amp}(z_k) = \frac{\pi}{2}$

(C) z_1, z_2 और z_3 से निर्मित त्रिभुज का केन्द्रक $\left(\frac{1}{\sqrt{3}}, \frac{-1}{3}\right)$ है

(D) z_1, z_2 और z_3 द्वारा निर्मित त्रिभुज का क्षेत्रफल $\frac{3\sqrt{3}}{2}$ है

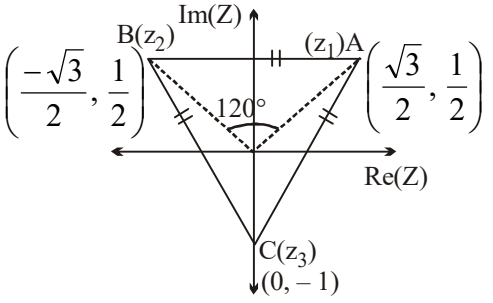
Ans. AB

Sol. We have $z^4 = iz \quad \Rightarrow \quad z^3 = i$

$\Rightarrow \quad z = e^{i(4k+1)\frac{\pi}{6}}$ (Using D.M.T.)

Put $k = 0, 1, 2$, we get

$z_1 = e^{i\frac{\pi}{6}}$, $z_2 = e^{i\frac{5\pi}{6}}$ and $z_3 = e^{i\frac{3\pi}{2}}$



Clearly triangle formed by z_1, z_2 and z_3 is equilateral.

$\therefore$ centroid of ΔABC is (0, 0) and Area $(\Delta ABC) = \dfrac{3\sqrt{3}}{4}$

49. If $z \in C$, which of the following relation(s) represents a circle on an Argand diagram?

यदि $z \in C$, तब निम्न में से कौन कौन से सम्मिश्र तल में वृत्त को दर्शाते हैं?

(A) $|z-1|+|z+1|=3$ (B) $(z-3+i)(\bar{z}-3-i)=5$

(C) $3|z-2+i|=7$ (D) $|z-3|=2$

Ans. BCD

Sol. (A) is obviously ellipse

(B) $(z-\alpha)(\bar{z}-\bar{\alpha})=5$ where $\alpha = 3-i$; $\bar{\alpha}=3+i$

$|z-\alpha|^2=5 \Rightarrow |z-\alpha|=\sqrt{5}$ circle with centre (3, – 1) and radius = $\sqrt{5} \Rightarrow$ (B) is correct

(C) $|z-(2-i)|=\dfrac{7}{3} \Rightarrow$ circle with centre (2, – 1) and radius = $\dfrac{7}{3} \Rightarrow$ (C) is correct

(D) $|z-3|=2 \Rightarrow$ circle with centre (3, 0) and radius = 2 $\Rightarrow$ (D) is correct

50. Equation of a straight line on the complex plane passing through a point P denoting the complex number α and perpendicular to the vector $\overrightarrow{OP}$ where 'O' is the origin can be written as :

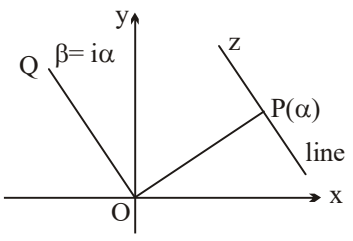
सम्मिश्र तल में P (α) से जाने वाली तथा सदिश $\overrightarrow{OP}$ (जहां 'O' मूल बिन्दु है) के लम्बवत सरल रेखा है :

(A) $\text{Im}\left(\dfrac{z-\alpha}{\alpha}\right)=0$ (B) $\text{Re}\left(\dfrac{z-\alpha}{\alpha}\right)=0$ (C) $\text{Re}(\bar{\alpha}z)=0$ (D) $\bar{\alpha}z+\alpha\bar{z}-2|\alpha|^2=0$

Ans. BD

Sol. Required line is passing through P(α) and parallel to the vector $\overrightarrow{OQ}$

hence $z = \alpha + i\lambda\alpha$, $\lambda \in R$



$\dfrac{z-\alpha}{\alpha}$ = purely imaginary

$\Rightarrow$ $\text{Re}\left(\dfrac{z-\alpha}{\alpha}\right)=0 \Rightarrow$ (B) (multiply N^r and D^r by $\bar{\alpha}$)

$\Rightarrow$ $\text{Re}\left((z-\alpha)\bar{\alpha}\right)=0 \Rightarrow \text{Re}\left(z\bar{\alpha}-|\bar{\alpha}|\right)=0$

also $\dfrac{z-\alpha}{\alpha}+\dfrac{\bar{z}-\bar{\alpha}}{\bar{\alpha}}=0$

$\overline{\alpha}(z-\alpha)+\alpha(\overline{z}-\overline{\alpha})=0$

$\overline{\alpha}\,z+\alpha\overline{z}-2|\alpha|^2=0 \quad \Rightarrow \qquad$ (D)

MATCH THE COLUMN

51. Match the statements in Column I with those in Column-II

[Note : Here z takes values in the complex plane and Im z and Re z denote, respectively, the imaginary part and the real part of z.] **[JEE-2010]**

Column I	**Column II**
(A) The set of points z satisfying $\left\|z-i\|z|\right\|=\left\|z+i\|z|\right\|$ is contained in or equal to	(p) an ellipse with eccentricity $\frac{4}{5}$
(B) The set of points z satisfying $\|z+4\|+\|z-4\|=10$ is contained in or equal to	(q) the set of points z satisfying Im z = 0
(C) If $\|w\|=2$, then the set of points $z=w-\frac{1}{w}$ is contained in or equal to	(r) the set of points z satisfying $\|\text{Im } z\| \le 1$
(D) If $\|w\|=1$, then the set of points $z=w+\frac{1}{w}$ is contained in or equal to	(s) the set of points z satisfying $\|\text{Re } z\| \le 2$
	(t) the set of points z satisfying $\|z\| \le 3$

स्तम्भ–I का स्तम्भ–II से मिलान कीजिये।

[Note : यहाँ z सम्मिश्र तल में स्थित है तथा Im z एवं Re z क्रमशः z के काल्पनिक एवं वास्तविक भाग है] **[JEE-2010]**

Column I	**Column II**
(A) $\left\|z-i\|z|\right\|=\left\|z+i\|z|\right\|$ को संतुष्ट करने वाले बिन्दु के समुच्चय बराबर है या निहित है।	(p) एक दीर्घवृत जिसकी उत्केन्द्रता $\frac{4}{5}$
(B) $\|z+4\|+\|z-4\|=10$ को संतुष्ट करने वाले बिन्दु किसमें निहित है या बराबर है	(q) Im z = 0 को सतुष्ट करने वाले बिन्दु z का समुच्चय
(C) यदि $\|w\|=2$, तो समुच्चयों जिनके लिये बिन्दु $z=w-\frac{1}{w}$ में निहित या बराबर है	(r) $\|\text{Im } z\| \le 1$ को सतुष्ट करने वाले बिन्दु z का समुच्चय
(D) यदि $\|w\|=1$, तो समुच्चयों जिनके लिये बिन्दु $z=w+\frac{1}{w}$ में निहित या बराबर है	(s) $\|\text{Re } z\| \le 2$ को सतुष्ट करने वाले बिन्दु z का समुच्चय
	(t) $\|z\| \le 3$ को सतुष्ट करने वाले बिन्दु z का समुच्चय

Ans. (A) q, r (B) p (C) p,s,t (D) q,r,s,t

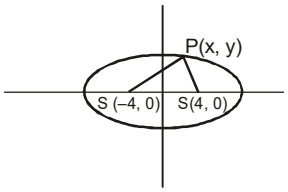
Sol. $\Rightarrow |x+iy-i\sqrt{x^2+y^2}|=|x+iy+i\sqrt{x^2+y^2}| \Rightarrow x^2+\left(y-\sqrt{x^2+y^2}\right)^2=x^2+\left(y+\sqrt{x^2+y^2}\right)^2$

$\Rightarrow 4y\sqrt{x^2+y^2}=0 \Rightarrow y=0 \Rightarrow \text{Im } z=0$

(B) $|z+4|+|z-4|=10$

Ellipse with $2a=10 \Rightarrow a=5$

$ae=4 \Rightarrow e=\frac{4}{5}$



(C) Let $w=2(\cos\theta+i\sin\theta)$

$z=2(\cos\theta+i\sin\theta)-\frac{(\cos\theta-i\sin\theta)}{2}$

$=\frac{3\cos\theta+5i\sin\theta}{2} \Rightarrow x=\frac{3\cos\theta}{2}, y=\frac{5\sin\theta}{2}$

$=\frac{x^2}{9/4}+\frac{y^2}{25/4}=1 \qquad e=\frac{4}{5}$

$|z|=\sqrt{\frac{9\cos^2\theta}{4}+\frac{25\sin^2\theta}{4}}=\sqrt{\frac{9+16\sin^2\theta}{4}}=\sqrt{\frac{9}{4}+4\sin^2\theta}\le\frac{5}{2}$

$|\text{Re } z|=\left|\frac{3}{2}\cos\theta\right|\le\frac{3}{2}$

(D) $z=\cos\theta+i\sin\theta+\cos\theta-i\sin\theta=2\cos\theta$

$\therefore |z|\le 2$

$\therefore \text{Im}(z)=0$

$(\text{Re } z) \Rightarrow |2\cos\theta|\le 2$

$|z|\le 2$

52. Match the following columns :

Column I	**Column II**
(A) Let ω be a non real cube root of unity then the number of distinct elements in the set $\left\{(1+\omega+\omega^2+.....+\omega^n)^m \mid m,n\in N\right\}$ is	(P) 4
(B) Let $1, \omega, \omega^2$ be the cube root of unity. The least possible degree of a polynomial with real coefficients having roots $2\omega, (2+3\omega), (2+3\omega^2), (2-\omega-\omega^2)$, is	(Q) 5

(C) $\alpha = 6 + 4i$ and $\beta = (2 + 4i)$ are two complex numbers on the complex plane.A complex number z satisfying $\text{amp}\left(\frac{z-\alpha}{z-\beta}\right) = \frac{\pi}{6}$ moves on the major segment of a circle whose radius is (R) 6 (S) 7

निम्न को सुमेलित कीजिए :

Column I	**Column II**
(A) माना ω एक अवास्तविक इकाई का घनमूल है तो समुच्चय $\left\{(1+\omega+\omega^2+.....+\omega^n)^m \mid m,n \in N\right\}$ में स्थित भिन्न अवयवों की संख्या होगी	(P) 4
(B) माना 1, ω, ω^2 इकाई के घनमूल है। बहुपद का न्यूनतम सम्भावित घात जिसके मूल 2ω, $(2 + 3\omega)$, $(2 + 3\omega^2)$, $(2 - \omega - \omega^2)$ है तथा गुणांक वास्तविक हैं।	(Q) 5
(C) $\alpha = 6 + 4i$ तथा $\beta = (2 + 4i)$ दो सम्मिश्र संख्यायें सम्मिश्र तल में स्थित है एक सम्मिश्र संख्या z जो $\text{amp}\left(\frac{z-\alpha}{z-\beta}\right) = \frac{\pi}{6}$ को संतुष्ट करती है वृत की दीर्घखण्ड में गमन कर रही है जिसकी त्रिज्या है	(R) 6 (S) 7

Ans. (A) S (B) Q (C) P

Sol.

53. Let z_1 lies on $|z| = 1$ and z_2 lies on $|z| = 2$. Match the following :

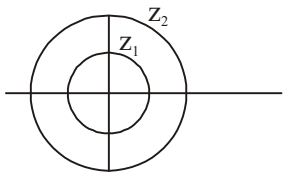
Column I	**Column II**
(A) Maximum value of $\lvert z_1 + z_2 \rvert$	(p) 3
(B) Minimum value of $\lvert z_1 - z_2 \rvert$	(q) 1
(C) Minimum value of $\lvert 2z_1 + 3z_2 \rvert$	(r) 4
(D) Maximum value of $\lvert z_1 - 2z_2 \rvert$	(s) 5

यदि z_1, $|z| = 1$ पर स्थित है तथा z_2, $|z| = 2$ पर स्थित है :

Column I	**Column II**
(A) $\lvert z_1 + z_2 \rvert$ का अधिकतम मान	(p) 3
(B) $\lvert z_1 - z_2 \rvert$ का न्यूनतम मान	(q) 1
(C) $\lvert 2z_1 + 3z_2 \rvert$ का न्यूनतम मान	(r) 4
(D) $\lvert z_1 - 2z_2 \rvert$ का अधिकतम मान	(s) 5

Ans. (A) p (B) q (C) r (D) s

Sol.



(A) $|z_1 + z_2| \le |z_1| + |z_2| \le 2 + 1 \le 3$

$(A) \to P$

(B) $|z_1 - z_2| \Rightarrow$ minimum distance b/w z_1 & $z_2 = 1$

$|z_1 - z_2| \Rightarrow$ z_1 तथा z_2 के मध्य दूरी $= 1$

$B \to q$

(C) $|2z_1 + 3z_2|$ minimum is न्यूनतम $= 6 - 2 = 4$

$(C) \to r$

(D) $|z_1 - 2z_2| \le |z_1| + |-2z_2|$

$1 + 4 \le 5$

$(D) \to s$

Exercise J-M

1. Let $p, q \in R$ and $\left(1-\sqrt{3}i\right)^{200} = 2^{199}\left(p+iq\right)$, $i = \sqrt{-1}$. Then $p + q + q^2$ and $p - q + q^2$ are roots of the equation : **[JEE(Main)Jan.2023]**

माना $p, q \in R$ तथा $\left(1-\sqrt{3}i\right)^{200} = 2^{199}\left(p+iq\right)$, $i = \sqrt{-1}$ तो $p + q + q^2$ तथा $p - q + q^2$ किस समीकरण के मूल है : **[JEE(Main)Jan.2023]**

(A) $x^2 - 4x - 1 = 0$ (B) $x^2 - 4x + 1 = 0$ (C) $x^2 + 4x + 1 = 0$ (D) $x^2 + 4x - 1 = 0$

Ans. B

Sol. $(1+\sqrt{3}i)^{200} = 2^{199}(p+iq)$

$$\Rightarrow 2^{200}\left(\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)^{200} = 2^{199}(p+iq)$$

$$\Rightarrow 2^{200}\left(\cos\left(\frac{\pi}{3}\right)+i\sin\left(\frac{\pi}{3}\right)\right)^{200} = 2^{199}(p+iq)$$

$$\Rightarrow 2^{200}\left(e^{i\frac{\pi}{2}}\right)^{200} = 2^{199}(p+iq)$$

$$\Rightarrow 2^{200}\left(\cos\left(\frac{200\pi}{3}\right)+i\sin\left(\frac{200\pi}{3}\right)\right) = 2^{199}(p+iq)$$

$$\Rightarrow 2^{200}\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right) = 2^{199}(p+iq)$$

$$\Rightarrow 2^{199}(-1+\sqrt{3}i) = 2^{199}(p+iq)$$

$$\Rightarrow p = -1, q = \sqrt{3}$$

$$p + q + q^2 = 2 + \sqrt{3}$$

$$p - q + q^2 = 2 - \sqrt{3}$$

2. The value of $\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$ is : **[JEE(Main)Jan.2023]**

$\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$ का मान है :

(A) $-\frac{1}{2}\left(\sqrt{3}-i\right)$ (B) $-\frac{1}{2}\left(1-i\sqrt{3}\right)$ (C) $\frac{1}{2}\left(1-i\sqrt{3}\right)$ (D) $\frac{1}{2}\left(\sqrt{3}+i\right)$

Ans. A

Sol. Let $\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}=z$

$$\left(\frac{1+z}{1+\bar{z}}\right)^3=\left(\frac{1+z}{1+\frac{1}{z}}\right)^3=z^3$$

$$\Rightarrow\left(i\left(\cos\frac{2\pi}{9}-i\sin\frac{2\pi}{9}\right)\right)^3$$

$$=-i\left(\cos\frac{2\pi}{3}-i\sin\frac{2\pi}{3}\right)=-i\left(\frac{-1}{2}-i\frac{\sqrt{3}}{2}\right)$$

$$\Rightarrow\frac{-1}{2}(\sqrt{3}-i)$$

3. Let $z_1=2+3i$ and $z_2=3+4i$. The set $S=\left\{z\in C:|z-z_1|^2-|z-z_2|^2=|z_1-z_2|^2\right\}$ represents a :

[JEE(Main)Jan.2023]

(A) straight line with the sum of its intercepts on the coordinate axes equals –18

(B) hyperbola with eccentricity 2

(C) straight line with the sum of its intercepts on the coordinate axes equals 14

(D) hyperbola with the length of the transverse axis 7

माना $z_1=2+3i$ तथा $z_2=3+4i$ है। तो समुच्चय $S=\left\{z\in C:|z-z_1|^2-|z-z_2|^2=|z_1-z_2|^2\right\}$ किस को निरूपित करता है :

[JEE(Main)Jan.2023]

(A) एक सरल रेखा जिसके निर्देशांक अक्षों पर अंतःखंडों का योग –18 है

(B) एक अतिपरवलय जिसकी उत्केन्द्रता 2 है

(C) एक सरल रेखा जिसके निर्देशांक अक्षों पर अंतःखंडों का योग 14 है

(D) अतिपरवलय जिसके अनुप्रस्थ अक्ष की लंबाई 7 है

Ans. C

Sol. $\left((x-2)^2+(y-3)^2\right)-\left((x-3)^2-(y-4)^2\right)=1+1$

$\Rightarrow x+y=7$

4. Let z be a complex number such that $\left|\frac{z-2i}{z+i}\right| = 2, z \neq -i$. Then z lies on the circle of radius 2 and centre :

[JEE(Main)Jan.2023]

माना z एक सम्मिश्र संख्या है तथा $\left|\frac{z-2i}{z+i}\right| = 2, z \neq -i$ है। तो z त्रिज्या 2 के एक वृत्त पर है जिसका केन्द्र है

[JEE(Main)Jan.2023]

(A) (2,0) (B) (0, –2) (C) (0,2) (D) (0,0)

Ans. B

Sol. $(z-2i)(\overline{z}+2i) = 4(z+i)(\overline{z}-i)$

$z\overline{z} + 4 + 2i(z-\overline{z}) = 4(z\overline{z} + 1 + i(\overline{z}-z))$

$3z\overline{z} - 6i(z-\overline{z}) = 0$

$x^2 + y^2 - 2i(2iy) = 0$

$x^2 + y^2 + 4y = 0$

5. For two non-zero complex numbers z_1 and z_2, if $\text{Re}(z_1z_2) = 0$ and $\text{Re}(z_1 + z_2) = 0$, then which of the following are possible?

A. $\text{Im}(z_1) > 0$ and $\text{Im}(z_2) > 0$

B. $\text{Im}(z_1) < 0$ and $\text{Im}(z_2) > 0$

C. $\text{Im}(z_1) > 0$ and $\text{Im}(z_2) < 0$

D. $\text{Im}(z_1) < 0$ and $\text{Im}(z_2) < 0$

Choose the correct answer from the options given below : **[JEE(Main)Jan.2023]**

(A) B and C (B) B and D (C) A and B (D) A and C

z_1 व z_2 दो शून्येतर सम्मिश्र संख्याऐं है, यदि $\text{Re}(z_1z_2) = 0$ और $\text{Re}(z_1 + z_2) = 0$ है, तब

A. $\text{Im}(z_1) > 0$ एवं $\text{Im}(z_2) > 0$

B. $\text{Im}(z_1) < 0$ एवं $\text{Im}(z_2) > 0$

C. $\text{Im}(z_1) > 0$ एवं $\text{Im}(z_2) < 0$

D. $\text{Im}(z_1) < 0$ एवं $\text{Im}(z_2) < 0$

नीचे दिये गये विकल्पों से सही उत्तर चुनिए : **[JEE(Main)Jan.2023]**

(A) B तथा C (B) B तथा D (C) A तथा B (D) A तथा C

Ans. A

Sol. Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

$\because$ $\text{Re}(z_1z_2) = 0$ and $\text{Re}(z_1 + z_2) = 0$,

$x_1x_2 - y_1y_2 = 0$ ...(i)

$x_1 + x_2 = 0$ ...(ii)

$x_1^2 + y_1y_2 = 0$

$y_1y_2 = -x_1^2$

$\Rightarrow$ Im (z_1) and Im (z_2) are of opposite sign.

6. For all $z \in C$ on the curve $C_1 : |z| = 4$, let the locus of the point $z + \frac{1}{z}$ be the curve C_2. Then :

[JEE(Main)Jan.2023]

(A) the curves C_1 and C_2 intersect at 4 points

(B) the curve C_2 lies inside C_1

(C) the curve C_1 lies inside C_2

(D) the curves C_1 and C_2 intersect at 2 points

माना वक्र $C_1 : |z| = 4$ पर सभी $z \in C$ के लिए बिंदु $z + \frac{1}{z}$ का बिंदुपथ वक्र C_2 है, तो : **[JEE(Main)Jan.2023]**

(A) वक्र C_1 तथा C_2 चार बिंदुओं पर मिलते हैं

(B) वक्र C_2, वक्र C_1 के अन्दर है

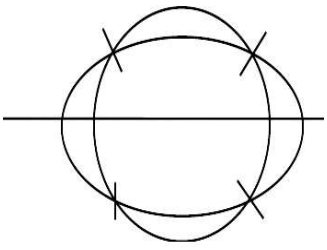
(C) वक्र C_1, वक्र C_2 के अन्दर है

(D) वक्र C_1 तथा C_2 दो बिंदुओं पर मिलते है

Ans. A

Sol. $|z| = 4$ is a circle $x^2 + y^2 = 16$ and $z + \frac{1}{z}$ is an ellipse $\frac{x^2}{\left(\frac{17}{4}\right)^2} + \frac{y^2}{\left(\frac{15}{4}\right)^2} = 1$

$\therefore$ the curves C_1 and C_2 intersect at 4 points



7. Let $z = 1 + i$ and $z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$. Then $\frac{12}{\pi}\arg(z_1)$ is equal to _____. **[JEE(Main)Jan.2023]**

माना $z = 1 + i$ तथा $z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$. है तो $\frac{12}{\pi}\arg(z_1)$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 9

Sol. $z = 1 + i$

$$z_1 = \frac{1 + i\bar{z}}{\bar{z}(1-z) + \frac{1}{z}}$$

$$z_1 = \frac{1 + i(1-i)}{(1-i)(1-1-i) + \frac{1}{1+I}}$$

$$= \frac{1 + i - i^2}{(1-i)(-i) + \frac{1-i}{2}}$$

$$= \frac{2+i}{\frac{-3i-1}{2}} = \frac{4+2i}{-3i-1}$$

$$= \frac{-(4+2i)(3i-1)}{(3i)^2 - (1)^2}$$

$$\text{Arg}(z_1) = \frac{3\pi}{4}$$

$$\therefore \frac{12}{\pi}\arg(z_1) = \frac{12}{\pi} \times \frac{3\pi}{4} = 9$$

8. The complex number $z = \dfrac{i-1}{\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}}$ is equal to : **[JEE(Main)Jan.2023]**

सम्मिश्र संख्या $z = \dfrac{i-1}{\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\cos\frac{\pi}{12} - i\sin\frac{\pi}{12}$ (B) $\sqrt{2}i\left(\cos\frac{5\pi}{12} - i\sin\frac{5\pi}{12}\right)$

(C) $\sqrt{2}\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$ (D) $\sqrt{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$

Ans. C

Sol. $$Z = \frac{i-1}{\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}} = \frac{\sqrt{2}\,e^{i3\pi/4}}{e^{i\pi/3}} = \sqrt{2}\,e^{i5\pi/12} = \sqrt{2}\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$$

9. If the center and radius of the circle $\left|\frac{z-2}{z-3}\right|=2$ are respectively (α, β) and γ, then $3(\alpha+\beta+\gamma)$ is equal to :

[JEE(Main)Jan.2023]

यदि वृत्त $\left|\frac{z-2}{z-3}\right|=2$ के केन्द्र एवं त्रिज्या क्रमशः (α, β) एवं γ है, तब $3(\alpha+\beta+\gamma)$ का मान है :

[JEE(Main)Jan.2023]

(A) 9 (B) 11 (C) 10 (D) 12

Ans. D

Sol. $\left|\frac{z-2}{z-3}\right|=2$

$(z-2)(\overline{z}-2)=4(\overline{z}-3)(z-3)$

$\Rightarrow z\overline{z}-2z-2\overline{z}+4=4\left(z\overline{z}-3z-3\overline{z}+9\right)$

$3z\overline{z}-10z-10\overline{z}+32=0$

$z\overline{z}-\frac{10}{3}z-\frac{10}{3}\overline{z}+\frac{32}{3}=0$

$\alpha=-\frac{10}{3}$ $\quad$ $c=\frac{32}{3}$

centre $\left(\frac{10}{3},0\right)$ $\quad$ $\gamma=\sqrt{|a|^2-c}=\sqrt{\frac{100}{9}-\frac{32}{3}}=\frac{2}{3}$

$\alpha=\frac{10}{3}, \beta=0, \gamma=\frac{2}{3}$

$\therefore \alpha+\beta+\gamma=\frac{10}{3}+\frac{2}{3}=\frac{12}{3}$ $\quad$ $\therefore 3(\alpha+\beta+\psi)=12$

10. Let a, b be two real numbers such that $ab<0$. If the complex number $\frac{1+ai}{b+i}$ is of unit modulus and $a+ib$ lies on the circle $|z-1|=|2z|$, then a possible value of $\frac{1+[a]}{4b}$, where $[t]$ is greatest integer :

[JEE(Main)Jan.2023]

माना a, b दो वास्तविक संख्याऐं है, जिन के लिए $ab < 0$ है। यदि सम्मिश्र संख्या $\frac{1+ai}{b+i}$ इकाई मापक की है तथा $a + ib$ वृत्त, $|z-1| = |2z|$ पर है, तो $\frac{1+[a]}{4b}$ जहाँ $[t]$ महत्तम पूर्णांक फलन है, का एक संभव मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 1 (D) –1

Ans. Drop

Sol. $ab < 0, \left|\frac{1+ai}{b+i}\right| = 1$

$|1 + ai| = |b + i|$

$a^2 + 1 = b^2 + 1 \Rightarrow a = \pm b \Rightarrow b = -a$ as $ab < 0$

(a, b) lies on $|z - 1| = |2z|$

$|a + ib - 1| = 2|a + ib|$

$(a - 1)^2 + b^2 = 4(a^2 + b^2)$

$(a - 1)^2 + a^2 = 4(2a^2)$

$1 - 2a = 6a^2 \Rightarrow 6a^2 + 2a - 1 = 0$

$$a = \frac{-2 \pm \sqrt{28}}{12} = \frac{-1 \pm \sqrt{7}}{6}$$

$$a = \frac{\sqrt{7} - 1}{6} \ \& \ b = \frac{1 - \sqrt{7}}{6}$$

$[a] = 0$

$$\therefore \frac{1 + [a]}{4b} = \frac{6}{4\left(1 - \sqrt{7}\right)} = -\left(\frac{1 + \sqrt{7}}{4}\right)$$

or $[a] = 0$

Similarly it is not matching with $a = \frac{-1 - \sqrt{7}}{6}$

No answer is matching.

11. Let $\alpha = 8 - 14i$, $A = \left\{z \in C : \frac{\alpha z - \bar{\alpha}\bar{z}}{z^2 - (\bar{z})^2 - 112i} = 1\right\}$ and $B = \{z \in C : |z + 3i| = 4\}$. Then $\sum_{z \in A \cap B} (\text{Re}\, z - \text{Im}\, z)$ is equal to _____. **[JEE(Main)Jan.2023]**

माना $\alpha = 8 - 14i$, $A = \left\{z \in C : \frac{\alpha z - \bar{\alpha}\bar{z}}{z^2 - (\bar{z})^2 - 112i} = 1\right\}$ तथा $B = \{z \in C : |z + 3i| = 4\}$ है। तो $\sum_{z \in A \cap B} (\text{Re}\, z - \text{Im}\, z)$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 14

Sol. $\alpha = 8 - 14i$

$z = x + iy$

$az = (8x + 14y) + i(-14x + 8y)$

$z + \bar{z} = 2x$ $z - \bar{z} = 2iy$

$$\text{Set A} : \frac{2i(-14x + 8y)}{i(4xy - 112)} = 1$$

$(x - 4)(y + 7) = 0$

$x = 4$ or $y = -7$

Set B : $x^2 + (y + 3)^2 = 16$

when $x = 4$ $y = -3$

when $y = -7$ $\quad x = 0$

$\therefore A \cap B = \{4 - 3i, 0 - 7i\}$

So, $\sum_{z \in A \cap B} (\text{Re}\, z - \text{Im}\, z) = 4 - (-3) + (0 - (-7)) = 14$

12. Let $a \neq b$ be two non-zero real numbers. Then the number of elements in the set $X = \left\{z \in \mathbb{C} : \text{Re}\left(az^2 + bz\right) = a \right.$ and $\left. \text{Re}\left(bz^2 + az\right) = b\right\}$ is equal to : **[JEE(Main)April,2023]**

माना $a \neq b$ दो शून्येतर वास्तविक संख्याएँ है। तो समुच्चय $X = \left\{z \in \mathbb{C} : \text{Re}\left(az^2 + bz\right) = a\right.$ तथा $\left. \text{Re}\left(bz^2 + az\right) = b\right\}$ में अवयवों की संख्या है : **[JEE(Main)April,2023]**

(A) 1 (B) 3 (C) 2 (D) 0

Ans. D

Sol. $\text{Re}\left(az^2 + bz\right) = a$

$az^2 + bz + a\bar{z}^2 + b\bar{z} = 2a$

$a\left(z^2 + \bar{z}^2\right) + b(z + \bar{z}) = 2a$ .....(1)

$\text{Re}\left(bz^2 + az\right) = b$

$bz^2 + az + b\bar{z}^2 + a\bar{z} = 2b$

$b\left(z^2 + \bar{z}^2\right) + a(z + \bar{z}) = 2b$ .....(2)

$(1) \times b - (2) \times (a)$

$\Rightarrow \left(b^2 - a^2\right)(z + \bar{z}) = 0$

$\Rightarrow \quad (z + \bar{z}) = 0 \quad \left(a^2 \neq b^2\right)$

$(1) \times a - (2) \times (b)$

$\Rightarrow \quad \left(a^2 - b^2\right)(z + \bar{z}) = 2\left(a^2 - b^2\right) \quad \left(a^2 \neq b^2\right)$

$z^2 + \bar{z}^2 = 2$

$\Rightarrow (z + \bar{z})^2 - 2z\bar{z} = 2$

$z\bar{z} = -1$

$\Rightarrow 1 + 1^2 = -1$

$\Rightarrow$ No solution

But when $a = -b$,

$\text{Re}(az - az) = a$

$\Rightarrow \quad \text{Re}\left(a\left(x^2 - y^2 + i2xy\right) - a(x + iy)\right) = a$

$\Rightarrow a\left(x^2 - y^2\right) - ax = a$

$\Rightarrow x^2 - y^2 - x = 1$

$\Rightarrow x^2 - x - 1 = y^2$

For any real values of y there two values of x, hence infinite complex numbers are possible.

13. If for $z = \alpha + i\beta, |z+2| = z + 4(1+i)$, then $\alpha + \beta$ and $\alpha\beta$ are the roots of the equation :

[JEE(Main)April,2023]

यदि $z = \alpha + i\beta$ के लिए $|z+2| = z + 4(1+i)$, तो $\alpha + \beta$ तथा $\alpha\beta$ किस समीकरण के मूल है ?

[JEE(Main)April,2023]

(A) $x^2 + 7x + 12 = 0$ (B) $x^2 + 2x - 3 = 0$ (C) $x^2 + x - 12 = 0$ (D) $x^2 + 3x - 4 = 0$

Ans. A

Sol. $z = \alpha + i\beta$

$|z+2| = z + 4(1+i)$

$|(\alpha+2) + i\beta| = \alpha + i\beta + 4 + 4i$

$\sqrt{(\alpha+2)^2 + \beta^2} = (\alpha+4) + i(\beta+4)$

compare real and imaginary part from both sides

$\beta + 4 = 0 \Rightarrow \beta = -4$

and $\sqrt{(\alpha+2)^2 + 16} = \alpha + 4$

$\Rightarrow \alpha^2 + 4 + 4\alpha + 16 = \alpha^2 + 16 + 8\alpha$

$\Rightarrow \alpha = 1, \beta = -4$

$\alpha + \beta = -3, \alpha\beta = -4$

Hence equation is $x^2 - (-3-4)x + (-3)(-4) = 0$

$x^2 + 7x + 12 = 0$

14. Let $A = \left\{\theta \in (0, 2\pi) : \dfrac{1 + 2i\sin\theta}{1 - i\sin\theta} \text{ is purely imaginary}\right\}$. Then the sum of the elements in A is :

[JEE(Main)April,2023]

माना $A = \left\{\theta \in (0, 2\pi) : \dfrac{1 + 2i\sin\theta}{1 - i\sin\theta} \text{ मात्र काल्पनिक है}\right\}$। तो A में अवयवों का योग है : **[JEE(Main)April,2023]**

(A) π (B) 4π (C) 2π (D) 3π

Ans. B

Sol. $\dfrac{1 + 2i\sin\theta}{1 - i\sin\theta}$ is purely imaginary

So, $\dfrac{1 - 2\sin^2\theta}{1 + \sin^2\theta} = 0$

$\Rightarrow \sin^2\theta = \dfrac{1}{2}$

$\therefore \quad \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \text{ for } \theta \in (0, 2\pi)$

$\therefore \quad$ Sum of all values $= \frac{16\pi}{4} = 4\pi$

15. Let the complex number $z = x + iy$ be such that $\frac{2z-3i}{2z+i}$ is purely imaginary. If $x + y^2 = 0$, then $y^4 + y^2 - y$ is equal to : **[JEE(Main)April,2023]**

माना सम्मिश्र संख्या $z = x + iy$ के लिए $\frac{2z-3i}{2z+i}$ मात्र काल्पनिक है। यदि $x + y^2 = 0$ है, तो $y^4 + y^2 - y$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{3}{2}$

Ans. C

Sol. $z = x + iy$

$\frac{(2z-3i)}{2z+i} = \text{purely imaginary}$

means $\text{Re}\left(\frac{2z-3i}{2z+i}\right) = 0$

$\Rightarrow \frac{(2z-3i)}{(2z+i)} = \frac{2(x+iy)-3i}{2(x+iy)+i}$

$= \frac{2x+2yi-3i}{2x+i2y+i}$

$= \frac{2x+i(2y-3)}{2x+i(2y+1)} \times \frac{2x-i(2y+1)}{2x-i(2y+1)}$

$\text{Re}\left[\frac{2z-3i}{2z+i}\right] = \frac{4x^2+(2y-3)(2y+1)}{4x^2+(2y+1)^2} = 0$

$\Rightarrow 4x^2 + (2y-3)(2y+1) = 0$

$\Rightarrow 4x^2 + \left[4y^2 + 2y - 6y - 3\right] = 0$

$\because x + y^2 = 0 \Rightarrow x = -y^2$

$\Rightarrow 4\left(-y^2\right)^2 + 4y^2 - 4y - 3 = 0$

$\Rightarrow 4y^4 + 4y^2 - 4y = 3$

$\Rightarrow y^4 + y^2 - y = \frac{3}{4}$

Therefore, correct answer is option (3).

16. Let $S=\left\{z=x+iy:\frac{2z-3i}{4z+2i} \text{ is a real number}\right\}$. Then which of the following is NOT correct :

[JEE(Main)April,2023]

माना $S=\left\{z=x+iy:\frac{2z-3i}{4z+2i} \text{ एक वास्तविक संख्या}\right\}$। तो निम्न में कौन सा सही नहीं है ? **[JEE(Main)April,2023]**

(A) $y+x^2+y^2\neq-\frac{1}{4}$ (B) $x=0$

(C) $(x,y)=\left(0,-\frac{1}{2}\right)$ (D) $y\in\left(-\infty,-\frac{1}{2}\right)\cup\left(-\frac{1}{2},\infty\right)$

Ans. C

Sol. $\because$ $z=x+iy$ and $\frac{2z-3i}{4z+2i}$ is real number

$\therefore$ $\frac{2x+(2y-3)i}{4x+(4y+2)i}$ is real number.

$\therefore$ $\frac{(2x+(2y-3)i)(4x-(4y+2)i)}{(4x)^2+(4y+2)^2}$ is real number

$\therefore$ $-2x(4y+2)+(2y-3)4x=0$

$\therefore$ $x=0$

Here $x=0$ and $y\in\mathbb{R}$ but $y\neq-\frac{1}{2}$ is not acceptable

in this case

Denominator will be zero.

17. Let w_1 be the point obtained by the rotation of $z_1=5+4i$ about the origin through a right angle in the anticlockwise direction, and w_2 be the point obtained by the rotation of $z_2=3+5i$ about the origin through a right angle in the clockwise direction. Then the principal argument of w_1-w_2 is equal to : **[JEE(Main)April,2023]**

माना $z_1=5+4i$ को मूल बिंदु के सापेक्ष घड़ी की विपरीत दिशा में एक समकोण तक घुमाने पर बिंदु w_1 प्राप्त होता है तथा $z_2=3+5i$ को मूलबिंदु के सापेक्ष घड़ी की दिशा में एक समकोण तक घुमाने पर बिंदु w_2 प्राप्त होता है। तो w_1-w_2 का मुख्य आयाम बराबर है :

[JEE(Main)April,2023]

(A) $-\pi+\tan^{-1}\frac{8}{9}$ (B) $\pi-\tan^{-1}\frac{33}{5}$ (C) $-\pi+\tan^{-1}\frac{33}{5}$ (D) $\pi-\tan^{-1}\frac{8}{9}$

Ans. D

Sol. $W_1=z_i=(5+4i)i=-4+5i\ldots$ (i)

$W_1=z_2(-i)=(3+5i)(-i)=5-3i\ldots(2)$

$W_1-W_2=-9+8i$

Principal argument $= \pi - \tan^{-1}\left(\frac{8}{9}\right)$

18. For $a \in \mathbb{C}$, let $A = \{z \in \mathbb{C} : Re(a + \bar{z}) > Im(\bar{a} + z)\}$ and $B = \{z \in \mathbb{C} : Re(a + \bar{z}) < Im(\bar{a} + z)\}$. Then among the two statements :
(S1) : If Re (a), Im (a) > 0, then the set A contains all the real numbers
(S2) : If Re (a), Im (a) < 0, then the set B contains all the real numbers **[JEE(Main)April,2023]**

(A) only (S1) is true (B) both are false (C) only (S2) is true (D) both are true

$a \in \mathbb{C}$ के लिए, माना $A = \{z \in \mathbb{C} : Re(a + \bar{z}) > Im(\bar{a} + z)\}$ तथा $B = \{z \in \mathbb{C} : Re(a + \bar{z}) < Im(\bar{a} + z)\}$ हैं। तो दो कथनों:
(S1) : यदि Re (a), Im (a) > 0 है, तो सभी वास्तविक संख्याएँ A में हैं
(S2) : यदि Re (a), Im (a) < 0 है, तो सभी वास्तविक संख्याएँ B में हैं **[JEE(Main)April,2023]**

(A) केवल (S1) सही है (B) दोनो गलत है (C) केवल (S2) सही है (D) दोनो सही है

Ans. B

Sol. Let $a = x_1 + iy_1, z = x + iy$

Now $Re(a + \bar{z}) > Im(\bar{a} + z)$

$\therefore x_1 + x > -y_1 + y$

$x_1 = 2, y_1 = 10, x = -12, y = 0$

Given inequality is not valid for these values.
S1 is false.

Now $Re(a + \bar{z}) < Im(\bar{a} + z)$

$x_1 + x < -y_1 + y$

$x_1 = -2, y_1 = -10, x = 12, y = 0$

Given inequality is not valid for these values.
S2 is false.

19. Let C be the circle in the complex plane with centre $z_0 = \frac{1}{2}(1 + 3i)$ and radius r = 1. Let $z_1 = 1 + i$ and the complex number z_2 be outside the circle C such that $|z_1 - z_0||z_2 - z_0| = 1$. If z_0, z_1 and z_2 are collinear, then the smaller value of $|z_2|^2$ is equal to : **[JEE(Main)April,2023]**

माना सम्मिश्र तल में वृत C का केन्द्र $z_0 = \frac{1}{2}(1 + 3i)$ तथा त्रिज्या r = 1 हैं। माना $z_1 = 1 + i$ है तथा वृत्त के बाहर सम्मिश्र संख्या z_2 इस प्रकार है कि $|z_1 - z_0||z_2 - z_0| = 1$ है। यदि z_0, z_1 तथा z_2 संरेख हैं, तो $|z_2|^2$ का छोटा मान बराबर है :

[JEE(Main)April,2023]

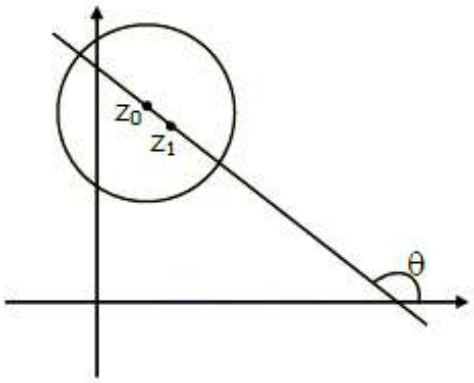
(A) $\frac{7}{2}$ (B) $\frac{13}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Ans. D

Sol. $|z_1 - z_0| = \left|\frac{1-i}{2}\right| = \frac{1}{\sqrt{2}}$

$\Rightarrow |z_2 - z_\circ| = \sqrt{2};$ centre $\left(\frac{1}{2}, \frac{3}{2}\right)$

$z_\circ\left(\frac{1}{2}, \frac{3}{2}\right)$ and $z_1(1,1)$



$\tan\theta = -1 \Rightarrow \theta = 135^\circ$

$z_2\left(\frac{1}{2} + \sqrt{2}\cos 135^\circ, \frac{3}{2} + \sqrt{2}\sin 135^\circ\right)$

or

$\left(\frac{1}{2} - \sqrt{2}\cos 135^\circ, \frac{3}{2} - \sqrt{2}\sin 135^\circ\right)$

$\Rightarrow z_2\left(-\frac{1}{2}, \frac{5}{2}\right)$ or $z_2\left(\frac{3}{2}, \frac{1}{2}\right)$

$\Rightarrow |z_2|^2 = \frac{26}{4}, \frac{5}{2}$

$\Rightarrow |z_2|_{\min^2} = \frac{5}{2}$

20. Let α, β be the roots of the equation $x^2 - \sqrt{2}x + 2 = 0$. Then $\alpha^{14} + \beta^{14}$ is equal to :

[JEE(Main)April,2023]

माना समीकरण $x^2 - \sqrt{2}x + 2 = 0$ के मूल α, β हैं। तो $\alpha^{14} + \beta^{14}$ बराबर है : **[JEE(Main)April,2023]**

(A) -64 (B) -128 (C) $-64\sqrt{2}$ (D) $-128\sqrt{2}$

Ans. B

Sol. $x^2 - \sqrt{2}x + 2 = 0$

$x = \frac{\sqrt{2} \pm \sqrt{2-8}}{2} = \frac{\sqrt{2} \pm \sqrt{6}i}{2}$

$\alpha = \frac{\sqrt{2} + \sqrt{6}i}{2} = \sqrt{2}e^{\frac{i\pi}{3}}$ & $\beta = \sqrt{2}e^{\frac{-i\pi}{3}}$

$\alpha^{14} = 2^7 e^{\frac{i14\pi}{3}} = 128\left[e^{\frac{i2\pi}{3}}\right]$

$\beta^{14} = 128\left[e^{\frac{-i2\pi}{3}}\right]$

$$\alpha^{14}+\beta^{14}=128(2)\cos\left(\frac{2\pi}{3}\right)=-128$$

21. Let $S=\left\{z\in\mathbb{C}:\overline{z}=i\left(z^2+\text{Re}(\overline{z})\right)\right\}$. The $\sum_{z\in S}|z|^2$ is equal to : **[JEE(Main)April,2023]**

माना $S=\left\{z\in\mathbb{C}:\overline{z}=i\left(z^2+\text{Re}(\overline{z})\right)\right\}$ है। तो $\sum_{z\in S}|z|^2$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{5}{2}$ (B) 4 (C) 3 (D) $\frac{7}{2}$

Ans. B

Sol. Let $z=x+iy$

$\overline{z}=i\left(z^2+\text{Re}\left(\overline{z}\right)\right)$

$\Rightarrow x-iy=i\,(x^2-y^2+2i\,xy+x)$

$x-iy=i\,(x^2-y^2+x)-2xy$

$x=-2xy\Rightarrow x(2y+1)=0$

$\Rightarrow x=0,y=\frac{-1}{2}$ ....(i)

$-y=x^2-y^2+x$ ....(ii)

Case (1) $x=0$

(ii) $\Rightarrow -y=-y^2\Rightarrow y^2-y=0\Rightarrow y=0,1$

$z=0,i$

Case (2) $y=\frac{-1}{2}$

(ii) $\Rightarrow \frac{1}{2}=x^2-\frac{1}{4}+x\Rightarrow x^2+x-\frac{3}{4}=0$

$4x+4x-3=0\Rightarrow(2x-1)(2x+3)=0$

$x=\frac{1}{2},\frac{-3}{2}$

$z=\frac{1}{2}-\frac{1}{2}i,\frac{-3}{2}-\frac{1}{2}i$

$\sum|z|^2=0+1+\frac{1}{2}+\frac{5}{2}=4$

22. If the set $\left\{\text{Re}\left(\frac{z-\overline{z}+z\overline{z}}{2-3z+5\overline{z}}\right):z\in\mathbb{C},\text{Re}(z)=3\right\}$ is equal to the interval $(\alpha,\beta]$, then $24(\beta-\alpha)$ is equal to :

[JEE(Main)April,2023]

यदि समुच्चय $\left\{\text{Re}\left(\frac{z-\overline{z}+z\overline{z}}{2-3z+5\overline{z}}\right):z\in\mathbb{C},\text{Re}(z)=3\right\}$ अन्तराल $(\alpha,\beta]$ के बराबर है, तो $24(\beta-\alpha)$ का मान होगा :

(A) 27 (B) 42 (C) 36 (D) 30

Ans. D

Sol. Let $z_1 = \left(\frac{z-\overline{z}+z\overline{z}}{2-3z+5\overline{z}}\right)$

Let $z = 3 + iy$

$\overline{z} = 3 - iy$

$$z_1 = \frac{2iy+\left(9+y^2\right)}{2-3\left(3+iy\right)+5\left(3-iy\right)}$$

$$= \frac{9+y^2+i\left(2y\right)}{8-8iy}$$

$$= \frac{\left(9+y^2\right)+i\left(2y\right)}{8\left(1-iy\right)}$$

$$\mathrm{Re}\left(z_1\right) = \frac{\left(9+y^2\right)-2y^2}{8\left(1+y^2\right)}$$

$$= \frac{9-y^2}{8\left(1+y^2\right)}$$

$$= \frac{1}{8}\left[\frac{10-\left(1+y^2\right)}{\left(1+y^2\right)}\right]$$

$$= \frac{1}{8}\left[\frac{10}{1+y^2}-1\right]$$

$1 + y^2 \in [1, \infty]$

$$\frac{1}{1+y^2} \in \left(0,1\right]$$

$$\frac{10}{1+y^2} \in \left(0,10\right]$$

$$\frac{10}{1+y^2}-1 \in \left(-1,9\right]$$

$$\mathrm{Re}\left(z_1\right) \in \left(\frac{-1}{8},\frac{9}{8}\right]$$

$\alpha = \frac{-1}{8}, \beta = \frac{9}{8}$

$24(\beta - \alpha) = 24\left(\frac{9}{8} + \frac{1}{8}\right) = 30$

23. For $\alpha, \beta, z \in \mathbb{C}$ and $\lambda > 1$, If $\sqrt{\lambda - 1}$ is the radius of the circle $|z - \alpha|^2 + |z - \beta|^2 = 2\lambda$, then $|\alpha - \beta|$ is equal to _____.

[JEE(Main)April,2023]

$\alpha, \beta, z \in \mathbb{C}$ तथा $\lambda > 1$ के लिए, यदि वृत्त $|z - \alpha|^2 + |z - \beta|^2 = 2\lambda$ की त्रिज्या $\sqrt{\lambda - 1}$ है, तो $|\alpha - \beta|$ बराबर है _____

[JEE(Main)April,2023]

Ans. 2

Sol. $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$

$z_1 = \alpha, z_2 = \beta$

$|\alpha - \beta|^2 = 2\lambda$

$|\alpha - \beta| = \sqrt{2\lambda}$

$2r = \sqrt{2\lambda}$

$2\sqrt{\lambda - 1} = \sqrt{2\lambda}$

$\Rightarrow 4(\lambda - 1) = 2\lambda$

$\lambda = 2$

$|\alpha - \beta = 2$

24. Let $S = \left\{z \in \mathbb{C} - \{i, 2i\} : \frac{z^2 + 8iz - 15}{z^2 - 3iz - 2} \in \mathbb{R}\right\}$. If $\alpha - \frac{13}{11}i \in S, \alpha \in \mathbb{R} - \{0\}$, then $242\alpha^2$ is equal to _____.

[JEE(Main)April,2023]

माना $S = \left\{z \in \mathbb{C} - \{i, 2i\} : \frac{z^2 + 8iz - 15}{z^2 - 3iz - 2} \in \mathbb{R}\right\}$ है। यदि $\alpha - \frac{13}{11}i \in S, \alpha \in \mathbb{R} - \{0\}$ है, तो $242\alpha^2$ बराबर है _____

[JEE(Main)April,2023]

Ans. 1680

Sol. $\left(\frac{z^2 + 8iz - 15}{z^2 - 3iz - 2}\right) \in R$

$\Rightarrow 1 + \frac{(1 \text{ liz } -13)}{\left(z^2 - 3iz - 2\right)} \in R$

Put $z = \alpha - \frac{13}{11}i$

$\Rightarrow \left(z^2 - 3iz - 2\right)$ is imaginary

Put $z = x + iy$

$\Rightarrow \left(x^2 - y^2 + 2xyi - 3ix + 3y - 2\right) \in$ Imaginary

$\Rightarrow \text{Re}\left(x^2 - y^2 + 3y - 2 + (2xy - 3x)i\right) = 0$

$\Rightarrow x^2 - y^2 + 3y - 2 = 0$

$x^2 = y^2 - 3y + 2$

$y^2 = (y - 1)(y - 2)$

$\therefore z = \alpha - \frac{13}{11}i$

Put $x = \alpha,\ y = \frac{-13}{11}$

$\alpha^2 = \left(\frac{-13}{11} - 1\right)\left(\frac{-13}{11} - 2\right)$

$\alpha^2 = \frac{(24 \times 35)}{121}$

$242\alpha^2 = 48 \times 35 = 1680$

25. Let $w = z\bar{z} + k_1 z + k_2 iz + \lambda(1+i), k_1, k_2 \in \mathbb{R}$. Let Re(w) = 0 be the circle C of radius 1 in the first quadrant touching the line y = 1 and the y-axis. If the curve Im(w) = 0 intersects C at A and B, then $30(AB)^2$ is equal to _____. **[JEE(Main)April,2023]**

माना $w = z\bar{z} + k_1 z + k_2 iz + \lambda(1+i), k_1, k_2 \in \mathbb{R}$ है। माना Re(w) = 0 प्रथम चतुर्थांश में इकाई त्रिज्या का एक वृत्त C है, जो रेखा y = 1 तथा y-अक्ष को स्पर्श करता है। यदि वक्र Im(w) = 0, वृत्त C को A तथा B पर काटता है, तो $30(AB)^2$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 24

Sol. $\omega = z\bar{z} + k_1 z + k_2 iz + \lambda(1+i)$

$\text{Re}(\omega) = x^2 + y^2 + k_1 x - x_2 y + \lambda = 0$

$\text{Centre} \equiv \left(\frac{-k_1}{2}, \frac{k_2}{2}\right) \equiv (1, 2)$

$\Rightarrow k_1 = -2,\ k_2 = 4$

radius = $1 \Rightarrow \lambda - 4 = 0$

$\text{Im} = k_1 y + k_2 x + \lambda = 0$

$\therefore\ 2x - y + 2 = 0$

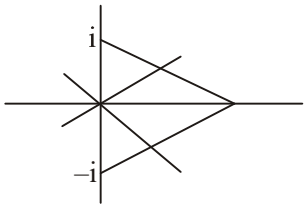
26. If $S = \{z \in C : |z - i| = |z + i| = |z - 1|\}$, then n(S) is : **[JEE(Main)Jan.2024]**

यदि If $S = \{z \in C : |z - i| = |z + i| = |z - 1|\}$ है, तो n(S) बराबर है : **[JEE(Main)Jan.2024]**

(A) 2 (B) 0 (C) 3 (D) 1

Ans. D

Sol.



27. If $z=\frac{1}{2}-2i$ is such that $|z+1|=\alpha z+\beta(1+i), i=\sqrt{-1}$ and $\alpha,\beta\in R$, then $\alpha+\beta$ is :

[JEE(Main)Jan.2024]

यदि $z=\frac{1}{2}-2i$ के लिए $|z+1|=\alpha z+\beta(1+i)$ हैं, जहाँ $i=\sqrt{-1}$ तथा $\alpha,\beta\in R$ है, तो $\alpha+\beta$ बराबर है :

[JEE(Main)Jan.2024]

(A) 2 (B) – 4 (C) 3 (D) –1

Ans. C

Sol. $\left|\frac{3}{2}-2i\right|=\left(\frac{\alpha}{2}-2\alpha i\right)+\beta i+\beta$

$2\alpha=\beta$ and $\frac{\alpha}{2}+\beta=\sqrt{\frac{9}{4}+4}$

$\frac{\alpha}{2}+2\alpha=\frac{5}{2}$

$\alpha = 1$

$\beta = 2$

$|\alpha+\beta|=3$

28. Let r and θ respectively be the modulus and amplitude of the complex number $z=2-i\left(2\tan\frac{5\pi}{8}\right)$, then (r, θ) is equal to :

[JEE(Main)Jan.2024]

यदि सम्मिश्र संख्या $z=2-i\left(2\tan\frac{5\pi}{8}\right)$ के मापांक तथा कोणांक क्रमशः r तथा θ हैं, तो (r, θ) बराबर है :

[JEE(Main)Jan.2024]

(A) $\left(2\sec\frac{11\pi}{8},\frac{11\pi}{8}\right)$ (B) $\left(2\sec\frac{3\pi}{8},\frac{5\pi}{8}\right)$ (C) $\left(2\sec\frac{3\pi}{8},\frac{3\pi}{8}\right)$ (D) $\left(2\sec\frac{5\pi}{8},\frac{3\pi}{8}\right)$

Ans. C

Sol. $z = 2 - i\left(2\tan\frac{5\pi}{8}\right) = x + iy$ (let)

$r = \sqrt{x^2 + y^2}$

$\theta = \tan^{-1}\frac{y}{x}$

$r = \sqrt{(2)^2 + \left(2\tan\frac{5\pi}{8}\right)^2}$

$= \left|2\sec\frac{5\pi}{8}\right| = \left|2\sec\left(\pi - \frac{3\pi}{8}\right)\right|$

$= 2\sec\frac{3\pi}{8}$

$\theta = \tan^{-1}$

$\frac{-2\tan\frac{5\pi}{8}}{2}$

$= \tan^{-1}\left(\tan\left(\pi - \frac{5\pi}{8}\right)\right) = \frac{3\pi}{8}$

29. If $z = x + iy, xy \neq 0$, satisfies the equation $z^2 + i\bar{z} = 0$, then $|z^2|$ is equal to : **[JEE(Main)Jan.2024]**

यदि $z = x + iy, xy \neq 0$ समीकरण $z^2 + i\bar{z} = 0$ को संतुष्ट करता है, तो $|z^2|$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 1 (B) 1/4 (C) 9 (D) 4

Ans. A

Sol. $z = x + iy, x \neq 0, y \neq 0$ $\Rightarrow |z| \neq 0$

$z^2 + i\bar{z} = 0$

$\Rightarrow z^2 = -i\bar{z}$

$\Rightarrow \left|z^2\right| = \left|-i\bar{z}\right| = \left|-i\right|\left|\bar{z}\right|$

$\Rightarrow |z|^2 = |z| \Rightarrow |z|(|z| - 1) = 0$

$\Rightarrow |z| = 1$

30. If z is a complex number, then the number of common roots of the equations $z^{1985} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$, is equal to : **[JEE(Main)Jan.2024]**

यदि z एक सम्मिश्र संख्या हैं, तो समीकरणों $z^{1985} + z^{100} + 1 = 0$ तथा $z^3 + 2z^2 + 2z + 1 = 0$ के उभयनिष्ठों मूलों की संख्या है : **[JEE(Main)Jan.2024]**

(A) 0 (B) 3 (C) 1 (D) 2

Ans. D

Sol. Given $z^{1985} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$

The roots of $z^{1985} + z^{100} + 1 = 0$ be ω and ω^2 and also satisfies $z^3 + 2z^2 + 2z + 1 = 0$

$\therefore$ ω and ω^2 are common solutions

(where ω is cube root of unity)

$\therefore$ 2 solutions

31. Let z_1 and z_2 be two complex numbers such that $z_1 + z_2 = 5$ and $z_1^3 + z_2^3 = 20 + 15i$. Then $|z_1^4 + z_2^4|$ equals :

[JEE(Main)Jan.2024]

माना दो साम्मिश्र संख्याओं z_1 तथा z_2 के लिए $z_1 + z_2 = 5$ तथा $z_1^3 + z_2^3 = 20 + 15i$ हैं। तो $|z_1^4 + z_2^4|$ बराबर है :

[JEE(Main)Jan.2024]

(A) $30\sqrt{3}$ (B) $25\sqrt{3}$ (C) $15\sqrt{15}$ (D) 75

Ans. D

Sol. $z_1 + z_2 = 5$

$z_1^3 + z_2^3 = 20 + 15i$

$z_1^3 + z_2^3 = (z_1 + z_2)^3 - 3z_1z_2(z_1 + z_2)$

$z_1^3 + z_2^3 = 125 - 3z_1.z_2(5)$

$\Rightarrow 20 + 15i = 125 - 15z_1z_2$

$\Rightarrow 3z_1z_2 = 25 - 4 - 3i$

$\Rightarrow 3z_1z_2 = 21 - 3i$

$\Rightarrow z_1 \cdot z_2 = 7 - i$

$\Rightarrow (z_1 + z_2)^2 = 25$

$\Rightarrow z_1^2 + z_2^2 = 25 - 2(7 - i)$

$\Rightarrow 11 + 2i$

$(z_1^2 + z_2^2)^2 = 121 - 4 + 44i$

$\Rightarrow z_1^4 + z_2^4 + 2(7 - i)^2 = 117 + 44i$

$\Rightarrow z_1^4 + z_2^4 = 117 + 44i - 2(49 - 1 - 14i)$

$\Rightarrow |z_1^4 + z_2^4| = 75$

32. Let $S = \{z \in C : |z - 1| = 1$ and $(\sqrt{2} - 1)(z + \bar{z}) - i(z - \bar{z}) = 2\sqrt{2}\}$. Let $z_1, z_2 \in S$ be such that $|z_1| = \max_{Z \in S} |z|$ and $|z_2| = \min_{z \in S} |z|$. Then $|\sqrt{2}z_1 - z_2|^2$ equals : **[JEE(Main)Jan.2024]**

माना $S = \{z \in C : |z-1| = 1$ तथा $\left(\sqrt{2}-1\right)\left(z+\bar{z}\right)-i\left(z-\bar{z}\right)=2\sqrt{2}\}$ है। माना $z_1, z_2 \in S$ के लिए $|z_1| = \max_{Z\in S}|z|$ तथा $|z_2| = \min_{Z\in S}|z|$ हैं। तो $\left|\sqrt{2}z_1 - z_2\right|^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $\left(\sqrt{2}-1\right)\left(z+\bar{z}\right)-i\left(z-\bar{z}\right)=2\sqrt{2}$

$\left(\sqrt{2}-1\right)(2x)-i(2iy)=2\sqrt{2}$

$\left(\sqrt{2}-1\right)x+y=\sqrt{2}$

and circle $|z-1|=1$

$(x-1)^2+y^2=1$

$(x-1)^2+\left(\sqrt{2}-\left(\sqrt{2}-1\right)x\right)^2=1$

$(x-1)^2+\left(\sqrt{2}(1-x)+x\right)^2=1$

$(x-1)^2+2(1-x)^2+x^2+2x\sqrt{2}(1-x)=1$

$3(x-1)^2+x^2+2\sqrt{2}x(1-x)=1$

$\left(-2\sqrt{2}+4\right)x^2+x\left(-6+2\sqrt{2}\right)+2=0$

$x = 1$ or $x=\dfrac{1}{2-\sqrt{2}}$

for $x = 1 \Rightarrow y = 1$

and $x=\dfrac{1}{2-\sqrt{2}} \Rightarrow y=\sqrt{2}-\dfrac{1}{\sqrt{2}}$

So, $\left|\sqrt{2}z_1 - z_2\right|^2 = \left|\dfrac{1}{\sqrt{2}-1}+i-1-i\right|^2 = 2$

33. If z is a complex number such that $|z| \geq 1$, then the minimum value of $\left|z+\dfrac{1}{2}(3+4i)\right|$ is :

[JEE(Main)Jan.2024]

यदि एक सम्मिश्र संख्या z के लिए $|z| \geq 1$ है, तो $\left|z+\dfrac{1}{2}(3+4i)\right|$ का न्यूनतम मान है : **[JEE(Main)Jan.2024]**

(A) 3 (B) $\dfrac{5}{2}$ (C) 2 (D) $\dfrac{3}{2}$

Ans. Bonus

Sol. p(z) is a point lying outside circle $x^2 + y^2 = 1$

$Q\left(\frac{-3}{2},-2\right)$

$\left|z+\frac{1}{2}(3+4i)\right| = PQ$

$(PQ)_{min} = 0$ when P coincides with Q.

34. If α satisfies the equation $x^2 + x + 1 = 0$ and $(1+\alpha)^7 = A + B\alpha + C\alpha^2, A, B, C \geq 0$, then $5(3A - 2B - C)$ is equal to _____. **[JEE(Main)Jan.2024]**

यदि α समीकरण $x^2 + x + 1 = 0$ को संतुष्ट करता है तथा $(1+\alpha)^7 = A + B\alpha + C\alpha^2, A, B, C \geq 0$ हैं, तो $5(3A - 2B - C)$ बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 5

Sol. $\alpha^2 + \alpha + 1 = 0 \left\langle \begin{matrix} \omega \\ \omega^2 \end{matrix} \right| \omega^3 = 1 \left(x^2 + x + 1 = 0\right) 0 \left\langle \begin{matrix} \omega \\ \omega^2 \end{matrix} \right| \omega^3 = 1$ let $\alpha = \omega$

$(1+\alpha)^7 = 1 + {}^7c_1\alpha + {}^7c_2\alpha^2 + {}^7c_3\alpha^3 + {}^7c_4\alpha^4 + {}^7c_5\alpha^5 + {}^7c_6\alpha^6 + {}^7c_7\alpha^7$

$= 1 + 7\alpha + 21\alpha^2 + 35\alpha^3 + 35\alpha^4 + 24\alpha^5 + 7\alpha^6 + \alpha^9$

$\because \alpha = \omega$

$= 1 + 7\omega + 21\omega^2 + 35\omega^3 + 35\omega^4 + 21\omega^5 + 7\omega^6 + \omega^7$

$\left(\because \omega^3 = 1\right)$

$= 1 + 7\omega + 21\omega^2 + 35 + 35\omega + 21\omega + 7 + \omega$

$= 43 + 43\omega + 42\omega^2$

$= 43 + 43\alpha + 42\alpha^2 = A + B\alpha + C\alpha^2$

$\Rightarrow A = 43$ |B = 43| C = 42

5(129 – 86 – 42)

=5(129 – 128) = 5

35. Let the complex numbers α and $\frac{1}{\overline{\alpha}}$ lie on the circle $|z - z_0|^2 = 4$ and $|z - z_0|^2 = 16$ respectively, where $z_0 = 1 + i$. Then, the value of $100\,|\alpha|^2$ is _____. **[JEE(Main)Jan.2024]**

माना सम्मिश्र संख्याएँ α तथा $\frac{1}{\overline{\alpha}}$ क्रमशः वृत्तों $|z - z_0|^2 = 4$ तथा $|z - z_0|^2 = 16$ पर हैं, जहाँ $z_0 = 1 + i$ है। तो $100\,|\alpha|^2$ का मान है _____। **[JEE(Main)Jan.2024]**

Ans. 20

Sol. $|\alpha - z_0|^2 = 4$

$\Rightarrow (\alpha - z_0)(\overline{\alpha} - \overline{z}_0) = 4$

$|\alpha|^2 - \alpha\bar{z}_0 - \bar{\alpha}z_0 + |z_0|^2 = 4$

$|\alpha|^2 - \alpha\bar{z}_0 - \bar{\alpha}z_0 = 2$ _____(i)

(ii) $\left|\frac{1}{\bar{\alpha}} - z_0\right|^2 = 16 \Rightarrow (1-\bar{\alpha}z_0)(1-\alpha\bar{z}_0) = 16|\alpha|^2$

$\Rightarrow 1 - \alpha\bar{z}_0 - \bar{\alpha}z_0 + |\alpha|^2 \cdot 2 = 16|\alpha|^2$ _____(ii)

from (i) and (ii) $-1 - |\alpha|^2 = 2 - 16|\alpha|^2 \Rightarrow 15|\alpha|^2 = 3 \Rightarrow 5|\alpha|^2 = 1$

$100|\alpha|^2 = 20$

36. Let α, β the roots of the equation $x^2 - x + 2 = 0$ with $\text{Im}(\alpha) > \text{Im}(\beta)$. Then $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ is equal to _____. **[JEE(Main)Jan.2024]**

माना समीकरण $x^2 - x + 2 = 0$ के मूल α, β हैं तथा $\text{Im}(\alpha) > \text{Im}(\beta)$ है। तो $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ बराबर है_____. **[JEE(Main)Jan.2024]**

Ans. 13

Sol. $x^2 + 2 = x$

$\Rightarrow x^4 + 4 + 4x^2 = x^2$

So,

$x^4 = -4 - 3x^2$

$\Rightarrow x^6 = -4x^2 - 3x^4$

$\Rightarrow x^6 = 12 + 5x^2$

Now,

$\alpha^6 - 5\alpha^2 = 12$

So $\alpha^4 + \beta^4 + 12$

$= 4 - 3(\alpha^2 + \beta^2)$

$= 4 - 3\,[1 - 4]$

$= 13$

37. Let α, β be the roots of the equation $x^2 - \sqrt{6}x + 3 = 0$ such that $\text{Im}(\alpha) > \text{Im}(\beta)$. Let a, b be integers not divisible by 3 and n be a natural number such that $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^n(a + ib)$, $i = \sqrt{-1}$. Then n + a + b is equal to **[JEE(Main)Jan.2024]**

माना समीकरण $x^2-\sqrt{6}x+3=0$ के मूल α, β हैं तथा Im (α) > Im (β) है। माना 3 से अभाज्य पूर्णांक a, b हैं तथा घन पूर्णांक n के लिए $\frac{\alpha^{99}}{\beta}+\alpha^{98}=3^n(a+ib)$, $i=\sqrt{-1}$ है। तो n + a + b बराबर है _____

[JEE(Main)Jan.2024]

Ans. 49

Sol. $x^2-\sqrt{6}x+3=0 \begin{matrix}\nearrow \alpha \\ \searrow \beta\end{matrix}$

$$x=\frac{\sqrt{6}\pm i\sqrt{6}}{2}=\frac{\sqrt{6}}{2}(1\pm i)$$

$$\alpha=\sqrt{3}\left(e^{i\frac{\pi}{4}}\right),\ \beta=\sqrt{3}\left(e^{-i\frac{\pi}{4}}\right)$$

$$\therefore \frac{\alpha^{99}}{\beta}+\alpha^{98}=\alpha^{98}\left(\frac{\alpha}{\beta}+1\right)$$

$$=\frac{\alpha^{98}(\alpha+\beta)}{\beta}=3^{49}\left(e^{i99\frac{\pi}{4}}\right)\times\sqrt{2}$$

$= 3^{49}(-1 + i)$
$= 3^n(a + ib)$
$\therefore$ n = 49, a = –1, b = 1
$\therefore$ n + a + b = 49 – 1 + 1 = 49

38. If α denotes the number of solutions of $|1-i|^x=2^x$ and $\beta=\left(\frac{|z|}{\arg(z)}\right)$, where $z=\frac{\pi}{4}(1+i)^4\left[\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i}+\frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right]$, $i=\sqrt{-1}$, then the distance of the point (α, β) from the line 4x – 3y = 7 is _____.

[JEE(Main)Jan.2024]

यदि $|1-i|^x = 2^x$ के हलों की संख्या α है तथा $\beta=\left(\frac{|z|}{\arg(z)}\right)$ है, जहाँ $z=\frac{\pi}{4}(1+i)^4\left[\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i}+\frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right]$, $i=\sqrt{-1}$ है, तो रेखा 4x – 3y = 7 से बिन्दु (α, β) की दूरी है _____.

[JEE(Main)Jan.2024]

Ans. 3

Sol. $$z=\frac{\pi}{4}(1+i)^4\left[\frac{1+\pi+\pi+1}{i(1+\pi)-\sqrt{\pi}+\sqrt{\pi}}\right]$$

$$=\frac{\pi}{4}(1+i)^4\frac{2}{i}=\frac{\pi}{2}\frac{(1+i)^4}{i}=\frac{-\pi}{2}i(1+i)^4$$

$$|z| = \left|\frac{-\pi}{2}\right||i||1+i|^4 = 2\pi$$

$$z = \frac{-\pi}{2}(-4i) = 2\pi i \Rightarrow \arg(z) = \frac{\pi}{2}$$

$$\Rightarrow \beta = \frac{2\pi}{\frac{\pi}{2}} = 4$$

Also $|1 - i|^x = 2^x$

$$\left(\sqrt{2}\right)^x = 2^x$$

$\Rightarrow x = 0$

$\Rightarrow$ 1 solution $\Rightarrow \alpha = 1$

Now perpendicular distance of $4x - 3y = 7$ from (1,4)

$$\left|\frac{4(1)-3(4)-7}{5}\right| = 3$$

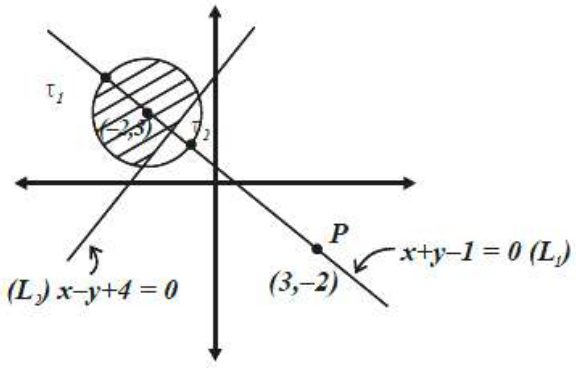
39. Let $P = \{z \in C : |z + 2 - 3i| \le 1\}$ and $Q = \{z \in C : z(1+i) + \bar{z}(1-i) \le -8\}$. Let in $P \cap Q$, $|z - 3 + 2i|$ be maximum and minimum at z_1 and z_2 respectively. If $|z_1|^2 + 2|z_2|^2 = \alpha + \beta\sqrt{2}$, where α, β are integers, then $\alpha + \beta$ equals _____. **[JEE(Main)Jan.2024]**

माना $P = \{z \in C : |z + 2 - 3i| \le 1\}$ तथा $Q = \{z \in C : z(1+i) + \bar{z}(1-i) \le -8\}$ हैं। माना $P \cap Q$ में, $|z - 3 + 2i|$, z_1 तथा z_2 पर क्रमशः अधिकतम तथा निम्नतम है। यदि $|z_1|^2 + 2|z_2|^2 = \alpha + \beta\sqrt{2}$ है, जहाँ α, β पूर्णांक हैं, तो $\alpha + \beta$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 36

Sol.



Clearly for the shaded region z_1 is the intersection of the circle and the line passing through P (L_1) and z_2 is intersection of line L_1 & L_2

Circle : $(x + 2)^2 + (y - 3)^2 = 1$

$L_1 : x + y - 1 = 0$

$L_2 : x - y + 4 = 0$

On solving circle & L_1 we get

$z_1 : \left(-2-\frac{1}{\sqrt{2}}, 3+\frac{1}{\sqrt{2}}\right)$

On solving L_1 and z_2 is intersection of line L_1 & L_2

we get $z_2 : \left(\frac{-3}{2}, \frac{5}{2}\right)$

$|z_1|^2 + 2|z_2|^2 = 14 + 5\sqrt{2} + 17$

$= 31 + 5\sqrt{2}$

So, $\alpha = 31$

$\beta = 5$

$\alpha + \beta = 36$

40. Let α and β be the sum and the product of all the non-zero solutions of the equation $(\bar{z})^2 + |z| = 0$, $z \in C$. Then $4(\alpha^2 + \beta^2)$ is equal to : **[JEE(Main)April,2024]**

माना समीकरण $(\bar{z})^2 + |z| = 0$, $z \in C$ के सभी शून्येत्तर हलों के योग और गुणनफल क्रमशः α और β हैं। तो $4(\alpha^2 + \beta^2)$ बराबर है: **[JEE(Main)April,2024]**

(A) 2 (B) 8 (C) 4 (D) 6

Ans. C

Sol. $z = x + iy$

$\bar{z} = x - iy$

$\bar{z}^2 = x^2 - y^2 - 2ixy$

$\Rightarrow x^2 - y^2 - 2ixy + \sqrt{x^2 + y^2} = 0$

$\Rightarrow x = 0$ or $y = 0$

$-y^2 + |y| = 0$ $x^2 + |x| = 0$

$|y| = |y|^2$ $x = 0$

$y = 0, \pm 1$

$\Rightarrow i, -i$ $\Rightarrow \alpha = i - i = 0$

are roots $\beta = i(-i) = 1$

$4(0 + 1) = 4$

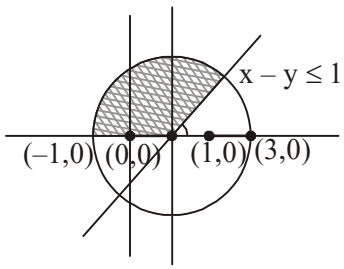
41. The area (in sq. units) of the region $S = \{z \in C : |z-1| \le 2; (z+\bar{z}) + i(z-\bar{z}) \le 2, \text{Im}(z) \ge 0\}$ is : **[JEE(Main)April,2024]**

क्षेत्र $S = \{z \in C : |z-1| \le 2; (z+\bar{z}) + i(z-\bar{z}) \le 2, \text{Im}(z) \ge 0\}$ का क्षेत्रफल (वर्ग इकाईयों में) है :

(A) $\frac{7\pi}{4}$ (B) $\frac{7\pi}{3}$ (C) $\frac{17\pi}{8}$ (D) $\frac{3\pi}{2}$

Ans. D

Sol.



$|z-1| \le 2 \Rightarrow (x-1)^2 + y^2 = 4$

$(z+\bar{z})+i(z-\bar{z}) \le 2 \Rightarrow x - y \le 1$

$\text{Im}\,|z| \ge 0 \Rightarrow y \ge 0$

$$\text{Area of region} = \frac{1}{2}(2)^2\left(\frac{3\pi}{4}\right) = \frac{3\pi}{2}$$

42. Consider the following two statements

Statement I : For any two non-zero complex numbers z_1, z_2.

$$(|z_1|+|z_2|)\left|\frac{z_1}{|z_1|}+\frac{z_2}{|z_2|}\right| \le 2(|z_1|+|z_2|) \text{ and}$$

Statement II : If x, y, z are three distinct complex numbers and a, b, c are three positive real numbers such that $\frac{a}{|y-z|} = \frac{b}{|z-x|} = \frac{c}{|x-y|}$, then $\frac{a^2}{y-z}+\frac{b^2}{z-x}+\frac{c^2}{x-y} = 1$.

Between the above two statemetns : **[JEE(Main)April,2024]**

(A) Statement I is correct but Statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are incorrect

(D) Both Statement I and Statement II are correct

कथनों

कथन I : किसी भी दो शून्येत्तर सम्मिश्र संख्याओं z_1, z_2 के लिए

$$\left(|z_1|+|z_2|\right)\left|\frac{z_1}{|z_1|}+\frac{z_2}{|z_2|}\right| \le 2\left(|z_1|+|z_2|\right)$$ तथा

कथन II : यदि x, y, z तीन भिन्न सम्मिश्र संख्याएँ हैं और a, b, c तीन धनात्मक वास्तविक संख्याएँ हैं ताकि $\frac{a}{|y-z|}=\frac{b}{|z-x|}=\frac{c}{|x-y|}$

है, तो $\frac{a^2}{y-z}+\frac{b^2}{z-x}+\frac{c^2}{x-y}=1$ है।

में से : **[JEE(Main)April,2024]**

(A) कथन I सही है, परन्तु कथन II गलत है

(B) कथन I गलत है परन्तु कथन II सही हैं

(C) कथन I तथा कथन II दोनों गलत है

(D) कथन I तथा कथन II दोनों सही है

Ans. A

Sol. **(I)** $\left(|Z_1|+|Z_2|\right)\left|\frac{Z_1}{|Z_1|}+\frac{Z_2}{|Z_2|}\right|$

$$\left|\frac{Z_1}{|Z_1|}+\frac{Z_2}{|Z_2|}\right| \le \left|\frac{Z_1}{|Z_1|}\right|+\left|\frac{Z_2}{|Z_2|}\right|=1+1=2$$

(II) $\frac{a}{|y-z|}=\frac{b}{|z-x|}=\frac{c}{|x-y|}$

$$\frac{a^2}{(y-z)}\times\left(\frac{\overline{y-z}}{y-z}\right)+\frac{b^2}{z-x}\times\frac{\overline{z-x}}{z-x}+\frac{c^2}{x-y}\times\frac{\overline{x-y}}{x-y}$$

$$=\frac{a^2}{|y-z|^2}\left(\bar{y}-\bar{z}+\bar{z}-\bar{x}+\bar{x}-\bar{y}\right)=0$$

So not true

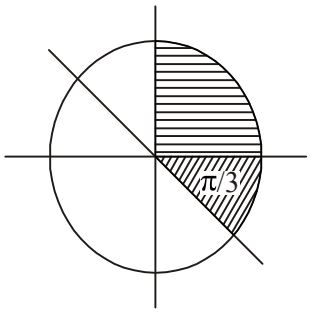
43. Let $S_1=\{z\in C:|z|\le 5\}, S_2\left\{z\in C:\text{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right)\ge 0\right\}$ and $S_3=\{z\in C:\text{Re}(z)\ge 0\}$. Then the area of the region $S_1\cap S_2\cap S_3$ is : **[JEE(Main)April,2024]**

माना $S_1 = \{z \in C : |z| \le 5\}, S_2 \left\{z \in C : \text{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right) \ge 0\right\}$ तथा $S_3 = \{z \in C : \text{Re}(z) \ge 0\}$ हैं। तो क्षेत्र $S_1 \cap S_2 \cap S_3$ का क्षेत्रफल है : **[JEE(Main)April,2024]**

(A) $\frac{125\pi}{24}$ (B) $\frac{125\pi}{4}$ (C) $\frac{125\pi}{12}$ (D) $\frac{125\pi}{6}$

Ans. C

Sol.



$S_1 : x^2 + y^2 \le 5^2 ; S_2 : \sqrt{3}\, x + y \ge 0 ; S_3 : x \ge 0$

$$\frac{z+1-i\sqrt{3}}{1-\sqrt{3}i} = \frac{\left((x+1)+i\left(y-\sqrt{3}\right)\left(1+i\sqrt{3}\right)\right)}{1+3}$$

$$I_m(z) = (x+1)\sqrt{3} + \left(y-\sqrt{3}\right) = \sqrt{3}x + y$$

$$S_1 \cap S_2 \cap S_3 = \frac{1}{4}\pi 5^2 + \frac{1}{2}5^2\frac{\pi}{3} = \frac{125\pi}{12}$$

44. If z_1, z_2 are two distinct complex number such that $\left|\frac{z_1 - 2z_2}{\frac{1}{2} - z_1\bar{z}_2}\right| = 2$, then : **[JEE(Main)April,2024]**

(A) both z_1 and z_2 lie on the same circle

(B) z_1 lies on a circle of radius $\frac{1}{2}$ and z_2 lies on a circle of radius 1

(C) either z_1 lies on a circle of radius 1 or z_2 lies on a circle of radius $\frac{1}{2}$

(D) either z_1 lies on a circle of radius $\frac{1}{2}$ or z_2 lies on a circle of radius 1

यदि z_1, z_2 दो भिन्न सम्मिश्र संख्याऐं इस प्रकार है कि $\left|\frac{z_1-2z_2}{\frac{1}{2}-z_1\bar{z}_2}\right|=2$ है, तो : **[JEE(Main)April,2024]**

(A) z_1 तथा z_2 दोनों समान वृत्त पर स्थित हैं

(B) z_1, $\frac{1}{2}$ त्रिज्या के वृत्त पर तथा z_2, 1 त्रिज्या के वृत्त पर स्थित है

(C) या तो z_1, 1 त्रिज्या के वृत्त पर या z_2, $\frac{1}{2}$ त्रिज्या के वृत्त पर स्थित है

(D) या तो z_1, $\frac{1}{2}$ त्रिज्या के वृत्त पर या z_2, 1 त्रिज्या के वृत्त पर स्थित है

Ans. C

Sol. $\left|\frac{z_1-2z_2}{\frac{1}{2}-z_1\bar{z}_2}\right|=2$

$\Rightarrow |z_1-2z_2|=|1-2z_1\bar{z}_2|$

$\Rightarrow |z_1-2z_2|^2|1-2z_1\bar{z}_2|^2$

$\Rightarrow |z_2|^2-2z_1\bar{z}_2-2\bar{z}_1z_2+4|z_2|^2=1-2\bar{z}_1z_2-2z_1\bar{z}_2+4|z_1|^2|z_2|^2$

$\Rightarrow |z_1|^2-1+4|z_2|^2-4|z_1|^2|z_2|^2=0$

$\Rightarrow \left(|z_1|^2-1\right)\left(1-4|z_2|^2\right)=0$

either $|z_1|^2-1=0$ or $1-4|z_2|^2=0$

$|z_1|=1$ or $|z_2|=\frac{1}{2}$

$\therefore$ either z_1 lies on circle of radius 1

or z_2 lies on circle of radius $\frac{1}{2}$

45. Let z be a complex number such that $|z+2|=1$ and $\text{Im}\left(\frac{z+1}{z+2}\right)=\frac{1}{5}$. Then the value of $\left|\text{Re}\left(\overline{z+2}\right)\right|$ is :

[JEE(Main)April,2024]

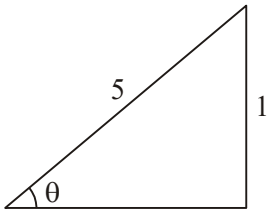
माना z एक समिश्र संख्या इस प्रकार है कि $|z+2|=1$ और $\text{Im}\left(\frac{z+1}{z+2}\right)=\frac{1}{5}$ तब $\left|\text{Re}\left(\overline{z+2}\right)\right|$ का मान होगा :

(A) $\frac{2\sqrt{6}}{5}$ (B) $\frac{24}{5}$ (C) $\frac{1+\sqrt{6}}{5}$ (D) $\frac{\sqrt{6}}{5}$

Ans. A

Sol. Let $z+2=e^{i\theta} \Rightarrow \left(\frac{z+1}{z+2}\right)=\frac{-2+e^{i\theta}+1}{e^{i\theta}}=1-\frac{1}{e^{i\theta}}$

$\Rightarrow \left(\frac{z+1}{z+2}\right)=1-\cos\theta+i\sin\theta$



$\sin\theta=\frac{1}{5}$

$\text{Re}(z+2)=\cos\theta \Rightarrow \left|\text{Re}\left(\overline{z+2}\right)\right|=\frac{2\sqrt{6}}{5}$

46. If the set $R=\{(a,b): a+5b=42, a,b\in N\}$ has m elements and $\sum_{n=1}^{m}\left(1-i^{n!}\right)=x+iy$, where $i=\sqrt{-1}$, then the value of $m+x+y$ is : **[JEE(Main)April,2024]**

यदि समुच्चय $R = \{(a, b) : a + 5b = 42, a, b \in N\}$ में m अवयव है और $\sum_{n=1}^{m}\left(1-i^{n!}\right)=x+iy$, जहाँ $i=\sqrt{-1}$, तब $m + x + y$ का मान होगा : **[JEE(Main)April,2024]**

(A) 8 (B) 12 (C) 5 (D) 4

Ans. B

Sol. $a + 5b = 42, a, b \in N$

$a = 42 - 5b, b = 1, a = 37$

$b = 2, a = 32$

$b = 3, a = 27$

$\vdots$

$b = 8, a = 2$

R has "8" elements $\Rightarrow m = 8$

$\sum_{n=1}^{8}\left(1-i^{n!}\right)=x+iy$

for $n \geq 4$, $i^{n!} = 1$

$\Rightarrow (1-i)+(1-i^{21})+(1-i^{31})$

$=1-i+2+1+1$

$=5-i=x+iy$

$m+x+y=8+5-1=12$

47. The sum of all possible values of $\theta \in [-\pi, 2\pi]$, for which $\frac{1+i\cos\theta}{1-2i\cos\theta}$ is purely imaginary, is equal to :

[JEE(Main)April,2024]

$\theta \in [-\pi, 2\pi]$ के सभी संभव मानों, जिसके लिए $\frac{1+i\cos\theta}{1-2i\cos\theta}$ शुद्धतः काल्पनिक है, का योग बराबर है:

[JEE(Main)April,2024]

(A) 2π (B) 5π (C) 3π (D) 4π

Ans. C

Sol. $z=\frac{1+i\cos\theta}{1-2i\cos\theta}$

$z=\bar{z} \Rightarrow \frac{1+i\cos\theta}{1-2i\cos\theta}=-\left(\overline{\frac{1+i\cos\theta}{1-2i\cos\theta}}\right)$

$(1+i\cos\theta)\left(\overline{1-2i\cos\theta}\right)=-(1-2i\cos\theta)\left(\overline{1+i\cos\theta}\right)$

$\Rightarrow 1+3i\cos\theta-2\cos^2\theta=-(1-3i\cos\theta-2\cos^2\theta)$

$\Rightarrow \cos^2\theta=\frac{1}{2} \Rightarrow \theta=\pm\frac{\pi}{4},\pm\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4}$

Sum $\Rightarrow 3\pi$

48. Let z be a complex number such that the real part of $\frac{z-2i}{z+2i}$ is zero. Then, the maximum value of $|z-(6+8i)|$ is equal to :

[JEE(Main)April,2024]

माना z एक सम्मिश्र संख्या है, जिसके लिए $\frac{z-2i}{z+2i}$ का वास्तविक भाग शून्य है। तो $|z-(6+8i)|$ का अधिकतम मान बराबर है :

[JEE(Main)April,2024]

(A) ∞ (B) 8 (C) 10 (D) 12

Ans. D

Sol. $\frac{z-2i}{z+2i}+\frac{\bar{z}+2i}{\bar{z}-2i}=0$

$z\bar{z} - 2i\bar{z} - 2iz + 4(-1) + z\bar{z} + 2zi + 2\bar{z}i + 4(-1) = 0$

$\Rightarrow 2|z|^2 = 8$

$\Rightarrow |z| = 2$

$|z - (6 + 8i)|_{\text{maximum}} = 10 + 2 = 12$

49. Let x_1, x_2, x_3, x_4 be the solution of the equation $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ and $\left(4 + x_1^2\right)\left(4 + x_2^2\right)\left(4 + x_3^2\right)\left(4 + x_4^2\right) = \frac{125}{16}m$. Then the value of m is _____.

[JEE(Main)April,2024]

माना x_1, x_2, x_3, x_4 समीकरण $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ के हल हैं तथा $\left(4 + x_1^2\right)\left(4 + x_2^2\right)\left(4 + x_3^2\right)\left(4 + x_4^2\right) = \frac{125}{16}m$ है। तो m का मान है _____ **[JEE(Main)April,2024]**

Ans. 221

Sol. $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$

x_1, x_2, x_3, x_4 are the solution of equation

$4x^4 + 8x^3 - 17x^2 - 12x + 9 = 4(x - x_1)(x - x_2)(x - x_3)(x - x_4)$

put $x = 2i$

$4(2i)^4 + 8(2i)^3 - 17(2i)^2 - 12(2i) + 9$

$= 4[(2i - x_1)(2i - x_2)(2i - x_3)(2i - x_4)]$

$\Rightarrow$ $64 - 64i + 68 - 24i + 9$

$\Rightarrow$ $|141 - 88i| = 4|(2i - x_1)(2i - x_2)(2i - x_3)(2i - x_4)|$

$\Rightarrow$ $(141)^2 + (88)^2 = 16[(4 + x_1^2)(4 + x_2^2)(4 + x_3^2)(4 + x_4^2)]$

$\Rightarrow$ $\frac{19881 + 7744}{16} = \left(4 + x_1^2\right)\left(4 + x_2^2\right)\left(4 + x_4^2\right)$

$\Rightarrow$ $\frac{27625}{16} = \frac{125}{16}m$

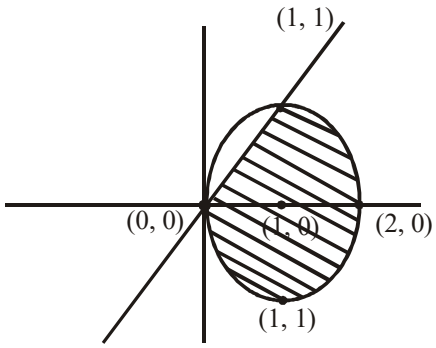
$m = 221$

50. The sum of the square of the modulus of the elements in the set $\{z = a + ib : a, b \in Z, z \in C, |z - 1| \le 1, |z - 5| \le |z - 5i|\}$ is _____. **[JEE(Main)April,2024]**

समुच्चय $\{z = a + ib : a, b \in Z, z \in C, |z - 1| \le 1, |z - 5| \le |z - 5i|\}$ के अवयवों के मापांक के वर्गों का योग है _____.

Ans. 9

Sol.



$|Z-1| \le 1$

$|Z-5| \le |Z-5i|$

(a, b) = {(0, 0), (1, 1), (1, 0), (1, –1), (2, 0)}

$|Z|^2$ = {0, 2, 1, 2, 4|

Sum = 9

Exercise J - A

1. Let z be a complex number such that the imaginary part of z is nonzero and $a = z^2 + z + 1$ is real. then a cannot take the value : **[JEE-2012]**

माना z एक सम्मिश्र संख्या है, इस प्रकार कि इसका काल्पनिक भाग अशून्य है तथा $a = z^2 + z + 1$ एक वास्तविक संख्या है तो a का मान निम्न में से क्या नहीं हो सकता है : **[JEE-2012]**

(A) –1 (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

Ans. D

Sol. Here $z^2 + z + 1 - a = 0$

$\Rightarrow$ $z = \frac{-1 \pm \sqrt{4a-3}}{2}$

Here $a \neq \frac{3}{4}$ otherwise z will be purely real.

2. Let $w = \frac{\sqrt{3}+i}{2}$ and $P = \{w^n : n = 1, 2, 3, ...\}$. Further $H_1 = \left\{z \in C : \text{Re } z > \frac{1}{2}\right\}$ and $H_2 = \left\{z \in C : \text{Re } z < \frac{-1}{2}\right\}$, where C is the set of all complex numbers. If $z_1 \in P \cap H_1$, $z_2 \in P \cap H_2$ and O represents the origin, then $\angle z_1 O z_2 =$ **[JEE-Adv.2013]**

माना कि $w = \frac{\sqrt{3}+i}{2}$ तथा $P = \{w^n : n = 1, 2, 3, ...\}$. इसके अतिरिक्त $H_1 = \left\{z \in C : \text{Re } z > \frac{1}{2}\right\}$ तथा $H_2 = \left\{z \in C : \text{Re } z < \frac{-1}{2}\right\}$, जहाँ C सम्मिश्र संख्याओं का समुच्चय है। यदि $z_1 \in P \cap H_1$, $z_2 \in P \cap H_2$ तथा O मूलबिन्दु प्रदर्शित करता है, तब $\angle z_1 O z_2 =$ **[JEE-Adv.2013]**

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{6}$ (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{6}$

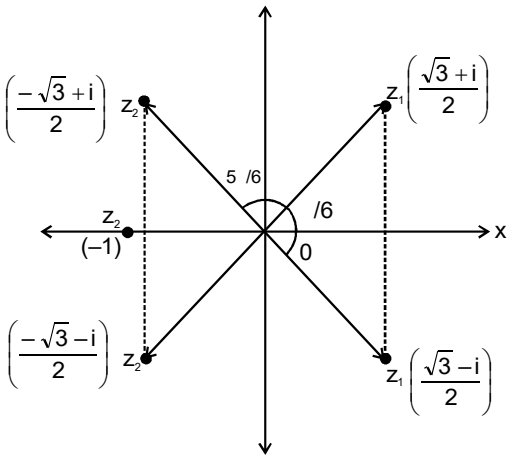
Ans. CD

Sol. $P = \omega^n = \cos\frac{n\pi}{6} + i\sin\frac{n\pi}{6}$, $H_1 = \text{Re}z > 1/2$

$z_1 = P \cap H_1 = \frac{\sqrt{3}+i}{2}, \frac{\sqrt{3}-i}{2}$

$z_2 = P \cap H_2 = -1, \frac{-\sqrt{3}+i}{2}, \frac{-\sqrt{3}-i}{2}$

$\angle z_1 O z_2 = \frac{2\pi}{3}, \frac{5\pi}{6}, \pi$



3. Let ω be a complex cube root of unity with $\omega \neq 1$ and $P = [p_{ij}]$ be a $n \times n$ matrix with $p_{ij} = \omega^{i+j}$. Then $P^2 \neq 0$, when n = **[JEE-Adv.2013]**

इकाई का एक सम्मिश्र घनमूल ω लीजिये, जहाँ $\omega \neq 1$ तथा $P = [p_{ij}]$ एक $n \times n$ आव्यूह लीजिये, जहाँ $p_{ij} = \omega^{i+j}$ । तब $P^2 \neq 0$, जब n = **[JEE-Adv.2013]**

(A) 57 (B) 55 (C) 58 (D) 56

Ans. BCD

Sol. $P^2 = [\alpha_{ij}]_{n \times n}$

$$\alpha_{ij} = \sum_{k=1}^{n} p_{ik}.p_{kj}$$

$$= \sum_{k=1}^{n} \omega^{i+k}.\omega^{k+j} = \omega^{i+j}\sum_{k=1}^{n} \omega^{2k}$$

$$= \omega^{i+j}(\omega^2 + \omega^4 + \omega^6 + ... + \omega^{2n})$$

If n is a multiple of 3 then $P^2 = \mathbf{0}$

$\Rightarrow$ n is not a multiple of 3

$\Rightarrow$ n can be 55, 58, 56

Paragraph for Question Nos. 4 to 5

Let $S = S_1 \cap S_2 \cap S_3$, where $S_1 = \{z \in C : |z| < 4\}, S_2 \left\{z \in C : \text{Im}\left[\frac{z-1+\sqrt{3}i}{1-\sqrt{3}i}\right] > 0\right\}$ and

$S_3 = \{z \in C : \text{Re}\, z > 0\}$

माना कि $S = S_1 \cap S_2 \cap S_3$, जहाँ $S_1 = \{z \in C : |z| < 4\}, S_2 \left\{z \in C : \text{Im}\left[\frac{z-1+\sqrt{3}i}{1-\sqrt{3}i}\right] > 0\right\}$ तथा

$S_3 = \{z \in C : \text{Re}\, z > 0\}$

4. Area of S = **[JEE-Adv.2013]**

S का क्षेत्रफल = **[JEE-Adv.2013]**

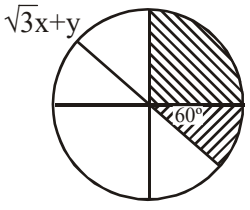
(A) $10\pi/3$ (B) $20\pi/3$ (C) $16\pi/3$ (D) $32\pi/3$

Ans. B

Sol. S_1 is interior of circle centred at (0,1) & radius = 4.

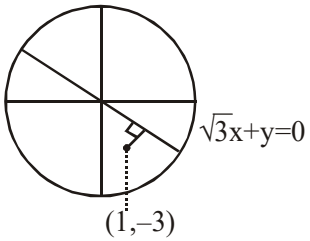
Re(z) > 0 is in I[st] & IV[th]quadrant.

$$\frac{\left(z-\left(1-i\sqrt{3}\right)\right)}{\left(1-i\sqrt{3}\right)} = \frac{\left((x-1)+i\left(y-\sqrt{3}\right)\right)}{\left(1-i\sqrt{3}\right)}$$



$$=\frac{\left((x-1)+i\left(y-\sqrt{3}\right)\right)\left(1+i\sqrt{3}\right)}{2}$$

$I_m(S_2)=\sqrt{3}x+y>0$



erpendicular distance from (1,–3) to the line is $P=\left|\frac{\sqrt{3}-3}{2}\right|=\left(\frac{3-\sqrt{3}}{2}\right)$

Area of $S=\frac{\pi(4)^2}{4}+\frac{1}{2}(4)^2\frac{\pi}{3}$

$\frac{8\pi}{3}+4\pi=\frac{20\pi}{3}$

5. $\min_{z\in S}|1-3i-z|=$ **[JEE-Adv.2013]**

(A) $\frac{2-\sqrt{3}}{2}$ (B) $\frac{2+\sqrt{3}}{2}$ (C) $\frac{3-\sqrt{3}}{2}$ (D) $\frac{3+\sqrt{3}}{2}$

Ans. C

Sol.

6. Let complex numbers α and $\frac{1}{\bar{\alpha}}$ lie on circles $(x-x_0)^2+(y-y_0)^2=r^2$ and $(x-x_0)^2+(y-y_0)^2=4r^2$, respectively. If $z_0=x_0+iy_0$ satisfies the equation $2|z_0|^2=r^2+2$, then $|\alpha|=$ **[JEE-Adv.2013]**

माना कि सम्मिश्र संख्याएँ α तथा $\frac{1}{\bar{\alpha}}$ क्रमशः वृत्त $(x-x_0)^2+(y-y_0)^2=r^2$ तथा $(x-x_0)^2+(y-y_0)^2=4r^2$, पर स्थित है। यदि $z_0=x_0+iy_0$ समीकरण $2|z_0|^2=r^2+2$ को सन्तुष्ट करता है, तब $|\alpha|=$ **[JEE-Adv.2013]**

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{7}}$ (D) $\frac{1}{3}$

Ans. C

Sol. $|z-z_0|=r$

$|z-z_0|=2r$

$|\alpha-z_0|=r$

$\left|\frac{1}{\bar{\alpha}}-z_0\right|=2r$ $\qquad \alpha\bar{\alpha}=|\alpha|^2$

$\left|\frac{\alpha}{|\alpha|^2}-z_0\right|=2r$

$(\alpha - z_0)\ (\bar{\alpha} - \bar{z}_0) = r^2 \quad \Rightarrow\ |\alpha|^2 - z_0\bar{\alpha} - \alpha\bar{z}_0 + |z_0|^2 = r^2$

$\left(\frac{\alpha}{|\alpha|^2} - z_0\right)\left(\frac{\bar{\alpha}}{|\alpha|^2} - \bar{z}_0\right) = 4r^2 \quad \Rightarrow\ \frac{|\alpha|^2}{|\alpha|^4} - \frac{z_0\bar{\alpha}}{|\alpha|^2} - \frac{\bar{z}_0\alpha}{|\alpha|^2} + |z_0|^2 = 4r^2$

$1 - z_0\bar{\alpha} - \bar{z}_0\alpha + |z_0|^2\,|\alpha|^2 = 4r^2|\alpha|^2 \quad \Rightarrow\ (|\alpha^2| - 1) + |z_0|^2\,(1 - |\alpha|^2) = r^2\,(1 - 4\alpha^2)$

$(|\alpha|^2 - 1)\left(1 - \frac{r^2+2}{2}\right) = r^2(1 - 4|\alpha|^2)$

$(|\alpha|^2 - 1)\left(\frac{-r^2}{2}\right) = r^2(1 - 4|\alpha|^2)$

$|\alpha|^2 - 1 = -2 + 8\,|\alpha|^2$

$1 = 7|\alpha|^2 \quad \Rightarrow\ |\alpha| = \frac{1}{\sqrt{7}}$

7. Let $z_k = \cos\left(\frac{2k\pi}{10}\right) + i\sin\left(\frac{2k\pi}{10}\right); k = 1, 2, ..., 9$. **[JEE-Adv.2014]**

Column-I	**Column-II**
P. For each z_k there exists a z_j such that $z_k \cdot z_j = 1$	1. True
Q. There exists a $k \in \{1, 2, ...9\}$ such that $z_1 \cdot z = z_k$ has no solution z in the set of complex numbers.	2. False
R. $\frac{\lvert 1 - z_1\rvert\,\lvert 1 - z_2\rvert ... \lvert 1 - z_9\rvert}{10}$ equals	3. 1
S. $1 - \sum_{k=1}^{9} \cos\left(\frac{2k\pi}{10}\right)$ equals	4. 2

माना $z_k = \cos\left(\frac{2k\pi}{10}\right) + i\sin\left(\frac{2k\pi}{10}\right); k = 1, 2, ..., 9$. **[JEE-Adv.2014]**

Column-I	**Column-II**
P. प्रत्येक z_k के लिये एक ऐसा z_j है जिसके लिये $z_k \cdot z_j = 1$	1. सत्य
Q. $\{1, 2, ... 9\}$ में एक ऐसा k है कि $z_1 \cdot z = z_k$ का कोई हल z सम्मिश्र संख्याओं (complex numbers) में नहीं है	2. असत्य
R. $\frac{\lvert 1 - z_1\rvert\,\lvert 1 - z_2\rvert ... \lvert 1 - z_9\rvert}{10}$ का मान है	3. 1
S. $1 - \sum_{k=1}^{9} \cos\left(\frac{2k\pi}{10}\right)$ का मान है	4. 2

Code :

	P	Q	R	S		P	Q	R	S
(A)	2	1	3	4	(B)	2	1	4	3
(C)	1	2	3	4	(D)	2	1	4	3

Ans. C

Sol. (P) $z_k z_j = 1 \quad \Rightarrow \quad z_j = z_{10-k}$

Hence for each $k \in \{1, 2, 3, \ldots, 9\}$ there exists z_j such that $z_k \cdot z_j = 1$ True

(Q) $z_1 \cdot z = z_k \quad \Rightarrow \quad z = z_{k-1}$ for $k = 2, 3, 4, \ldots, 9$ &

$z = 1$ for $k = 1$ False

(R) $z_1, z_2, \ldots, z_9$ are roots of the equation $z^{10} = 1$ other then unity, hence

$$\frac{z^{10}-1}{z-1} = 1 + z + \ldots + z^9 = (z - z_1)(z - z_2) \ldots (z - z_9)$$

Substituting $z = 1$, we get $\dfrac{(1-z_1)(1-z_2)\ldots(1-z_9)}{10} = \dfrac{10}{10} = 1$

(S) $1 - \sum_{k=1}^{9} \cos\left(\dfrac{2k\pi}{10}\right) = 1 - \{$sum of real parts of roots of $z^{10} = 1$ except $1\}$

$= 1 - (-1) = 2$

(as $1 + z_1 + z_2 + \ldots + z_9 = 0$) $\quad \Rightarrow \quad \sum \mathrm{Re}(z_k) + 1 = 0$

Hindi. (P) $z_k z_j = 1 \quad \Rightarrow \quad z_j = z_{10-k}$

अतः प्रत्येक k के लिए $\{1, 2, 3, \ldots, 9\}$ इस प्रकार विद्यमान z_j इस प्रकार है कि $z_k \cdot z_j = 1$ सत्य

(Q) $z_1 \cdot z = z_k \quad \Rightarrow \quad z = z_{k-1}$ $k = 2, 3, 4, \ldots, 9$ & के लिए तथा

$z = 1$, $k = 1$ के लिए गलत

(R) $z_1, z_2, \ldots, z_9$ समीकरण $z^{10} = 1$ के मूल है अतः

$$\frac{z^{10}-1}{z-1} = 1 + z + \ldots + z^9 = (z - z_1)(z - z_2) \ldots (z - z_9)$$

$z = 1$ के प्रतिस्थापन से $\dfrac{(1-z_1)(1-z_2)\ldots(1-z_9)}{10} = \dfrac{10}{10} = 1$

(S) $1 - \sum_{k=1}^{9} \cos\left(\dfrac{2k\pi}{10}\right) = 1 - \{1$ को छोड़कर $z^{10} = 1$ के मूलों के वास्तविक भागों का योगफल$\}$

$= 1 - (-1) = 2$

(चूंकि $1 + z_1 + z_2 + \ldots + z_9 = 0$) $\quad \Rightarrow \quad \left(\sum \mathrm{Re}(z_k) + 1 = 0\right)$

8. Match the following columns : **[JEE-Adv.2015]**

Column I		**Column II**	
(A) In R^2, if the magnitude of the projection vector of the vector $\alpha\hat{i}+\beta\hat{j}$ on $\sqrt{3}\,\hat{i}+\hat{j}$ is $\sqrt{3}$ and if $\alpha = 2+\sqrt{3}\,\beta$, then possible value(s) of $\lvert\alpha\rvert$ is (are)	(P)	1
(B) Let a and b be real numbers such that the function $f(x)=\begin{cases}-3ax^2-2, & x<1\\ bx+a^2, & x\geq 1\end{cases}$ is differentiable for all $x \in R$. Then possible value(s) of a is (are)	(Q)	2
(C) Let $\omega \neq 1$ be a complex cube root of unity. If $(3-3\omega+2\omega^2)^{4n+3}+(2+3\omega-3\omega^2)^{4n+3}+(-3+2\omega+3\omega^2)^{4n+3}=0$, then possible value(s) of n is (are)	(R)	3
(D) Let the harmonic mean of two positive real numbers a and b be 4. If q is a positive real number such that a, 5, q, b is an arithmetic progression, then the value(s) of $\lvert q-a\rvert$ is (are)	(S)	4
	(T)	5

निम्न कॉलम को सुमेलित कीजिए : **[JEE-Adv.2015]**

Column I		**Column II**	
(A) माना कि R^2 में, यदि सदिश $\alpha\hat{i}+\beta\hat{j}$ का सदिश $\sqrt{3}\,\hat{i}+\hat{j}$ पर प्रक्षेप सदिश का परिमाण $\sqrt{3}$ हो और यदि $\alpha = 2+\sqrt{3}\,\beta$ हो, तब $\lvert\alpha\rvert$ के संभव मान है	(P)	1
(B) माना कि वास्तविक संख्याएं a और b इस प्रकार है कि फलन $f(x)=\begin{cases}-3ax^2-2, & x<1\\ bx+a^2, & x\geq 1\end{cases}$ सभी $x \in R$ के लिए अवकलनीय है। तब α के संभव मान है	(Q)	2
(C) माना कि $\omega \neq 1$ इकाई का एक सम्मिश्र घनमूल है। यदि $(3-3\omega+2\omega^2)^{4n+3}+(2+3\omega-3\omega^2)^{4n+3}+(-3+2\omega+3\omega^2)^{4n+3}=0$, तब n के संभव मान है	(R)	3
(D) माना कि दो धनात्मक वास्तविक संख्याएं a और b का हरात्मक माध्य 4 है। यदि एक धनात्मक वास्तविक संख्या q इस प्रकार है कि a, 5, q, b एक समान्तर श्रेणी है। तब $\lvert q-a\rvert$ का मान है	(S)	4
	(T)	5

Ans. (A) PQ (B) PQ (C) PQST (D) QT

Sol. (A) $\left|\left(\alpha\hat{i}+\beta\hat{j}\right).\left(\frac{\sqrt{3}\hat{i}+\hat{j}}{2}\right)\right|=\sqrt{3} \quad \Rightarrow \quad \sqrt{3}\alpha+\beta=\pm 2\sqrt{3}$

$$\sqrt{3}\alpha+\left(\frac{\alpha-2}{\sqrt{3}}\right)=\pm 2\sqrt{3}$$

$\Rightarrow 3\alpha+\alpha-2=\pm 6 \quad \Rightarrow 4\alpha=8,-4 \quad \Rightarrow \alpha=2,-1$

(A – P, Q)

(B) Continuous $\Rightarrow -3a-2=b+a^2$

Differentiable $\Rightarrow -6a=b \quad \Rightarrow \quad 6a=a^2+3a+2$

$\Rightarrow a^2-3a+2=0 \quad \Rightarrow \quad a=1,\ 2$

(B → P, Q)

(D) $\frac{2ab}{a+b}=4$ and $2q=5+b$ _____(i)

$q=10-a$ and $2q=5+b$

$\Rightarrow 20-2a=5+b \Rightarrow 15=2a+b$ _____(ii)

From (i) and (ii) $a(15-2a)=2a+2(15-2a)$

$\Rightarrow 15a-2a^2=-2a+30 \Rightarrow 2a^2-17a+30=0 \Rightarrow a=6,\frac{5}{2}$

$\Rightarrow q=4,\frac{15}{2} \Rightarrow |q-a|=2,5$

(D → Q, T)

(C) Let $a=3-3\omega+2\omega^2$

$a\omega=3\omega-3\omega^2+2$

$a\omega^2=3\omega^2-3+2\omega$

Now $a^{4n+3}\left(1+\omega^{4x+3}+\left(\omega^2\right)^{4n+3}\right)=0$

$\Rightarrow$ n should not be a multiple of 3

hence P, Q, S, T

9. For any integer k, let $\alpha_k=\cos\left(\frac{k\pi}{7}\right)+i\sin\left(\frac{k\pi}{7}\right)$, where $i=\sqrt{-1}$. The value of the expression $\frac{\sum_{k=1}^{12}|\alpha_{k+1}-\alpha_k|}{\sum_{k=1}^{3}|\alpha_{4k-1}-\alpha_{4k-2}|}$

is. **[JEE-Adv.2015]**

किसी भी पूर्णांक k के लिए, $\alpha_k=\cos\left(\frac{k\pi}{7}\right)+i\sin\left(\frac{k\pi}{7}\right)$, जहाँ $i=\sqrt{-1}$ है। तब व्यंजक $\frac{\sum_{k=1}^{12}|\alpha_{k+1}-\alpha_k|}{\sum_{k=1}^{3}|\alpha_{4k-1}-\alpha_{4k-2}|}$ का मान है।

[JEE-Adv.2015]

Ans. 4

Sol. $\alpha_k=\cos\frac{2k\pi}{14}+i\sin\frac{2k\pi}{14}=e^{i\frac{2k\pi}{14}}$

Now $\frac{\sum_{k=1}^{12}\left|e^{\frac{i2(k+1)\pi}{14}}-e^{\frac{i2k\pi}{14}}\right|}{\sum_{k=1}^{3}\left|e^{\frac{i(4k-1)\pi}{14}}-e^{\frac{i(4k-2)\pi}{14}}\right|}=\frac{\sum_{k=1}^{12}\left|e^{\frac{i2\pi}{14}}-1\right|}{\sum_{k=1}^{3}\left|e^{\frac{i2\pi}{14}}-1\right|}=\frac{12}{3}=4$

10. Let $z=\frac{-1+\sqrt{3}i}{2}$, where $i=\sqrt{-1}$ and $r, s \in \{1,2,3\}$. Let $P=\begin{bmatrix}(-z)^r & z^{2s}\\ z^{2s} & z^r\end{bmatrix}$ and I be the identity matrix of order 2. Then the total number of ordered pairs (r, s) for which $P^2=-I$ is. **[JEE-Adv.2016]**

माना कि $z=\frac{-1+\sqrt{3}i}{2}$ है, जहाँ $i=\sqrt{-1}$ और $r, s \in \{1, 2, 3\}$ हैं। माना कि $P=\begin{bmatrix}(-z)^r & z^{2s}\\ z^{2s} & z^r\end{bmatrix}$ और I दो कोटि का तत्समक आव्यूह है। तब वे सभी क्रमित युग्म (r, s), जिनके लिए $P^2=-I$ है, की कुल संख्या है। **[JEE-Adv.2016]**

Ans. 1

Sol. $z=\frac{-1\sqrt{3}i}{2}=\omega$

$$P^2=\begin{bmatrix}(-\omega)^2 & (\omega)^{2s}\\ (\omega)^{2s} & \omega^r\end{bmatrix}\begin{bmatrix}(-\omega)^r & \omega^{2s}\\ \omega^{2s} & \omega^r\end{bmatrix}$$

$P^2=-I$

$$=\begin{bmatrix}(-\omega)^{2r}+\omega^{4s} & \omega^{2s}(\omega^2+(-\omega)^0\\ \omega^{2s}((-\omega)^r+\omega^r) & \omega^{2r}+\omega^{4s}\end{bmatrix}=\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$$

$\Rightarrow \omega^{2r}+\omega^{4s}=-1$ and $\omega^{2s}(\omega^r+(-\omega)^r)=0$

$\Rightarrow 1+\omega^s+\omega^{2r}=0$ and $r \neq$ odd

$\Rightarrow r=s=1$

11. Let $a, b \in \mathbb{R}$ and $a^2+b^2 \neq 0$. Suppose $S=\left\{z \in \mathbb{C}: z=\frac{1}{a+ibt}, t \in \mathbb{R}, t \neq 0\right\}$, where $i=\sqrt{-1}$. If $z=x+iy$ and $z \in S$, then (x, y) lies on. **[JEE-Adv.2016]**

(A) the circle with radius $\frac{1}{2a}$ and centre $\left(\frac{1}{2a}, 0\right)$ for $a>0, b \neq 0$

(B) the circle with radius $-\frac{1}{2a}$ and centre $\left(-\frac{1}{2a}, 0\right)$ for $a<0, b \neq 0$

(C) the x-axis for $a \neq 0, b=0$

(D) the y-axis for $a=0, b \neq 0$

माना कि $a, b \in \mathbb{R}$ और $a^2+b^2 \neq 0$ है। मान लीजिए कि $S=\left\{z \in \mathbb{C}: z=\frac{1}{a+ibt}, t \in \mathbb{R}, t \neq 0\right\}$, जहाँ $i=\sqrt{-1}$ है। यदि $z=x+iy$ और $z \in S$ है, तब (x, y) **[JEE-Adv.2016]**

(A) उस वृत पर है जिसकी त्रिज्या $\frac{1}{2a}$ और केंद्र बिन्दु $\left(\frac{1}{2a},0\right)$ है जब $a > 0, b \neq 0$

(B) उस वृत है कि जिसकी त्रिज्या $-\frac{1}{2a}$ और केन्द्र बिन्दु $\left(-\frac{1}{2a},0\right)$ for $a < 0, b \neq 0$

(C) x-अक्ष पर है जब $a \neq 0, b = 0$

(D) y-अक्ष पर है जब $a = 0, b \neq 0$

Ans. ACD

Sol. $x + iy = \frac{a - ibt}{a^2 + b^2t^2}$

$\Rightarrow x = \frac{a}{a^2 + b^2t^2}, \; y = \frac{-bt}{a^2 + b^2t^2}$

(C) for $b = 0 \Rightarrow x = \frac{1}{a}, y = 0$

(D) for $a = 0 \Rightarrow x = 0, \; y = \frac{-1}{bt}$

(A) Eliminating t we get

$$x^2 + y^2 - \frac{x}{a} = 0$$

12. Let a, b, x and y be real numbers such that $a - b = 1$ and $y \neq 0$. If the complex number $z = x + iy$ satisfies $\text{Im}\left(\frac{az+b}{z+1}\right) = y$, then which of the following is(are) possible value(s) of x? **[JEE-Adv.2017]**

माना कि a, b, x तथा y इस प्रकार की वास्तविक संख्यायें (real numbers) है, कि $a - b = 1$ तथा $y \neq 0$ हैं। यदि सम्मिश्र संख्या (complex number) $z = x + iy$, $\text{Im}\left(\frac{az+b}{z+1}\right) = y$ को सन्तुष्ट करती है, तब निम्न में से कौनसा/से x का/के संभावित मान है/हैं?

[JEE-Adv.2017]

(A) $1 + \sqrt{1 + y^2}$ (B) $-1 + \sqrt{1 - y^2}$ (C) $1 - \sqrt{1 + y^2}$ (D) $-1 - \sqrt{1 - y^2}$

Ans. BD

Sol. $\text{Im}\left(\frac{a(x+iy)+b}{x+iy+1}\right) = y$

$\text{Im}\left(\frac{(ax+b)+iay)((x+1)-iy)}{(x+1)^2+y^2}\right) = y$

$$\frac{ay(x+1)-y(ax+b)}{(x+1)^2+y^2}=y$$

$a-b=(x+1)^2+y^2$

$1=(x+1)^2+y^2$

$(x+1)^2=1-y^2$

$(x+1)=\pm\sqrt{1-y^2}$

$x=-1\pm\sqrt{1-y^2}$

13. For a non-zero complex number z let arg(z) denote the principal argument with $-\pi<\arg(z)\le\pi$. Then, which of the following statement(s) is (are) FALSE? **[JEE-Adv.2018]**

(A) $\arg(-1-i)=\frac{\pi}{4}$, where $i=\sqrt{-1}$

(B) The function $f:R\to(-\pi,\pi)]$ defined by $f(t)=\arg(-1+it)$ for all $t\in\mathbb{R}$, is continuous at all points of $\mathbb{R}$, where $i=\sqrt{-1}$

(C) For any two non-zero complex numbers z_1 & z_2, $\arg\left(\frac{z_1}{z_2}\right)-\arg(z_1)+\arg(z_2)$ is an integer multiple of 2π

(D) For any three given distinct complex numbers z_1, z_2 and z_3, the locus of the point z satisfying the condition $\arg\left(\frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}\right)=\pi$, lies on a straight line

किसी शून्येत्तर (non-zero) सम्मिश्र संख्या z के लिये, माना कि arg(z) इसके मुख्य कोणांक को दर्शाता है, जहाँ $-\pi<\arg(z)\le\pi$ । तब निम्नलिखित में से कौनसा (से) कथन असत्य है (हैं)? **[JEE-Adv.2018]**

(A) $\arg(-1-i)=\frac{\pi}{4}$, जहाँ $i=\sqrt{-1}$

(B) फलन $f:R\to(-\pi,\pi)]$, जो सभी $t\in\mathbb{R}$ के लिये $f(t)=\arg(-1+it)$ के द्वारा परिभाषित है, $\mathbb{R}$ के सभी बिन्दुओं पर संतत है, जहाँ $i=\sqrt{-1}$

(C) किन्ही भी दो शून्योत्तर सम्मिश्र संख्याओं z_1 और z_2 के लिए $\arg\left(\frac{z_1}{z_2}\right)-\arg(z_1)+\arg(z_2)$ 2π का एक पूर्णांक गुणज है

(D) किन्ही भी तीन दी गयी भिन्न सम्मिश्र संख्याओं z_1, z_2 तथा z_3 के लिये, प्रतिबंध $\arg\left(\frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}\right)=\pi$, को सन्तुष्ट करने वाले बिन्दु z का बिन्दुपथ एक सरल रेखा पर स्थित है

Ans. ABD

Sol. (A) $\text{Arg}(-1-i)=-3\frac{\pi}{3}$

(B) $f(t) = \text{Arg}(-1 + it)$

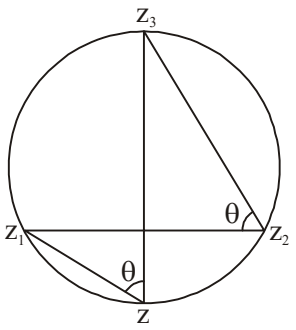
$$\begin{cases} \pi - \tan^{-1} t & t \geq 0 \\ -\left(\pi + \tan^{-1} t\right) & t < 0 \end{cases}$$

If is discontinuous at $t = 0$

(C) $\text{Arg}\left(\dfrac{z_1}{z_2}\right) - \text{Arg}\, z_1 + \text{Arg}\, z_2$

$$\text{Arg}\left(\frac{z_1}{z}\right) = \text{Arg}\, z_1 + \text{Arg}\, z_2 + 2n\pi$$

so the expression becomes $2n\pi$



(D) $\text{Arg}\left(\dfrac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}\right) = \pi$

If is circle

14. Let s, t, r be non-zero complex numbers and L be the set of solutions $z = x + iy$ $\left(x, y \in R, i = \sqrt{-1}\right)$ of the equation $sz + t\overline{z} + r = 0$, where $\overline{z} = x - iy$. Then, which of the following statement(s) is (are) TRUE? **[JEE-Adv.2018]**

(A) If L has exactly one element, then $|s| \neq |t|$

(B) If $|s| = |t|$, then L has infinitely many elements

(C) The number of elements in $L \cap \{z : |z - 1 + i| = 5\}$ is at most 2

(D) If L has more than one element, then L has infinitely many elements

माना कि s, t, r शून्येत्तर सम्मिश्र संख्यायें हैं और L समीकरण $sz + t\overline{z} + r = 0$ के हलों $z = x + iy$ $\left(x, y \in R, i = \sqrt{-1}\right)$ का समुच्चय है, जहाँ $\overline{z} = x - iy$ । तब निम्नलिखित में से कौन सा (से) कथन सत्य है (हैं)? **[JEE-Adv.2018]**

(A) यदि L में ठीक एक अवयव है, तब $|s| \neq |t|$

(B) यदि $|s| = |t|$, तब L में अनन्त अवयव हैं

(C) $L \cap \{z : |z-1+i|=5\}$ में अवयवों की अधिकतम संख्या 2 है

(D) यदि L में एक से ज्यादा अवयव हैं, तब L में अनन्त अवयव हैं

Ans. ACD

Sol. $sz + t\overline{z} + r = 0,\ \overline{z} = x - iy$

$\overline{s}\,\overline{z} + \overline{t}z + r = 0$

(1) + (2)

$\left(t + \overline{s}\right)\overline{z} + \left(s + \overline{t}\right)z + \left(r + \overline{r}\right) = 0$

$\left(t - \overline{s}\right)\overline{z} + \left(s - \overline{t}\right)z + \left(r - \overline{r}\right) = 0$

For unique solution

$$\frac{t+\bar{s}}{t-\bar{s}} \neq \frac{s+\bar{t}}{s-\bar{t}}$$

On solving the above equation we get

$|t| \neq |s|$

$\therefore$ option (A) is correct

Lines overlap if

$$\frac{t+\bar{s}}{t-\bar{s}} \neq \frac{\bar{t}+s}{s-\bar{t}} = \frac{r+\bar{r}}{r-\bar{r}}$$

$|t| = |s|$ $\qquad\qquad$ $\bar{t}r - \bar{t}\,\bar{r} + sr - s\bar{r} = sr + s\bar{r} - \bar{t}r - \bar{t}\,\bar{r}$

$2\bar{t}r - 2s\bar{r}$

$\bar{t}r = s\bar{r}$

$\therefore |\bar{t}||r| = |s||r|$

$\therefore |t| = |s|$

$\therefore$ If $|t| = |s|$, lines will be parallel for sure but it may not be coincident

For option (C) if element of set L represent line, then this line and given circle can have maximum two common points so option (C) is correct

15. Let S be the set of all complex numbers z satisfying $|z-2+i| \geq \sqrt{5}$ if the complex number z_0 is such that $\frac{1}{|z_0-1|}$ is the maximum of the set $\left\{\frac{1}{|z-1|} : z \in S\right\}$ then the principal argument of $\frac{4-z_0-\bar{z}_0}{z_0-\bar{z}_0+2i}$ is : **[JEE-Adv.2019]**

माना कि S उन सभी सम्मिश्र संख्याओं z का समूच्चय है जो $|z-2+i| \geq \sqrt{5}$ को संतुष्ट करती है। यदि एक सम्मिश्र संख्या z_0 ऐसी है जिससे $\frac{1}{|z_0-1|}$ समुच्चय $\left\{\frac{1}{|z-1|} : z \in S\right\}$ का उच्चतम है, तब $\frac{4-z_0-\bar{z}_0}{z_0-\bar{z}_0+2i}$ का मुख्य कोणांक है। **[JEE-Adv.2019]**

(A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{2}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$

Ans. B

Sol.

16. Let $\omega \neq 1$ be a cube root of unity. Then the minimum of the set $\{|a+b\omega+c\omega^2|^2 : a,b,c \text{ distinct no-zero integers}\}$ equals. **[JEE-Adv.2019]**

माना कि $\omega \neq 1$ का एक घनमूल है। तब समुच्चय $\{|\alpha + b\omega+c\omega^2|^2 : a,b,c$ भिन्न अशून्य पूर्णांक है।$\}$ का निम्नतम बराबर ____ **[JEE-Adv.2019]**

Ans. 3

Sol. $|a+b\omega+c\omega^2|^2$

$= (a+b\omega+c\omega^2)\ (a+b\omega^2+c\omega)$

$= a^2 + b^2 + c^2 - ab - bc - ca$

$= \frac{1}{2}\ ((a-b)^2 + (b-c)^2 + (c-a)^2)$

for minimum put $a = 1, b = 2\ c = 3$

minimum $= 3$

17. Let S be the set of all complex numbers z satisfying $\left|z^2 + z + 1\right| = 1$. Then which of the following statements is/are TRUE? **[JEE-(Adv)2020]**

(A) $\left|z + \frac{1}{2}\right| \le \frac{1}{2}$ for all $z \in S$

(B) $|z| \le 2$ for all $z \in S$

(C) $\left|z + \frac{1}{2}\right| \ge \frac{1}{2}$ for all $z \in S$

(D) Then set S has exactly four elements

माना S उन सभी समिश्र संख्या z का समुचय है जो $\left|z^2 + z + 1\right| = 1$ को संतुष्ट करती है तो निम्न में से कौनसा/से कथन सत्य है/हैं? **[JEE-(Adv)2020]**

(A) सभी $z \in S$ के लिए $\left|z + \frac{1}{2}\right| \le \frac{1}{2}$

(B) सभी $z \in S$ के लिए $|z| \le 2$

(C) सभी $z \in S$ के लिए $\left|z + \frac{1}{2}\right| \ge \frac{1}{2}$

(D) समुच्चय S में ठीक चार अवयव है

Ans. BC

Sol. $\left|z^2 + z + 1\right| = 1$

$z^2 + z + 1 = e^{i\theta}$

$$z = \frac{-1 \pm \sqrt{4e^{i\theta} - 3}}{2}$$

$$z + \frac{1}{2} = \pm\frac{1}{2}\sqrt{(4\cos\theta - 3) + (4\sin\theta)i}$$

$$\left|z + \frac{1}{2}\right| = \frac{1}{2}\left((4\cos\theta - 3)^2 + (4\sin\theta)^2\right)^{\frac{1}{4}}$$

$$\left|z + \frac{1}{2}\right| = \frac{1}{2}\left(25 - 24\cos\theta\right)^{\frac{1}{4}}$$

$$\left|z+\frac{1}{2}\right| \in \left[\frac{1}{2}, \frac{\sqrt{7}}{2}\right]$$

$$\left|z+\frac{1}{2}\right| \geq \frac{1}{2}$$

$$2z = -1 \pm \sqrt{(4\cos\theta - 3) + (4\sin\theta)i}$$

$|2z| \leq 1+\sqrt{7}$ (triangle inequality)

$$|z| \leq \frac{1+\sqrt{7}}{2}$$

$|z| \leq 2$

18. For a complex number z, let Re(z) denote the real part of z. Let S be the set of all complex numbers z satisfying $z^4 - |z|^4 = 4\,i\,z^2$, where $i = \sqrt{-1}$. Then the minimum possible value of $|z_1 - z_2|^2$, where $z_1, z_2 \in S$ with $\text{Re}(z_1) > 0$ and $\text{Re}(z_2) < 0$, is _____ **[JEE-(Adv)2020]**

एक सम्मिश्र संख्या z के लिए, माना Re(z), z के वास्तविक भाग को दर्शाता है। माना S सभी सम्मिश्र संख्यओं z का समुच्च है जो $z^4 - |z|^4 = 4\,i\,z^2$ को संतुष्ट करती है, जहाँ $i = \sqrt{-1}$, तब $|z_1 - z_2|^2$ का न्यूनतम संभावित मान, जहाँ $z_1, z_2 \in S$ तथा $\text{Re}(z_1) > 0$ एवं $\text{Re}(z_2) < 0$, है _____ **[JEE-(Adv)2020]**

Ans. 8

Sol. $z^4 - |z|^4 = 4i\,z^2$

$z^4 - z^2\bar{z}^2 = 4i\,z^2 \qquad z^2 \neq 0$

$z^2 - \bar{z}^2 = 4i$

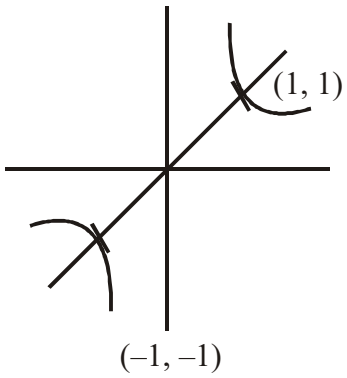
$\left(z+\bar{z}\right)\left(z-\bar{z}\right) = 4i \qquad (1)$

$z = x + iy$

then by (1)

$4ixy = 4i$

$xy = 1$



It is a hyberbola, z_1 and z_2 are two points on this hyberbola

$|z_1 - z_2|_{min}$ = T.A. of hyberbola = $2\sqrt{2}$

$|z_1 - z_2|^2_{min} = 8$

19. For any complex number $w = c + id$, let $\arg(w) \in (-\pi, \pi]$ where $i = \sqrt{-1}$. Let α and β be real numbers such that for all complex numbers $z = x + iy$ satisfying $\arg\left(\frac{z+\alpha}{z+\beta}\right) = \frac{\pi}{4}$, the ordered pair (x, y) lies on the circle $x^2 + y^2 + 5x - 3y + 4 = 0$. Then which of the following statements is (are) TRUE? **[JEE-(Adv)2021]**

किसी सम्मिश्र संख्या $w = c + id$ के लिए माना $\arg(w) \in (-\pi, \pi]$ जहाँ $i = \sqrt{-1}$ माना α तथा β वास्तविक संख्याऐं हैं जबकि सभी सम्मिश्र संख्याओं $z = x + iy$ के लिए $\arg\left(\frac{z+\alpha}{z+\beta}\right) = \frac{\pi}{4}$ को संतुष्ट करती हैं, क्रमित युग्म (x, y) वृत्त $x^2 + y^2 + 5x - 3y + 4 = 0$ पर स्थित है। तब निम्न में से कौनसा/कौनसे कथन सही है/है? **[JEE-(Adv)2021]**

(A) $\alpha = -1$ (B) $\alpha\beta = 4$ (C) $\alpha\beta = -4$ (D) $\beta = 4$

Ans. BD

Sol. $\arg\left(\frac{z+\alpha}{z+\beta}\right) = \frac{\pi}{4}$; α, β are real numbers

$\Rightarrow \arg(z + \alpha) - \arg(z + \beta) = \frac{\pi}{4}$

$\Rightarrow \arg(x + \alpha + iy) - \arg(x + \beta + iy) = \frac{\pi}{4}$

$\Rightarrow \tan^{-1}\left(\frac{y}{\alpha + x}\right) - \tan^{-1}\left(\frac{y}{\beta + x}\right) = \frac{\pi}{4}$

$\Rightarrow \tan^{-1}\left[\frac{\frac{y}{\alpha+x} - \frac{y}{\beta+x}}{1 + \left(\frac{y}{\alpha+x}\right)\left(\frac{y}{\beta+x}\right)}\right] = \frac{\pi}{4}$

$\Rightarrow \frac{\beta y + xy - \alpha y - xy}{\alpha\beta + x(\alpha+\beta) + x^2 + y^2} = \tan\frac{\pi}{4} = 1$

$\beta y - \alpha y = \alpha\beta + x(\alpha+\beta) + x^2 + y^2$

$\Rightarrow x^2 + y^2 + x(\alpha+\beta) + y(\alpha-\beta) + \alpha\beta = 0$

$x^2 + y^2 + 5x - 3y + 4 = 0$

$\Rightarrow \alpha + \beta = 5 \qquad \alpha\beta = 4$

$\alpha - \beta = -3$

$\alpha = 1$

$\beta = 4$

20. Let $\theta_1, \theta_2, ..., \theta_{10}$ be positive valued angles (in radian) such that $\theta_1 + \theta_2 + ... + \theta_{10} = 2\pi$. Define the complex numbers $z_1 = e^{i\theta_1}$, $z_k = z_{k-1}e^{i\theta_k}$ for $k = 2,3, ..., 10$, where $i = \sqrt{-1}$. Consider the statements P and Q given below:

$P: |z_2 - z_1| + |z_3 - z_2| + ... + |z_{10} - z_9| + |z_1 - z_{10}| \le 2\pi$

$Q: |z_2^2 - z_1^2| + |z_3^2 - z_2^2| + ... + |z_{10}^2 - z_9^2| + |z_1^2 - z_{10}^2| \le 4\pi$

Then: **JEE(Adv.)2021]**

(A) P is TRUE and Q is FALSE (B) Q is TRUE and P is FALSE

(C) both P and Q are TRUE (D) both P and Q are FALSE

माना $\theta_1, \theta_2, ..., \theta_{10}$ धनात्मक मान वाले कोण (रेडियन में) हैं जबकि $\theta_1 + \theta_2 + ...+\theta_{10} = 2\pi$. k = 2,3, ..., 10 के लिए सम्मिश्र संख्याओं $z_1 = e^{i\theta_1}$, $z_k = z_{k-1}e^{i\theta_k}$ को परिभाषित कीजिए, जहाँ $i = \sqrt{-1}$ नीचे दिये गये कथनों P व Q पर विचार कीजिए :

$$P : |z_2 - z_1| + |z_3 - z_2| + ... + |z_{10} - z_9| + |z_1 - z_{10}| \le 2\pi$$

$$Q : |z_2^2 - z_1^2| + |z_3^2 - z_2^2| + ... + |z_{10}^2 - z_9^2| + |z_1^2 - z_{10}^2| \le 4\pi$$

तो :

(A) P सही है तथा Q गलत है (B) Q सही है तथा P गलत है

(C) P तथा Q दोनों सही हैं (D) P तथा Q दोनों गलत हैं

Ans. (C)

Sol. $\theta_1 + \theta_2 + ... \theta_{10} = 2\pi$

$z_1 = e^{i\theta_1}$, $z_2 = z_1e^{i\theta_2}$, $z_3 = z_2e^{i\theta_3}$ .....

$$|z_k - z_{k-1}| = |z_{k-1}e^{i\theta_k} - z_{k-1}|$$

$$= |z_{k-1}||e^{i\theta_k} - 1| = \left|2\sin\left(\frac{\theta_k}{2}\right)\right|$$

$$|z_k^2 - z_{k-1}^2| = |z_{k-1}^2e^{i2\theta_k} - z_{k-1}^2| = |z_{k-1}^2||e^{i2\theta_{k-1}}| = |2\sin\theta_k|$$

Statement P

$$|z_2 - z_1| + |z_3 - z_2| + ... + |z_{10} - z_9| + |z_1 - z_{10}|$$

$$\left|2\sin\frac{\theta_2}{2}\right| + \left|2\sin\frac{\theta_3}{2}\right| + ... + \left|2\sin\frac{\theta_{10}}{2}\right| + \left|2\sin\left(\frac{\theta_2 + ... + \theta_{10}}{2}\right)\right|$$

$$\left|2\sin\frac{\theta_2}{2}\right| + \left|2\sin\frac{\theta_3}{2}\right| + ... + \left|2\sin\frac{\theta_{10}}{2}\right| + \left|2\sin\left(\frac{2\pi - \theta_1}{2}\right)\right|$$

$$= \left|2\sin\frac{\theta_1}{2}\right| + \left|2\sin\frac{\theta_2}{2}\right| + ... + \left|2\sin\frac{\theta_{10}}{2}\right|$$

Maximum value will occur when $\theta_1 = \theta_2 ... = \theta_{10} = \frac{2\pi}{10}$

$$LHS_{max} = 10\left(2\sin\frac{\pi}{10}\right) = 20\sin(18°) = 20\left(\frac{\sqrt{5}-1}{4}\right) \approx 6.18 < 2\pi$$

$\Rightarrow$ P is true

Statement Q

LHS = $|2 \sin \theta_1| + |2 \sin \theta_2| + ... + |2 \sin \theta_{10}|$

Maximum Value will occur when $\theta_1 = \theta_2 = ... \theta_{10} = \frac{\pi}{5}$.

$\text{LHS} = 10\left(2\text{Sin}\frac{\pi}{5}\right) = 20\text{Sin}\,36° < 4\pi$

$\Rightarrow$ Statement Q is true.

21. Let z be a complex number with non-zero imaginary part. If $\frac{2+3z+4z^2}{2-3z+4z^2}$ is a real number, then the value of $|z|^2$ is _____. **[JEE-(Adv)2022]**

माना अशून्य काल्पनिक भाग वाली एक सम्मिश्र संख्या z है। यदि $\frac{2+3z+4z^2}{2-3z+4z^2}$ वास्तविक संख्या है, तब $|z|^2$ का मान ज्ञात कीजिए। **[JEE-(Adv)2022]**

Ans. (0.50)

Sol. Let $w = \frac{4z^2+3z+2}{4z^2-3z+2} = 1+\frac{6z}{4z^2-3z+2}$

$\Rightarrow \quad w = 1+\frac{6}{2\left(2z+\frac{1}{z}\right)-3}$

$\because\ w \in R$ then $2z+\frac{1}{z} \in R$

$\Rightarrow \quad 2z+\frac{1}{z} = 2\overline{z}+\frac{1}{\overline{z}}$

$\Rightarrow \quad 2(z-\overline{z})-\frac{z-\overline{z}}{|z|^2} = 0$

$\Rightarrow \quad (z-\overline{z})\left(2-\frac{1}{|z|^2}\right) = 0$

$\because \quad z \neq \overline{z}$ (given)

So $|z|^2 = \frac{1}{2}$

Let $\overline{z}$ denote the complex conjugate of a complex number z and let $i = \sqrt{-1}$. In the set of complex numbers, the number of distinct roots of the equation $\overline{z}-z^2 = i\left(\overline{z}+z^2\right)$ is_____. **[JEE-(Adv)2022]**

माना $\overline{z}$, किसी सम्मिश्र संख्या z के सम्मिश्र संयुग्मी को दर्शाता है तथा माना $i = \sqrt{-1}$ है। सम्मिश्र संख्याओं के समुच्चय में समीकरण $\overline{z}-z^2 = i\left(\overline{z}+z^2\right)$ के भिन्न मूलों की संख्या ज्ञात कीजिए। **[JEE-(Adv)2022]**

Ans. (4.00)

Sol. $\bar{z}-z^2=i\left(\bar{z}+z^2\right)$

$\Rightarrow (1-i)\bar{z}=(1+i)z^2$

$\Rightarrow \dfrac{(1-i)}{(1+i)}\bar{z}=z^2$

$\Rightarrow \left(-\dfrac{2i}{2}\right)\bar{z}=z^2$

$\therefore z^2=-i\bar{z}$

Let $z = x + iy$,

$\therefore (x^2-y^2)+i(2xy)=-i(x-iy)$

so, $x^2-y^2+y=0$ _____(1)

and $(2y+1)x=0$ _____(2)

$\therefore x=0$ or $y=-\dfrac{1}{2}$

Case I : When $x = 0$

$\therefore (1) \Rightarrow y(1-y)=0 \ \therefore y=0,1$

$\therefore (0,0), (0,1)$

Case II : When $y=-\dfrac{1}{2}$

$\therefore (1) \Rightarrow x^2-\dfrac{1}{4}-\dfrac{1}{2}=0 \Rightarrow x^2=\dfrac{3}{4} \Rightarrow x=\pm\dfrac{\sqrt{3}}{2}$

$\therefore \left(\dfrac{\sqrt{3}}{2},-\dfrac{1}{2}\right),\left(-\dfrac{\sqrt{3}}{2},-\dfrac{1}{2}\right)$

$\Rightarrow$ Number of distinct 'z' is equal to 4.

23. Let $\bar{z}$ denote the complex conjugate of a complex number z. If z is a non-zero complex number for which both real and imaginary parts of $\left(\bar{z}\right)^2+\dfrac{1}{z^2}$ are integers, then which of the following is/are possible value(s) of $|z|$?

[JEE-(Adv)2022]

माना एक सम्मिश्र संख्या z के सम्मिश्र संयुग्मी को $\bar{z}$ द्वारा दर्शाया जाता है। यदि z एक अशून्य सम्मिश्र संख्या है जिसके लिए $\left(\bar{z}\right)^2+\dfrac{1}{z^2}$ के वास्तविक और काल्पनिक दोनों भाग पूर्णांक है, तब निम्नलिखित में से $|z|$ के संभावित मान कौनसे है/हैं ?

[JEE-(Adv)2022]

(A) $\left(\dfrac{43+3\sqrt{205}}{2}\right)^{\frac{1}{4}}$ (B) $\left(\dfrac{7+\sqrt{33}}{4}\right)^{\frac{1}{4}}$ (C) $\left(\dfrac{9+\sqrt{65}}{4}\right)^{\frac{1}{4}}$ (D) $\left(\dfrac{7+\sqrt{13}}{6}\right)^{\frac{1}{4}}$

Ans. (A)

Sol. Let $z = r(\cos\theta + i\sin\theta)$

$$\bar{z}^2 + \frac{1}{z^2} = r^2(\cos 2\theta - i\sin 2\theta)$$

$$+\frac{1}{r^2}(\cos 2\theta - i\sin 2\theta)$$

$$= \left(r^2 + \frac{1}{r^2}\right)(\cos 2\theta - i\sin 2\theta)$$

How

$$\left.\begin{aligned}\left(r^2 + \frac{1}{r^2}\right)\cos 2\theta = \lambda \Rightarrow \left(r^4 + \frac{1}{r^4} + 2\right)\cos^2 2\theta = \lambda^2 \\ \left(r^2 + \frac{1}{r^2}\right)\sin 2\theta = \mu \Rightarrow \left(r^4 + \frac{1}{r^4} + 2\right)\sin^2 2\theta = \mu^2\end{aligned}\right\} \Rightarrow r^4 + \frac{1}{r^4} + 2 = \lambda^2 + \mu^2$$

So, $r^4 + \frac{1}{r^4} \in I$

(A) $r^4 = \frac{43 + 3\sqrt{205}}{2} \Rightarrow \frac{1}{r^4} = \frac{2(43 - 3\sqrt{205})}{1849 - 1845} = \frac{43 - 3\sqrt{205}}{2} \Rightarrow r^4 + \frac{1}{r^4} = 43 \in I$

(B) $r^4 = \frac{7 + \sqrt{33}}{4} \Rightarrow \frac{1}{r^4} = \frac{4(7 - \sqrt{33})}{49 - 33} = \frac{7 - \sqrt{33}}{4} \Rightarrow r^4 + \frac{1}{r^4} \notin I$

(C) $r^4 = \frac{9 + \sqrt{65}}{4} \Rightarrow \frac{1}{r^4} = \frac{4(9 - \sqrt{65})}{81 - 65} = \frac{9 - \sqrt{65}}{4} \Rightarrow r^4 + \frac{1}{r^4} \notin I$

(D) $r^4 = \frac{7 + \sqrt{13}}{6} \Rightarrow \frac{1}{r^4} = \frac{6(7 - \sqrt{13})}{49 - 13} = \frac{6(7 - \sqrt{13})}{36} = \frac{7 - \sqrt{13}}{6} \Rightarrow r^4 + \frac{1}{r^4} \notin I$

24. Let $A = \left\{\frac{1967 + 1686i\sin\theta}{7 - 3i\cos\theta} : \theta \in R\right\}$. If A contains exactly one positive integer n, then the value of n is_____

[JEE-(Adv)2023].

माना कि $A = \left\{\frac{1967 + 1686i\sin\theta}{7 - 3i\cos\theta} : \theta \in R\right\}$ है। यदि A में केवल एक धनात्मक पूर्णांक (positive integer) n है, तब n का मान है_____

[JEE-(Adv)2023]

Ans. 281

Sol. $A = \frac{1967 + 1686i\sin\theta}{7 - 3i\cos\theta}$

$$= \frac{281(7 + 6i\sin\theta)}{7 - 3i\cos\theta} \times \frac{7 + 3i\cos\theta}{7 + 3i\cos\theta}$$

$$= \frac{281\left(49 - 18\sin\theta\cos\theta + i\left(21\cos\theta + 42\sin\theta\right)\right)}{49 + 9\cos^2\theta}$$

for positive integer

Im(A) = 0

21 cos θ + 42 sin θ = 0

$$\tan\theta = \frac{-1}{2}; \sin 2\theta = \frac{-4}{5}, \cos^2\theta = \frac{4}{5}$$

$$\mathrm{Re}(A) = \frac{281\left(49 - 9\sin 2\theta\right)}{49 + 9\cos^2\theta}$$

$$= \frac{281\left(49 - 9\times\frac{-4}{5}\right)}{49 + 9\times\frac{4}{5}} = 281(+\text{ve integer})$$

25. Let z be a complex number satisfying $|z|^3 + 2z^2 + 4\bar{z} - 8 = 0$, where $\bar{z}$ denotes the complex conjugate of z. Let the imaginary part of z be non-zero.

Match each entry in **List-I** to the correct entries in **List-II**. **[JEE-(Adv)2023]**

List-I	**List-II**
(P) $\|z\|^2$ is equal to	(1) 12
(Q) $\|z - \bar{z}\|^2$ is equal to	(2) 4
(R) $\|z\|^2 + \|z + \bar{z}\|^2$ is equal to	(3) 8
(S) $\|z + 1\|^2$ is equal to	(4) 10
	(5) 7

The correct option is :

माना कि $|z|^3 + 2z^2 + 4\bar{z} - 8 = 0$ को संतुष्ट करने वाली एक सम्मिश्र संख्या (complex number) z है, जहाँ $\bar{z}$ सम्मिश्र संख्या z का संयुग्मी (conjugate) है। माना कि z का काल्पनिक भाग (imaginary part) अशून्य (nonzero) है।

Match each entry in **List-I** to the correct entries in **List-II**. **[JEE-(Adv)2023]**

List-I	**List-II**
(P) $\|z\|^2$	(1) 12 के बराबर है
(Q) $\|z - \bar{z}\|^2$	(2) 4 के बराबर है
(R) $\|z\|^2 + \|z + \bar{z}\|^2$	(3) 8 के बराबर है
(S) $\|z + 1\|^2$	(4) 10 के बराबर है
	(5) 7 के बराबर है

सही विकल्प है :

(A) (P) $\rightarrow$ (1); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (5); (S) $\rightarrow$ (4)
(B) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (3); (S) $\rightarrow$ (5)
(C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (5); (S) $\rightarrow$ (1)
(D) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (5); (S) $\rightarrow$ (4)

Ans. B

Sol. $|z|^3 + 2z^2 + 4\bar{z} - 8 = 0$ _____(1)

taking conjugate

$|z|^3 + 2\bar{z}^2 + 4z - 8 = 0$ _____(2)

(1) – (2)

$2\left(z^2 - \bar{z}^2\right) + 4\left(\bar{z} - z\right) = 0$

$\left(z - \bar{z}\right)\left(z + \bar{z}\right) + 2\left(\bar{z} - z\right) = 0$

$\left(z - \bar{z}\right)\left(z + \bar{z} - 2\right) = 0$

$\therefore z - \bar{z} \neq 0$ (Otherwise Img part of z will be zero)

$\therefore z + \bar{z} - 2 = 0$

$x + iy + x - iy - 2 = 0$

$2x - 2 = 0$

$x = 1$

$\therefore$ z will be of the form $z = 1 + iy$

$|z| = \sqrt{1 + y^2}$; $z^2 = 1 - y^2 + 2iy$; $\bar{z} = 1 - iy$

Substitute in equation (1)

$\left(1 + y^2\right)^{\frac{3}{2}} + 2\left(1 - y^2 + 2iy\right) + 4\left(1 - iy\right) - 8 = 0$

$\left(1 + y^2\right)^{\frac{3}{2}} + 2 - 2y^2 + 4 - 8 + i\left(4y - 4y\right) = 0$

$\left(1 + y^2\right)^{\frac{3}{2}} + \left(-2y^2 - 2\right) = 0$

$\left(1 + y^2\right)^{\frac{3}{2}} - 2\left(1 + y^2\right) = 0$

$\left(1 + y^2\right)\left\{\sqrt{1 + y^2} - 2\right\} = 0$

$1 + y^2 = 4$

$y^2 = 3$

(P) $|z|^2 = 1 + y^2$

$= 4$

(Q) $\left|z-\bar{z}^2\right| \Rightarrow \left|(1+iy)-(1-iy)\right|^2$

$= |1 + iy - 1 + iy|^2$

$= |2iy|^2$

$= 4y^2$

$= 4 \cdot 3$

$= 12$

(R) $|z|^2 + \left|z+\bar{z}\right|^2$

$= 4 + |1 + iy + 1 - iy|^2$

$\Rightarrow 4 + 4$

$\Rightarrow 8$

(S) $|z + 1|^2 = |1 + iy + 1|^2$

$= |2 + iy|^2$

$= \left(\sqrt{4+y^2}\right)^2$

$= 4 + 3$

$= 7$

26. Let $A_1, A_2, A_3, ... A_8$ be the vertices of a regular octagon that lie on a circle of radius 2. Let P be a point on the circle and let PA_i denote the distance between the poibnts P and A_i for i = 1,2,...,8. If P varies over the circle, then the maximum value of the product $PA_1 \cdot PA_2 \cdot ..... \cdot PA_8$, is _____ **[JEE-(Adv)2023]**

माना कि $A_1, A_2, A_3, ... A_8$ एक सम अष्टभुज (regular octagon) के शीर्ष (vertices) हैं जो एक वृत्त जिसकी त्रिज्या 2 है, पर स्थित हैं। माना कि P वृत्त पर एक बिन्दु है और बिन्दुओं P एवं A, के बीच की दूरी PA_i, i = 1,2,....,8 है। यदि P वृत्त के ऊपर विचरित (varies) करता है तब गुणनफल $PA_1 \cdot PA_2 \cdot ..... \cdot PA_8$ का उच्चतम मान (maximum value) है _____ **[JEE-(Adv)2023]**

Ans. 512

Sol. $z_r = 2e^{i\left(\frac{r\pi}{8}\right)}$ $\qquad r = \{0, 1, 2, ..., 7\}$

$PA_1.PA_2.....PA_8$

$= |z - z_1||z - z_2|.....|z - z_8|$

$= \left|(z - z_1)(z - z_2).....(z - z_8)\right|$ .....(1)

Also

$z^8 - 2^8 = (z - z_1)(z - z_2).....(z - z_8)$ $\qquad \{|z| = 2\}$

$\Rightarrow \left|z^8 - 2^8\right| = \left|(z-z_1)(z-z_2).....(z-z_8)\right|$

by triangle inequality

$$\left|z^8 + \left(-2^8\right)\right| \leq |z|^8 + 2^8$$

$$\leq 2^8 + 2^8$$

$$\leq 2^9$$

So max $\left(PA_1 . PA_2.....PA_8\right) = 2^9$

$= 2^9$

Ans. $= 512$

27. Let $S = \left\{a + b\sqrt{2} : a, b \in Z\right\}$, $T_1 = \left\{\left(-1+\sqrt{2}\right)^n : n \in N\right\}$ and $T_2 = \left\{\left(1+\sqrt{2}\right)^n : n \in N\right\}$. Then which of the following statements is(are) TRUE ? **[JEE-(Adv)2024]**

(A) $Z \cup T_1 \cup T_2 = S$

(B) $T_1 \cap \left(0, \frac{1}{2024}\right) = \phi$, where ϕ denotes the empty set.

(C) $T_2 \cap (2024, \infty) \neq \phi$

(D) For any given $a, b \in Z$, $\cos\left(\pi\left(a + b\sqrt{2}\right)\right) + i\sin\left(\pi\left(a + b\sqrt{2}\right)\right) \in Z$ if and only if $b = 0$, where $i = \sqrt{-1}$.

माना कि $S = \left\{a + b\sqrt{2} : a, b \in Z\right\}$, $T_1 = \left\{\left(-1+\sqrt{2}\right)^n : n \in N\right\}$ और $T_2 = \left\{\left(1+\sqrt{2}\right)^n : n \in N\right\}$ हैं। तब निम्नलिखित कथनों में से कौनसा(से) सत्य है(हैं) ? **[JEE-(Adv)2024]**

(A) $Z \cup T_1 \cup T_2 = S$

(B) $T_1 \cap \left(0, \frac{1}{2024}\right) = \phi$, जहाँ ϕ रिक्त समुच्चय को दर्शाता है।

(C) $T_2 \cap (2024, \infty) \neq \phi$

(D) किन्हीं दिये गए a, b $\in$ Z के लिए, $\cos\left(\pi\left(a + b\sqrt{2}\right)\right) + i\sin\left(\pi\left(a + b\sqrt{2}\right)\right) \in Z$ यदि और यदि केवल b = 0, जहाँ $i = \sqrt{-1}$ है।

Ans. ACD

Sol. $T_1 = \left(-1+\sqrt{2}\right)^n = \dfrac{1}{\left(\sqrt{2}+1\right)^n}$

$\left(T_1\right)_{max} = \dfrac{1}{\sqrt{2}+1} = 0.4142$

$\left(T_1\right)_{min} \to 0$

Option (B) is incorrect

$T_2 = \left(1+\sqrt{2}\right)^n$

$\left(T_2\right)_{min} = 1+\sqrt{2} = 2.4142$

$\left(T_2\right)_{max} \to \infty$

Option (C) is correct

For $\cos(\pi\left(a+b\sqrt{2}\right)+i\sin\left(\pi\left(a+b\sqrt{2}\right)\right) \in Z$

$\sin\left(\pi\left(a+b\sqrt{2}\right)\right) = 0$

$\pi\left(a+b\sqrt{2}\right) = n\pi$

$a+b\sqrt{2} = n \in Z$

$\Rightarrow b = 0$ and $a \in Z$

Option (D) is correct

Both T_1 and T_2 are of the type $a+b\sqrt{2}$

hence $Z \cup T_1 \cup T_2 \subset S$

Option (A) is correct

28. Let R denote the set of all real numbers. Let $z_1 = 1 + 2i$ and $z_2 = 3i$ be two complex numbers, where $i = \sqrt{-1}$. Let

$S = \{(x, y) \in R \times R : |x + iy - z_1| = 2|x + iy - z_2|\}$.

The which of the following statements is (are) TRUE ? **[JEE-(Adv)2025]**

(A) S is a circle with centre $\left(-\dfrac{1}{3}, \dfrac{10}{3}\right)$ (B) S is a circle with centre $\left(\dfrac{1}{3}, \dfrac{8}{3}\right)$

(C) S is a circle with radius $\frac{\sqrt{2}}{3}$ (D) S is a circle with radius $\frac{2\sqrt{2}}{3}$

मान लीजिए कि R सभी वास्तविक संख्याओं के समुच्चय को दर्शाता है। मान लीजिए कि $z_1 = 1 + 2i$ और $z_2 = 3i$ दो सम्मिश्र संख्याएँ हैं, जहाँ $i = \sqrt{-1}$ है। मान लीजिए कि

$S = \{(x, y) \in R \times R : |x + iy - z_1| = 2|x + iy - z_2|\}$.

है।

तब निम्नलिखित कथनों में से कौन सा (से) सत्य है (हैं) ? **[JEE-(Adv)2025]**

(A) S एक वृत्त है, जिसका केन्द्र $\left(-\frac{1}{3}, \frac{10}{3}\right)$ है (B) S एक वृत्त है, जिसका केन्द्र $\left(\frac{1}{3}, \frac{8}{3}\right)$ है

(C) S एक वृत्त है, जिसकी त्रिज्या $\frac{\sqrt{2}}{3}$ है (D) S एक वृत्त है, जिसकी त्रिज्या $\frac{2\sqrt{2}}{3}$ है

Ans. AD

Sol. $|x + iy - (1 + 2i)| = 2|x + iy - 3i|$

$\Rightarrow (x - 1)^2 + (y - 2)^2 = 4(x^2 + (y - 3)^2)$

$\Rightarrow x^2 + y^2 - 2x - 4y + 5 = 4x^2 + 4y^2 - 24y + 36$

$\Rightarrow 3x^2 + 3y^2 + 2x - 20y + 31 = 0$

$\Rightarrow x^2 + y^2 + \frac{2}{3}x - \frac{20}{3}y + \frac{31}{3} = 0$

Centre $\left(-\frac{1}{3}, \frac{10}{3}\right)$

radius $= \sqrt{\frac{1}{9} + \frac{100}{9} - \frac{31}{3}} = \sqrt{\frac{8}{9}} = \frac{2\sqrt{2}}{3}$

29. For a non-zero complex number z, let arg(z) denote the principal argument of z, with $-\pi < \arg(z) \le \pi$. Let ω be the cube root of unity for which $0 < \arg(\omega) < \pi$. Let $\alpha = \arg\left(\sum_{n=1}^{2025} (-\omega)^n\right)$.

Then the value of $\frac{3\alpha}{\pi}$ is _____. **[JEE-(Adv)2025]**

किसी शून्येतर (non-zero) सम्मिश्र संख्या (complex number) z के लिए, मान लीजिए कि arg(z), z के मुख्य कोणांक (principal argument) को दर्शाता है, जहाँ $-\pi < \arg(z) \le \pi$ है। मान लीजिए कि ω, एकक (unity) का वह घनमूल (cube root) है, जिसके लिए

$0 < \arg(\omega) < \pi$ है। मान लीजिए कि $\alpha = \arg\left(\sum_{n=1}^{2025}(-\omega)^n\right)$.

है। तब $\frac{3\alpha}{\pi}$ का मान_____ है। **[JEE-(Adv)2025]**

Ans. –2

Sol. Clearly $= \omega = e^{+i2\pi/3}$

$$\sum_{n=1}^{2025}(-\omega)^n = \left[-\omega + \omega^2 - \omega^3 + \omega^4 - \omega^5 + \omega^6 ..... + \omega^{2024} - \omega^{2025}\right]$$

$$= \frac{(-\omega)(1-(-\omega)^{2025})}{1+\omega}$$

$$= \frac{-\omega}{-\omega^2}\left(1+\omega^{2025}\right) = \frac{1}{\omega}(2) = \frac{2}{\omega} = 2\omega^2$$

$$\arg(z) = \frac{-2\pi}{3} = \alpha$$

$$\Rightarrow \frac{3\alpha}{\pi} = -2$$

30. Let R denote the set of all real numbers and let $i = \sqrt{-1}$. Consider the matrices $S = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

Let a, b, c, d be real numbers such that $ST = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Let $H = \{x + iy : x, y \in R \text{ and } y > 0\}$. Then which of the following statements is (are) **TRUE** : **[JEE-(Adv)2026]**

(A) $\frac{b+ia}{d+ic} = i$

(B) If $\omega = \frac{-1+i\sqrt{3}}{2}$, then $\frac{a\omega+b}{c\omega+d} = \omega$

(C) If m is an integer greater than 2 such that $(ST)^2 = (ST)^m$, then m is an integer multiple of 8

(D) If $z \in H$, then $\frac{az+b}{cz+d} \in H$

मान लीजिए कि R सभी वास्तविक संख्याओं के समुच्चय को दर्शाता है, और मान लीजिए कि $i=\sqrt{-1}$ है। आव्यूहों $S=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$ और $T=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ पर विचार कीजिए। मान लीजिए कि a, b, c, d इस प्रकार की वास्तविक संख्याएँ हैं कि $ST=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$ है। मान लीजिए कि $H=\{x+iy : x, y \in R \text{ तथा } y>0\}$ है। तब निम्नलिखित कथनों में से कौन सा (से) **सत्य** है (हैं) :

[JEE-(Adv)2026]

(A) $\frac{b+ia}{d+ic}=i$

(B) यदि $\omega=\frac{-1+i\sqrt{3}}{2}$ है, तब $\frac{a\omega+b}{c\omega+d}=\omega$ है

(C) यदि m, 2 से बड़ा एक ऐसा पूर्णांक है कि $(ST)^2=(ST)^m$ है, तब m, 8 का एक पूर्णांक गुणज है

(D) यदि $z \in H$ है, तब $\frac{az+b}{cz+d} \in H$ है

Ans. BD

Sol. $ST=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}=\begin{bmatrix}0 & -1\\ 1 & 1\end{bmatrix}=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$

$a=0, b=-1, c=1, d=1$

(A) $\frac{b+ia}{d+ic}=\frac{-1}{1+i}\times\frac{1-i}{1-i}=\frac{i-1}{2}\left(\text{Incorrect}\right)$

(B) $\frac{a\omega+b}{c\omega+d}=\frac{-1}{\omega+1}=\frac{-1}{-\omega^2}=\omega\left(\text{correct}\right)$

(C) $\left(ST\right)^2=\begin{bmatrix}0 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}0 & -1\\ 1 & 1\end{bmatrix}=\begin{bmatrix}-1 & -1\\ 1 & 0\end{bmatrix}$

$\left(ST\right)^3=\begin{bmatrix}-1 & -1\\ 1 & 0\end{bmatrix}\begin{bmatrix}0 & -1\\ 1 & 1\end{bmatrix}=\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}=-I$

$(ST)^6=I$

$(ST)^2=(ST)^M$

where $M=6\lambda+2$ (Incorrect)

(D) $\frac{az+b}{cz+d}=\frac{-1}{z+1}=\frac{-1}{(x+1)+iy}\times\frac{(x+1)-iy}{(x+1)-iy}$

$$= -\frac{(x+1)+iy}{(x+1)^2+y^2}$$

Imaginary part $= \frac{y}{(x+1)^2+y^2} > 0$ as $y > 0$

31. Match each entry in **List-I** to the correct entry in **List-II** and choose the correct option.

[JEE-(Adv)2026]

List-I	**List-II**
(P) If α and β are the distinct roots of the equation $x^2 + x + 1 = 0$, then the quadratic equation with roots $\frac{1}{(\alpha+1)^{2026}}$ and $\frac{1}{(\beta+1)^{2026}}$	(1) $x^2 + x + 1 = 0$
(Q) If α and β are the distinct roots of the equation $x^2 + x + 1 = 0$, then the quadratic equation with roots $\frac{1}{(\alpha+1)^{2027}}$ and $\frac{1}{(\beta+1)^{2027}}$	(2) $x^2 - x + 1 = 0$
(R) If γ and δ are the distinct roots of the equation $x^2 - x + 1 = 0$, then the value of $\frac{1}{(\gamma-1)^{2026}} + \frac{1}{(\delta-1)^{2026}}$ is	(3) $x^2 + x - 1 = 0$
(S) If p and r are the distinct roots of the equation $x^2 + x - 1 = 0$, then the value of $\frac{1}{(p+1)^3} + \frac{1}{(r+1)^3}$ is	(4) -1
	(5) -4

The **correct** option is :

सूची-I की प्रत्येक प्रविष्टि का **सूची-II** की सही प्रविष्टि से मिलान कर सही विकल्प चुनिए। **[JEE-(Adv)2026]**

सूची-I	**सूची-II**
(P) यदि α और β, समीकरण $x^2 + x + 1 = 0$ के भिन्न मूल हैं, तब वह द्विघातीय समीकरण जिसके मूल $\frac{1}{(\alpha+1)^{2026}}$ और $\frac{1}{(\beta+1)^{2026}}$ हैं।	(1) $x^2 + x + 1 = 0$ है
(Q) यदि α और β समीकरण $x^2 + x + 1 = 0$ के भिन्न मूल हैं,	(2) $x^2 - x + 1 = 0$ है

तब वह द्विघातीय समीकरण जिसके मूल

$\frac{1}{(\alpha+1)^{2027}}$ और $\frac{1}{(\beta+1)^{2027}}$ हैं,

(R) यदि γ और δ, समीकरण $x^2 - x + 1 = 0$ के भिन्न मूल हैं, (3) $x^2 + x - 1 = 0$ है

तब $\frac{1}{(\gamma-1)^{2026}} + \frac{1}{(\delta-1)^{2026}}$ का मान

(S) यदि p और r, समीकरण $x^2 + x - 1 = 0$ के भिन्न मूल हैं, (4) -1 है

तब $\frac{1}{(p+1)^3} + \frac{1}{(r+1)^3}$ का मान

(5) -4 है

सही विकल्प का चयन कीजिए :

(A) (P) $\rightarrow$ (1); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (5); (S) $\rightarrow$ (4)

(B) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (5)

(C) (P) $\rightarrow$ (1); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (4); (S) $\rightarrow$ (5)

(D) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (5); (S) $\rightarrow$ (4)

Ans. C

Sol. P] Let $\alpha = w$ & $\beta = w^2$

So $\frac{1}{(\alpha+1)^{2026}} = \frac{1}{(1+w)^{2026}} = \frac{1}{\left(-w^2\right)^{2026}} = \frac{1}{w^{4052}} = \frac{1}{w^2} = w$

& $\frac{1}{(\beta+1)^{2026}} = \frac{1}{\left(1+w^2\right)^{2026}} = \frac{1}{(-w)^{2026}} = \frac{1}{w^{2026}} = \frac{1}{w} = w^2$

$\therefore$ Quadratic will be $x^2 + x + 1 = 0$

$P \rightarrow 1$

Q] Again $\alpha = w$ & $\beta = w^2$

$\frac{1}{(\alpha+1)^{2027}} = \frac{1}{(w+1)^{2027}} = \frac{1}{(-w^2)^{2027}} = -\frac{1}{w^{4054}} = -\frac{1}{w} = -w^2$

& $\frac{1}{(\beta+1)^{2027}} = \frac{1}{\left(1+w^2\right)^{2027}} = \frac{1}{(-w)^{2027}} = -\frac{1}{w^{2027}} = -\frac{1}{w^2} = -w$

$\therefore$ Quartratic Equation: $x^2 - x + 1 = 0$

$Q \rightarrow 2$

R] Let $\gamma = -w^2$ & $\delta = -w$

$\therefore \ \dfrac{1}{(\gamma-1)^{2026}} = \dfrac{1}{\left(-w^2\right)^{2026}} = w$

$\dfrac{1}{(\delta-1)^{2026}} = \dfrac{1}{(-w)^{2026}} = w^2$

$w + w^2 = -1$

$R \to 4$

S] $x^2 + x - 1 = 0$

find $\dfrac{1}{(p+1)^3} + \dfrac{1}{(r+1)^3} = \dfrac{(r+1)^3 + (p+1)^3}{\{(p+1)(r+1)\}^3} = \dfrac{r^3 + 1 + 3r^2 + 3r + p^3 + 1 + 3p^2 + 3p}{(-1)^3}$

Now $p + r = -1$; $pr = -1$

$p^2 + r^2 = 3 \Rightarrow p^3 + r^3 = -4$

$S \to 5$

Exercise C - I

1. $\sin^{-1}\left\{\frac{1}{i}(z-1)\right\}$, where z nonreal, can be the angle of a triangle if :

(A) $\text{Re}(z) = 1, \text{Im}(z) = 2$ (B) $\text{Re}(z) = 1, 0 < \text{Im}(z) \le 1$

(C) $\text{Re}(z) + \text{Im}(z) = 0$ (D) None of these

$\sin^{-1}\left\{\frac{1}{i}(z-1)\right\}$, जहाँ z अवास्तविक है, त्रिभुज का कोण हो सकता है यदि :

(A) $\text{Re}(z) = 1, \text{Im}(z) = 2$ (B) $\text{Re}(z) = 1, 0 < \text{Im}(z) \le 1$

(C) $\text{Re}(z) + \text{Im}(z) = 0$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\sin^{-1}\left(\frac{x-1}{i} + y\right)$ should exists and must be positive $\Rightarrow$ $x = 1 \quad 0 < y \le 1$

2. Let A, B, C represent the complex numbers z_1, z_2, z_3 respectively on the complex plane. If the circumcentre of the triangle ABC lies at the origin, then the orthocentre is represented by the complex number :

माना सम्मिश्र समतल पर A, B, C क्रमशः सम्मिश्र संख्याओं z_1, z_2, z_3 को प्रदर्शित करते है। यदि त्रिभुज ABC का परिकेन्द्र मूलबिन्दु पर हो, तो लम्ब केन्द्र को प्रदर्शित करने वाली सम्मिश्र संख्या है :

(A) $z_1 + z_2 - z_3$ (B) $z_2 + z_3 - z_1$ (C) $z_3 + z_1 - z_2$ (D) $z_1 + z_2 + z_3$

Ans. D

Sol. $G \to \text{Centroid of } \Delta = \frac{z_1 + z_2 + z_3}{3}$

$H \to \text{Orthocentre} = z \text{ say}, O \to \text{Circum centre} = 0$

$\because$ G divides HO in ratio 2 : 1 reckening from

$$\frac{z_1 + z_2 + z_3}{3} = \frac{2 \cdot 0 + 1 \cdot z}{2 + 1} \quad \Rightarrow \quad z = z_1 + z_2 + z_3$$

3. If $(a + ib)^5 = \alpha + i\beta$ then $(b + ia)^5$ is equal to :

यदि $(a + ib)^5 = \alpha + i\beta$ हो, तो $(b + ia)^5$ बराबर है :

(A) $\beta + i\alpha$ (B) $\alpha - i\beta$ (C) $\beta - i\alpha$ (D) $-\alpha - i\beta$

Ans. A

Sol. $(b + ia)^5 = i(-ib + a)^5 = i(\alpha - i\beta) = \beta + i\alpha$

4. If $|z| = \max\{|z-1|, |z+1|\}$ then :

यदि $|z| =$ अधिकतम $\{|z-1|, |z+1|\}$ हो, तो :

(A) $|z + \bar{z}| = \frac{1}{2}$ (B) $z + \bar{z} = 1$ (C) $|z + \bar{z}| = 1$ (D) $z \in \phi$

Ans. D

Sol. $\max\{|z-1|, |z+1|\} = |z-1|$ if $\text{Re}(z) \le 0$

$|z+1|$ if $\text{Re}(z) > 0$

for $\text{Re}(z) > 0$ $|z| = |z+1| \quad \Rightarrow \quad \text{Re}(z) = -\frac{1}{2}$ Not possible संभव नहीं है।

for $\text{Re}(z) \le 0$ $|z| = |z-1| \Rightarrow \text{Re}(z) = \frac{1}{2}$ i.e. not possible संभव नहीं है।

5. If $|z_1 - 1| < 1, |z_2 - 2| < 2, |z_3 - 3| < 3$ then $|z_1 + z_2 + z_3|$:

(A) is less than 6 (B) is more than 3 (C) is less than 12 (D) lies between 6 and 12

यदि $|z_1 - 1| < 1, |z_2 - 2| < 2, |z_3 - 3| < 3$ हो, तब $|z_1 + z_2 + z_3|$ का मान है :

(A) 6 से कम (B) 3 से अधिक (C) 12 से कम (D) 6 व 12 के मध्य

Ans. C

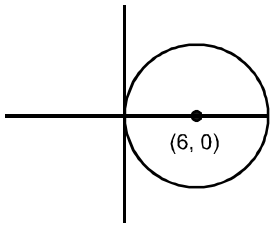
Sol. $|(z_1 - 1) + (z_2 - 2) + (z_3 - 3)| \le |z_1 - 1| + |z_2 - 2| + |z_3 - 3|$

$\Rightarrow \quad |z_1 + z_2 + z_3 - 6| < 6$

Let $z = z_1 + z_2 + z_3 = 3$

then $|z - 6| < 6$ is circular disc

Clearly $|z| < 12$



6. The expression $\left[\frac{1+i\tan\alpha}{1-i\tan\alpha}\right]^n - \frac{1+i\tan n\alpha}{1-i\tan n\alpha}$ when simplified reduces to :

(A) zero (B) $2\sin n\alpha$ (C) $2\cos n\alpha$ (D) None

व्यंजक $\left[\frac{1+i\tan\alpha}{1-i\tan\alpha}\right]^n - \frac{1+i\tan n\alpha}{1-i\tan n\alpha}$ को सरलीकृत करने पर मान है :

(A) zero (B) $2\sin n\alpha$ (C) $2\cos n\alpha$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\left(\frac{e^{i\alpha}}{e^{-i\alpha}}\right)^n - \left(\frac{e^{in\alpha}}{e^{-in\alpha}}\right) \quad \Rightarrow \quad e^{i2n\alpha} - e^{i2n\alpha} = 0$

7. The equation of the radical axis of the two circles represented by the equations, $|z-2| = 3$ and $|z-2-3i| = 4$ on the complex plane is :

सम्मिश्र समतल में समीकरणों $|z-2| = 3$ और $|z-2-3i| = 4$ से प्रदर्शित दो वृत्तों की मूलाक्ष का समीकरण हैं :

(A) $3iz - 3i\bar{z} - 2 = 0$ (B) $3iz - 3i\bar{z} + 2 = 0$ (C) $iz - i\bar{z} + 1 = 0$ (D) $2iz - 2i\bar{z} + 3 = 0$

Ans. B

Sol. square both the sides , use $z\bar{z}=|z|^2$ and subtract

8. If $z=(3+7i)(p+iq)$ where $p, q \in I-\{0\}$, is purely imaginary then minimum value of $|z|^2$ is :

यदि $z=(3+7i)(p+iq)$ जहां $p, q \in I-\{0\}$, पूर्णतः काल्पनिक है, तब $|z|^2$ का न्यूनतम मान है :

(A) 0 (B) 58 (C) $\frac{3364}{3}$ (D) 3364

Ans. D

Sol. $z=(3p-7q)+i(3q+7p)$
for purely imaginary $3p=7q \Rightarrow p=7$ or $q=3$ (for least value)
$|z|=|3+7i|\,|p+iq| \Rightarrow |z|^2=58(p^2+q^2)=58[7^2+9]=58^2 \Rightarrow$ (D)

9. Let $z_r\,(1 \le r \le 4)$ be complex numbers such that $|z_r|=\sqrt{r+1}$ and
$|30z_1+20z_2+15z_3+12z_4|=k\,|z_1z_2z_3+z_2z_3z_4+z_3z_4z_1+z_4z_1z_2|$. Then the value of k equals :

माना $z_r\,(1 \le r \le 4)$ सम्मिश्र संख्याऐं हैं इस प्रकार है कि $|z_r|=\sqrt{r+1}$
और $|30z_1+20z_2+15z_3+12z_4|=k\,|z_1z_2z_3+z_2z_3z_4+z_3z_4z_1+z_4z_1z_2|$. तब k बराबर :
(A) $|z_1z_2z_3|$ (B) $|z_2z_3z_4|$ (C) $|z_3z_4z_1|$ (D) $|z_4z_1z_2|$

Ans. D

Sol. We have $\left|\frac{z_1}{2}+\frac{z_2}{3}+\frac{z_3}{4}+\frac{z_4}{5}\right| = \frac{k}{60}|z_1z_2z_3z_4|\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}+\frac{1}{z_4}\right|$

Now, $z_1\,\bar{z}_1=2$, $z_2\,\bar{z}_2=3$, $z_3\,\bar{z}_3=4$ and $z_4\,\bar{z}_4=5$

So, $k=\frac{60}{|z_1z_2z_3z_4|}=\frac{60}{\sqrt{2}\sqrt{3}\sqrt{4}\sqrt{5}}=\sqrt{30}=|z_4z_1z_2|$ **Ans.**

Note for objective take $z_1=\sqrt{2}$; $z_2=\sqrt{3}$; $z_3=2$; $z_4=\sqrt{5}$

Aliter: Given $|z_1|=\sqrt{2}$; $|z_2|=\sqrt{3}$; $|z_3|=\sqrt{4}$; $|z_4|=\sqrt{5}$

Hence Given that

$$\left|\frac{z_1}{2}+\frac{z_2}{3}+\frac{z_3}{4}+\frac{z_4}{5}\right| = \frac{k}{60}|z_1z_2z_3z_4|\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}+\frac{1}{z_4}\right|$$

$$\therefore \left|\frac{z_1}{2}+\frac{z_2}{3}+\frac{z_3}{4}+\frac{z_4}{5}\right| = \frac{k}{60}\sqrt{120}\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}+\frac{1}{z_4}\right| = \frac{k}{\sqrt{30}}\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}+\frac{1}{z_4}\right|$$

Hence $k = \sqrt{30} = |z_1 z_2 z_3 z_4| \Rightarrow D$

using $z_1\bar{z}_1 = 2$; $z_2\bar{z}_2 = 3$; $z_3\bar{z}_3 = 4$; $z_4\bar{z}_4 = 5$

10. Consider the equation $10z^2 - 3iz - k = 0$, where z is a complex variable and $i^2 = -1$. Which of the following statements is True?

(A) For all real positive numbers k, both roots are pure imaginary

(B) For negative real numbers k, both roots are pure imaginary

(C) For all pure imaginary numbers k, both roots are real and irrational

(D) For all complex numbers k, neither root is real

माना $10z^2 - 3iz - k = 0$, जहां z चर सम्मिश्र संख्या और $i^2 = -1$. तब निम्न में से सत्य कथन है?

(A) धनात्मक वास्तविक संख्या k के लिए, दोनों मूल पूर्णतः काल्पनिक होंगे

(B) ऋणात्मक वास्तविक संख्या k के लिए, दोनों मूल पूर्णतः काल्पनिक होंगे

(C) सभी पूर्णतः काल्पनिक संख्या k के लिए, दोनों मूल वास्तविक और अपरिमेय होंगे

(D) सभी सम्मिश्र संख्याओं k के लिए, मूल वास्तविक नहीं होंगे

Ans. B

Sol. Use the quadratic formula to obtain $z = \dfrac{3i \pm \sqrt{-9 + 40k}}{20}$

which has discriminant $D = -9 + 40k$. If $k = 1$, then $D = 31$, so (A) is false.

If k is a negative real number, then D is a negative real number, so (B) is true.

If $k = i$, then $D = -9 + 40i = 16 + 40i + 25i^2 = (4 + 5i)^2$, and the roots are $\dfrac{1}{5} + \dfrac{2}{5}i$ and $-\dfrac{1}{5} - \dfrac{1}{10}i$, so (C) is false.

If $k = 0$ (which is a complex number), then the roots are 0 and $\dfrac{3}{10}i$, so (D) is false.

11. If $z^3 + (3 + 2i)z + (-1 + ia) = 0$ has one real root, then the value of 'a' lies in the interval : $(a \in R)$

यदि $z^3 + (3 + 2i)z + (-1 + ia) = 0$ का एक मूल वास्तविक है, तब 'a' का मान किस अन्तराल में होगा : (जहां $a \in R$)

(A) $(-2, -1)$ (B) $(-1, 0)$ (C) $(0, 1)$ (D) $(1, 2)$

Ans. B

Sol. Let $z = \alpha$ be a real root

$\alpha^3 + (3 + 2i)\alpha + (-1 + ia) = 0 \Rightarrow (\alpha^3 + 3\alpha - 1) + i(a + 2\alpha) = 0$

$\therefore$ $\alpha^3 + 3\alpha - 1 = 0$ and $\alpha = -a/2$

$\therefore$ $-\dfrac{a^3}{8} - \dfrac{3a}{2} - 1 = 0$

$a^3 + 12a + 8 = 0$

Let $f(a) = a^3 + 12a + 8$

$\therefore \quad f(-1) < 0 \quad$ and $\quad f(0) > 0$

hence $a \in (-1, 0)$

12. If $|z| = 1$ and $|\omega - 1| = 1$ where $z, \omega \in C$, then the largest set of values of $|2z - 1|^2 + |2\omega - 1|^2$ equals :

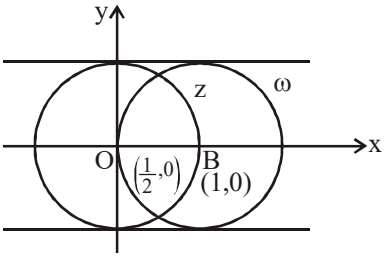
यदि $|z| = 1$ और $|\omega - 1| = 1$ जहां $z, \omega \in C$, तब $|2z - 1|^2 + |2\omega - 1|^2$ के मानों का दीर्घतम समुच्चय है :

(A) [1, 9] (B) [2, 6] (C) [2, 12] (D) [2, 18]

Ans. D

Sol. $E = |2z - 1|^2 + |2\omega - 1|^2$; $E = \left[\left| z - \frac{1}{2} \right|^2 + \left| \omega - \frac{1}{2} \right|^2 \right]$

$\left| z - \frac{1}{2} \right|$ distance of z tan $\left(\frac{1}{2}, 0 \right)$ and $\left| \omega - \frac{1}{2} \right|$ distance of ω from $\left(\frac{1}{2}, 0 \right)$.



$$E_{max} = 4\left[\left(-1, -\frac{1}{2}\right)^2 + \left(2 - \frac{1}{2}\right)^2\right] = 4\left[\frac{9}{4} + \frac{9}{4}\right] = 18$$

$$E_{min} = 4\left[\left(1 - \frac{1}{2}\right)^2 + \left(2 - \frac{1}{2}\right)^2\right] = 4\left[\frac{1}{4} + \frac{1}{4}\right] = 2$$

rajected E = [2, 18]

Alternatively :

$(2z - 1)\ (2\bar{z} - 1) + (2\omega - 1)\ (2\bar{\omega} - 1)$

$\left[4|z|^2 - 2(z + \bar{z}) + 1\right] + \left[4|\omega|^2 - 2(\omega + \bar{\omega}) + 1\right]$

$E = 6 - 4R + z + 4\,\omega\bar{\omega} - 4\,Re\omega$

Now $|\omega - 1|^2 = 1 \Rightarrow (\omega - 1)\,(\bar{\omega} - 1) = 1$

$\omega\bar{\omega} - (\omega + \bar{\omega}) = 0$

$\omega\bar{\omega} = \omega + \bar{\omega}$

$\omega\bar{\omega} = 2\,Re\,\omega$

Hence $E = 6 - 4Rez + 4\,Re\omega$

$= 6 + 4\,(Re\,\omega - Re\,z)$

$= 6 + 4\,(x_2 - x_1)$

$E_{max} = 6 + 4\,(2 + 1) = 18$

$E_{min} = 6 + 4\,(0 - 1) = 2 \Rightarrow$ say [2, 18]

13. If $Arg\,(z + a) = \frac{\pi}{6}$ and $Arg\,(z - a) = \frac{2\pi}{3}$; $a \in R^+$, then :

(A) z is independent of a (B) $|a| = |z + a|$

(C) $z = a\,\text{Cis}\,\frac{\pi}{6}$ (D) $z = a\,\text{Cis}\,\frac{\pi}{3}$

यदि $\text{Arg}(z + a) = \frac{\pi}{6}$ और $\text{Arg}(z - a) = \frac{2\pi}{3}$; $a \in R^+$, तब :

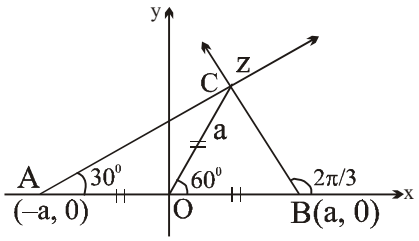
(A) z, a से स्वतन्त्र है (B) $|a| = |z + a|$

(C) $z = a\,\text{Cis}\,\frac{\pi}{6}$ (D) $z = a\,\text{Cis}\,\frac{\pi}{3}$

Ans. D

Sol. Refer the figure z lies on the point of intersection of the rays from A and B. ΔACB is a right angle and OBC is an equilateral triangle

$\Rightarrow OC = a \Rightarrow z = a\,\text{Cis}\,\frac{\pi}{3} \Rightarrow$ (D)



14. Number of imaginary complex numbers satisfying the equation, $z^2 = \bar{z}\,2^{1-|z|}$ is :

समीकरण $z^2 = \bar{z}\,2^{1-|z|}$ को सन्तुष्ट करने वाले काल्पनिक सम्मिश्र हलों की संख्या :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $z^2 = \bar{z} \cdot 2^{1-|z|}$

$z^3 = |z|^2\, 2^{1-|z|}$ ....(1) $\Rightarrow$ $|z| = 2^{1-|z|}$

hence z^3 is purely +ve real (as $z \neq 0$) $\Rightarrow$ z is + ve real

hence $z = r\,e^{i\frac{2k\pi}{3}}$ $k = 0, 1, 2$

we therefore need to solve

$r = 2^{1-r} \Rightarrow$ $2^r = \frac{2}{r}$ $\Rightarrow$ $r = 1$

$\therefore$ $z = e^{i\frac{2k\pi}{3}}$

hence $z = 1, \omega, \omega^2$

but 1 is not imaginary

hence $z = w$ or w^2 $\Rightarrow$ (C)

15. If $\cos\theta + i\sin\theta$ is a root of the equation $x^n + a_1x^{n-1} + a_2x^{n-2} + ... + a_{n-1}x + a_n = 0$ then the value of $\sum_{r=1}^{n} a_r \cos r\theta$ equals : (where all coefficient are real)

(A) 0 (B) 1 (C) -1 (D) None

यदि $\cos\theta + i\sin\theta$ समीकरण $x^n + a_1x^{n-1} + a_2x^{n-2} + ... + a_{n-1}x + a_n = 0$ का मूल है तब $\sum_{r=1}^{n} a_r \cos r\theta$ का मान : (जहां सभी गुणांक वास्तविक है)

(A) 0 (B) 1 (C) -1 (D) कोई नहीं

Ans. C

Sol. Divide the equation by x^n and put $x = \cos\theta + i\sin\theta$.
Equate real and imaginary part

16. Given $f(z)$ = the real part of a complex number z. For example, $f(3 - 4i) = 3$. If $a \in N, n \in N$ then the value of $\sum_{n=1}^{6a} \log_2 \left| f\left(\left(1+i\sqrt{3}\right)^n\right)\right|$ has the value equal to :

दिया है $f(z)$ = सम्मिश्र संख्या z का वास्तविक भाग इस प्रकार कि $f(3 - 4i) = 3$. यदि $a \in N, n \in N$ तब $\sum_{n=1}^{6a} \log_2 \left| f\left(\left(1+i\sqrt{3}\right)^n\right)\right|$ का मान है :

(A) $18a^2 + 9a$ (B) $18a^2 + 7a$ (C) $18a^2 - 3a$ (D) $18a^2 - a$

Ans. D

Sol. $\left(1+i\sqrt{3}\right)^n = \left[2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)\right]^n = 2^n\left(\cos\frac{n\pi}{3} + i\sin\frac{n\pi}{3}\right)$

$f\left(\left(1+i\sqrt{3}\right)^n\right)$ = real part of $z = 2^n\cos\frac{n\pi}{3}$

$\therefore \quad \sum_{n=1}^{6a} \log_2\left|2^n\cos\frac{n\pi}{3}\right| = \sum_{n=1}^{6a}\left(n + \log_2\left|\cos\frac{n\pi}{3}\right|\right) = \frac{6a(6a+1)}{2} + \underbrace{(-1-1+0-1-1+0)}_{\text{a such term}}$

$= 3a(6a + 1) - 4a = 18a^2 - a$ **Ans.**

17. All complex numbers 'z' which satisfy the relation $|z-|z+1|| = |z+|z-1||$ on the complex plane lie on the :

(A) line $y = 0$ or an ellipse with foci $(-1, 0)$ and $(1, 0)$
(B) radical axis of the circles $|z - 1| = 1$ and $|z + 1| = 1$
(C) circle $x^2 + y^2 = 1$
(D) line $x = 0$ or on a line segment joining $(-1, 0)$ to $(1, 0)$

सम्बन्ध $\left|z-|z+1|\right| = \left|z+|z-1|\right|$ को सन्तुष्ट करने वाले सभी सम्मिश्र संख्याऐं :

(A) रेखा $y = 0$ या दीर्घवृत्त जिनके ध्रुव $(-1, 0)$ और $(1, 0)$ है पर

(B) वृत्तों $|z-1| = 1$ और $|z+1| = 1$ के मूलाक्ष पर

(C) वृत्त $x^2 + y^2 = 1$ पर

(D) रेखा $x = 0$ या $(-1, 0)$ से $(1, 0)$ को जोडने वाली रेखा खण्ड पर

Ans. D

Sol. Given $|z-|z+1||^2 = |z+|z-1||^2$

$\therefore \quad (z-|z+1|)(\overline{z}-|z+1|) = (z+|z-1|)(\overline{z}+|z-1|)$

$z\overline{z}-z|z+1|-\overline{z}|z+1|+|z+1|^2 = z\overline{z}+z|z-1|+\overline{z}|z-1|+|z-1|^2$

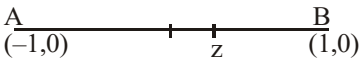
$|z+1|^2-|z-1|^2 = (z+\overline{z})\left[|z-1|+|z+1|\right]$

$(z+1)(\overline{z}+1)-(z-1)(\overline{z}-1) = (z+\overline{z})\left[|z-1|+|z+1|\right]$

$(z\overline{z}+z+\overline{z}+1)-(z\overline{z}-z-\overline{z}+1) = (z+\overline{z})\left[|z-1|+|z+1|\right]$

$2(z+\overline{z}) = (z+\overline{z})\left[|z+1|+|z-1|\right]$

$(z+\overline{z})\left[|z+1|+|z-1|-2\right] = 0$



$\Rightarrow$ either $z+\overline{z} = 0 \quad \Rightarrow$ z is purely imaginary

$\Rightarrow$ z lies on y axis $\Rightarrow \quad x = 0$

or $|z+1|+|z-1| = 2$

$\Rightarrow$ z lie on the line segment joining $(-1, 0)$ and $(1, 0) \quad \Rightarrow$ **(D)**

18. The locus of z which lies in shaded region is best represented by :

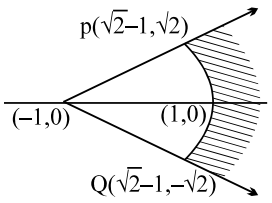
z का बिन्दुपथ जो छायांकित क्षेत्र (सीमाओं रहित) में स्थित हो, का श्रेष्ठ निरूपण है :

(A) $z : |z+1| > 2, |\arg(z+1)| < \pi/4$

(B) $z : |z-1| > 2, |\arg(z-1)| < \pi/4$

(C) $z : |z+1| < 2, |\arg(z+1)| < \pi/2$

(D) $z : |z-1| < 2, |\arg(z-1)| < \pi/2$



Ans. A

Sol. Obvious (A) $|z| \le 1$ is region interior of circle of radius 1 unit and centre (0, 0) and arg. of given region is $-\pi/4 \le \arg z \le \pi/4$

Hindi. स्पष्टतया तथा (A) $|z| \le 1$ इकाई त्रिज्या के वृत्त का आन्तरिक भाग है जिसका केन्द्र (0, 0) तथा दिए गए क्षेत्र का कोणांक $-\pi/4 \le$ कोणांक $z \le \pi/4$ है।

19. If $w = \alpha + i\beta$ where $\beta \neq 0$ and $z \neq 1$, satisfies the condition that $\frac{w - \overline{w}z}{1-z}$ is purely real, then the set of values of z is :

माना $w = \alpha + i\beta$, $\beta \neq 0$ और $z \neq 1$ यदि $\frac{w - \overline{w}z}{1-z}$ विशुद्ध वास्तविक हो, तो z के मानों का समुच्चय है :

(A) $\{z : |z| = 1\}$ (B) $\{z : z = \bar{z}\}$ (C) $\{z : z \neq 1\}$ (D) $\{z : |z| = 1, z \neq 1\}$

Ans. D

Sol. $\frac{w - \overline{w}z}{1-z} = \frac{\overline{w} - w\bar{z}}{1-\bar{z}}$

$w - w\bar{z} - \overline{w}z + \overline{w}z\bar{z} = \overline{w} - w\bar{z} - z\overline{w} + wz\bar{z}$

$\left(w - \overline{w}\right) = \left(w - \overline{w}\right)z\bar{z} \Rightarrow w - \overline{w} = 0$ or $|z^2| = 1$

$\frac{w - \overline{w}z}{1-z} = \frac{w(1-z)}{1-z}$ $z = 1$

$\Rightarrow z \neq 1$

ONE OR MORE THAN ONE CORRECT TYPE

20. If $|z_1 + z_2|^2 = |z_1|^2 + |z_z|^2$ then :

(A) $\frac{z_1}{z_2}$ is purely real (B) $\frac{z_1}{z_2}$ is purely imaginary

(C) $z_1\bar{z}_2 + z_2\bar{z}_1 = 0$ (D) amp $\frac{z_1}{z_2}$ may be equal to $\frac{\pi}{2}$

यदि $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$ हो, तो :

(A) $\frac{z_1}{z_2}$ विशुद्ध वास्तविक है (B) $\frac{z_1}{z_2}$ विशुद्ध काल्पनिक है

(C) $z_1\bar{z}_2 + z_2\bar{z}_1 = 0$ (D) amp $\frac{z_1}{z_2}$ का मान $\frac{\pi}{2}$ के बराबर हो सकता है

Ans. BCD

Sol. $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$

$z_1\bar{z}_2 + \bar{z}_1 z_2 = 0$

$\Rightarrow \quad \frac{z_1}{z_2} = -\frac{\bar{z}_1}{\bar{z}_2}$

$$\frac{z_1}{z_2}+\overline{\left(\frac{z_1}{z_2}\right)}=0 \Rightarrow \quad \frac{z_1}{z_2} \text{ is purely imaginary}$$

$$\text{so amp}\left(\frac{z_1}{z_2}\right) \text{ is may be } \frac{\pi}{2} \quad \text{or} \quad -\frac{\pi}{2}$$

Hindi $|z_1+z_2|^2=|z_1|^2+|z_2|^2$

$$z_1\bar{z}_2+\bar{z}_1z_2=0 \qquad \Rightarrow \qquad \frac{z_1}{z_2}=-\frac{\bar{z}_1}{\bar{z}_2}$$

$$\frac{z_1}{z_2}+\overline{\left(\frac{z_1}{z_2}\right)}=0 \qquad \Rightarrow \qquad \frac{z_1}{z_2} \text{ विशुद्ध काल्पनिक है।}$$

इसलिए $\text{amp}\left(\frac{z_1}{z_2}\right)$, $\frac{\pi}{2}$ के बराबर हो सकता है।

21. If $2\cos\theta = x+\frac{1}{x}$ and $2\cos\varphi = y+\frac{1}{y}$, then :

(A) $x^n+\frac{1}{x^n}=2\cos(n\theta)$ (B) $\frac{x}{y}+\frac{y}{x}=2\cos(\theta-\varphi)$

(C) $xy+\frac{1}{xy}=2\cos(\theta+\varphi)$ (D) None of these

यदि $2\cos\theta = x+\frac{1}{x}$ तथा $2\cos\varphi = y+\frac{1}{y}$ है, तो :

(A) $x^n+\frac{1}{x^n}=2\cos(n\theta)$ (B) $\frac{x}{y}+\frac{y}{x}=2\cos(\theta-\varphi)$

(C) $xy+\frac{1}{xy}=2\cos(\theta+\varphi)$ (D) इनमें से कोई नहीं

Ans. ABC

Sol. (A) $x=e^{i\theta}$ $y=e^{i\phi}$

$$x^n+\frac{1}{x^n}=2\cos n\theta$$

(B) $$\frac{x}{y}+\frac{y}{x}=e^{i(\theta-\phi)}+e^{-i(\theta-\phi)}=2\cos(\theta-\phi)$$

(C) $$xy+\frac{1}{xy}=2\cos(\theta+\varphi)$$

22. The locii of a point P(z) in the complex plane satisfying $\left|z+\frac{1}{z}\right|=2$ are two circles C_1 and C_2. These circles :

(A) have centres on real axis
(B) cut each other orthogonally
(C) are congruent
(D) have exactly two common tangents

बिन्दु P(z) का बिन्दुपथ, जोकि $\left|z+\frac{1}{z}\right|=2$ को सन्तुष्ट करता है, दो वृत्त C_1 और C_2 है तब ये वृत्तों :

(A) के केन्द्र वास्तविक अक्ष पर है
(B) एक दूसरे के लम्बवत प्रतिच्छेद करते है
(C) समान है
(D) की दो उभयनिष्ठ स्पर्श रेखाऐं है

Ans. BCD

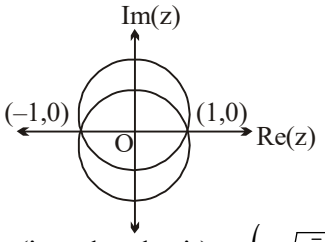
Sol. We have $\left|z+\frac{1}{z}\right|=2 . \Rightarrow \left|z+\frac{1}{z}\right|^2=4 \Rightarrow \left(z+\frac{1}{z}\right)\left(\bar{z}+\frac{1}{\bar{z}}\right)=4 \Rightarrow z\bar{z}+\frac{\bar{z}}{z}+\frac{z}{\bar{z}}+\frac{1}{z\bar{z}}=4$

$\Rightarrow z\bar{z}+\frac{\bar{z}^2+z^2}{z\bar{z}}+\frac{1}{z\bar{z}}=4 \quad \Rightarrow (z\bar{z})^2-4(z\bar{z})+z^2+\bar{z}^2+1=0$

$\Rightarrow (z\bar{z})^2-2z\bar{z}+1-2z\bar{z}+z^2+\bar{z}^2=0$

$\Rightarrow (z\bar{z}-1)^2+(z-\bar{z})^2=0 \Rightarrow (z\bar{z}-1)^2-i^2(z-\bar{z})^2=0$

$\Rightarrow \left(z\bar{z}-1+i(z-\bar{z})\right)\left(z\bar{z}-1-i(z-\bar{z})\right)=0$



Each of the above factors represents a circle with centre and radius (in ordered pair) as $\left(i,\sqrt{i\bar{i}+1}\right)$ and $\left(-i,\sqrt{i\bar{i}+1}\right)$ i.e. ((0, 1), $\sqrt{2}$) and ((0, – 1), $\sqrt{2}$) respectively.

The union of the circle with equations $x^2+y^2-2y-1=0$ & $x^2+y^2+2y-1=0$

Note : These circles cut each other orthogonally.

23. If 1, ω , ω^2,..... ω^{n-1} are the nth roots of unity, then $(2-\omega)(2-\omega^2).....(2-\omega^{n-1})$ equals :

(A) 2^n-1
(B) ${}^nC_1+{}^nC_2+...+{}^nC_n$
(C) $[{}^{2n+1}C_0+{}^{2n+1}C_1+....+{}^{2n+1}C_n]^{1/2}-1$
(D) None of these

यदि 1, ω , ω^2,..... ω^{n-1} इकाई के nवें मूल हैं, तो $(2-\omega)(2-\omega^2).....(2-\omega^{n-1})$ बराबर होगा :

(A) 2^n-1
(B) ${}^nC_1+{}^nC_2+...+{}^nC_n$
(C) $[{}^{2n+1}C_0+{}^{2n+1}C_1+....+{}^{2n+1}C_n]^{1/2}-1$
(D) इनमें से कोई नहीं

Ans. ABC

Sol. $1+x+x^2+...+x^{n-1}=(x-\omega)(x-\omega^2).....(x-\omega^{n-1})$

Put x = 2

$(2-\omega)(2-\omega^2).....(2-\omega^{n-1})=1+2+...+2^{n-1}$

$$= \frac{2^n - 1}{2-1} = 2^n - 1$$

$$= {}^nC_0 + {}^nC_1 + ... + {}^nC_n$$

(C) We know,

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + ... + {}^{2n+1}C_n = \frac{2^{2n+1}}{2}$$

$$= 2^{2n}$$

24. Which of the following represents a point on an argands' plane, equidistant from the roots of the equation $(z + 1)^4 = 16z^4$?

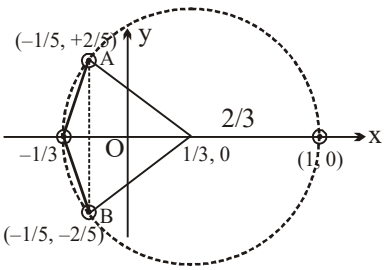
सम्मिश्र तल में निम्न में से कौनसा/से बिन्दु समीकरण $(z + 1)^4 = 16z^4$ के मूलों से समान दूरी पर है?

(A) $(0, 0)$ (B) $\left(-\frac{1}{3}, 0\right)$ (C) $\left(\frac{1}{3}, 0\right)$ (D) $\left(0, \frac{2}{\sqrt{5}}\right)$

Ans. C

Sol. $\left(\frac{z+1}{z}\right)^4 = 16 \Rightarrow \frac{z+1}{z} = 2 \text{ or } -2 \text{ or } 2i \text{ or } -2i$

Roots are $1 ; -\frac{1}{3} ; \left(-\frac{1}{5} - \frac{2}{5}i\right)$ and $\left(-\frac{1}{5} + \frac{2}{5}i\right)$



Note that $\left(-\frac{1}{3}, 0\right)$ and $(1, 0)$ are equidistant from $\left(\frac{1}{3}, 0\right)$ and since it lies on the perpendicular bisector of AB, it will be equidistant from A and B also.

Alternatively: $|z + 1| = 2|z|$

$$(z + 1)(\bar{z} + 1) = 4(z\bar{z}) \quad .....(1)$$

This is the equation of circle with centre (1/3, 0) which is equidistant from the root of the equation.

25. Let z_1, z_2, z_3 are the coordinates of the vertices of the triangle $A_1A_2A_3$. Which of the following statements are equivalent :

(A) $A_1A_2A_3$ is an equilateral triangle.

(B) $(z_1 + \omega z_2 + \omega^2 z_3)(z_1 + \omega^2 z_2 + \omega z_3) = 0$, where ω is the imaginary cube root of unity.

(C) $\frac{z_2 - z_1}{z_3 - z_2} = \frac{z_3 - z_2}{z_1 - z_3}$

(D) $\begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ z_2 & z_3 & z_1 \end{vmatrix} = 0$

माना z_1, z_2, z_3, त्रिभुज $A_1A_2A_3$ के शीर्ष है तब निम्न में से कौनसे कथन समरूप है :

(A) $A_1A_2A_3$ एक समबाहु त्रिभुज

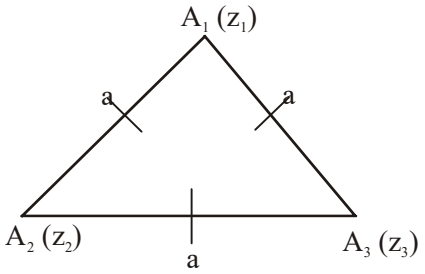
(B) $(z_1 + \omega z_2 + \omega^2 z_3)(z_1 + \omega^2 z_2 + \omega z_3) = 0$, जहाँ ω इकाई का काल्पनिक धनमूल हैं

(C) $\dfrac{z_2 - z_1}{z_3 - z_2} = \dfrac{z_3 - z_2}{z_1 - z_3}$

(D) $\begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ z_2 & z_3 & z_1 \end{vmatrix} = 0$

Ans. ABCD

Sol.



$$\frac{z_2 - z_1}{a} = \frac{z_3 - z_1}{a} e^{\frac{i\pi}{3}}$$

$$z_2 - z_1 = (z_3 - z_2) e^{\frac{i\pi}{3}} \qquad ...(1)$$

Similarly, $z_3 - z_2 = (z_1 - z_3) e^{\frac{i\pi}{3}}$ ...(2)

divide (1) by (2)

$$\frac{z_2 - z_1}{z_3 - z_2} = \frac{z_3 - z_2}{z_1 - z_3}$$

(B) (D)

For equilateral Δ, we know

$$z_1^2 + z_2^2 + z_3^2 - z_1z_2 - z_2z_3 - z_1z_3 = 0$$

26. If $\alpha_1, \alpha_2, \alpha_3, ..., \alpha_{n-1}$ are the imaginary n^{th} roots of unity then the product $\prod_{r=1}^{n-1}(i-\alpha_r)$ (where $i=\sqrt{-1}$) can take the value equal to :

यदि $\alpha_1, \alpha_2, \alpha_3, ..., \alpha_{n-1}$ काल्पनिक तथा साथ ही इकाई के n वें मूल तब युग्म $\prod_{r=1}^{n-1}(i-\alpha_r)$ (जहां $i=\sqrt{-1}$) का मान हो सकता है :

(A) 0 (B) 1 (C) i (D) $(1+i)$

Ans. ABCD

Sol. $\frac{z^n-1}{z-1} = (z-\alpha_1)(z-\alpha_2).....(z-\alpha_{n-1})$

put $z=i$

$$\prod_{r=1}^{n-1}(i-\alpha_r)=\frac{i^n-1}{i-1}=\begin{cases}0 & \text{if} \quad n=4k\\ 1 & \text{if} \quad n=4k+1\\ 1+i & \text{if} \quad n=4k+2\\ i & \text{if} \quad n=4k+3\end{cases}$$

27. Let tangents at $A(z_1)$ and $B(z_2)$ are drawn to the circle $|z|=2$. Then which of the following is/are CORRECT?

(A) The equation of tangent at A is given by $\frac{z}{z_1}+\frac{\bar{z}}{\bar{z}_1}=2$.

(B) If tangents at $A(z_1)$ and $B(z_2)$ intersect at $P(z_p)$, then $z_p=\frac{2z_1z_2}{z_1+z_2}$.

(C) Slope of tangent at $A(z_1)$ is $\frac{1}{i}\left(\frac{z_1+\bar{z}_1}{z_1-\bar{z}_1}\right)$.

(D) If points $A(z_1)$ and $B(z_2)$ on the circle $|z|=2$ are such that $z_1+z_2=0$, then tangents intersect at $\frac{\pi}{2}$

यदि वृत्त $|z|=2$ पर $A(z_1)$ और $B(z_2)$ से स्पर्श रेखाऐं डाली जाती है तब निम्न में से सत्य कथन चुनिये?

(A) A पर स्पर्श रेखा का समीकरण $\frac{z}{z_1}+\frac{\bar{z}}{\bar{z}_1}=2$

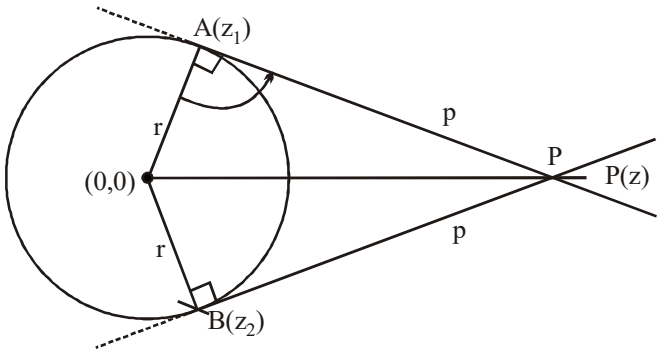
(B) यदि $A(z_1)$ और $B(z_2)$ पर स्पर्श रेखाऐं $P(z_p)$ पर प्रतिच्छेद करती है, तब $z_p=\frac{2z_1z_2}{z_1+z_2}$

(C) $A(z_1)$ पर स्पर्श रेखा की प्रवणता $\frac{1}{i}\left(\frac{z_1+\bar{z}_1}{z_1-\bar{z}_1}\right)$ हैं

(D) यदि $|z|=2$ पर बिन्दु $A(z_1)$ और $B(z_2)$ इस प्रकार है कि $z_1+z_2=0$ तब स्पर्श रेखा $\frac{\pi}{2}$ पर कट करती है

Ans. ABC

Sol. **(A)** $\arg\left(\frac{z-z_1}{0-z_1}\right)=\frac{\pi}{2}$



$\Rightarrow \frac{z-z_1}{z_1}$ is purely imaginary.

So, $\frac{z_1-z}{z_1}+\frac{\overline{z}_1-\overline{z}}{\overline{z}_1}=0$

$\Rightarrow \quad \frac{z}{z_1}+\frac{\overline{z}}{\overline{z}_1}=2$

(B) By applying rotation, we get (AP = p)

$\frac{z-z_1}{0-z_1}=\frac{p}{r}e^{\frac{i\pi}{2}}$ ....(1)

Also $\frac{0-z_2}{z-z_2}=\frac{r}{p}e^{\frac{i\pi}{2}}$ ....(2)

$\therefore$ On multiplying (1) and (2), we get $z_p=\frac{2z_1z_2}{z_1-z_2}$

(C) As equation of tangent at $A(x_1, y_1)$ is
$xx_1+yy_1=4$

$\therefore$ Slope of tangent $=\frac{-x_1}{y_1}=\frac{-2x_1}{2y_1}=-i\left(\frac{z_1+\overline{z}_1}{z_1-\overline{z}_1}\right)=\frac{1}{i}\left(\frac{z_1+\overline{z}_1}{z_1-\overline{z}_1}\right)$

(D) Clearly tangents are parallel lines.
(As $A(z_1)$ and $B(z_2)$ are ends of diameter of circle.)

Exercise C - II

1. Let z_1, z_2, z_3 are three pair wise distinct complex numbers and t_1, t_2, t_3 are non-negative real numbers such that $t_1 + t_2 + t_3 = 1$. Prove that the complex number $z = t_1z_1 + t_2z_2 + t_3z_3$ lies inside a triangle with vertices z_1, z_2, z_3 or on its boundry.

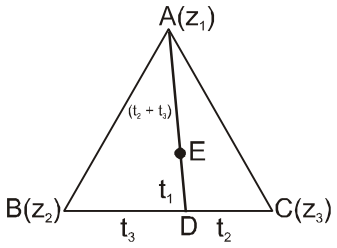
माना z_1, z_2, z_3 तीन भिन्न सम्मिश्र संख्यायें है तथा t_1, t_2, t_3 अऋणात्मक वास्तविक संख्यायें इस प्रकार है कि $t_1 + t_2 + t_3 = 1$ है। सिद्ध कीजिये सम्मिश्र संख्या $z = t_1z_1 + t_2z_2 + t_3z_3$ त्रिभुज z_1, z_2, z_3 के अन्दर अथवा उसकी सीमाओं पर स्थित है।

Ans.

Sol. Affixes of a point D which divides z_2, z_3 in the ratio $t_3 : t_2$ is $\frac{t_2z_2 + t_3z_3}{t_2 + t_3}$ (Internal division)

Affies of a point E which divides. AD in the ratio

$$(t_2 + t_3) : t_1 \text{ is } \left(\frac{t_1z_1 + t_2z_2 + t_3z_3}{t_1 + t_2 + t_3}\right) \quad \text{(Internal division)}$$



Hence E always lies in or on the ΔABC

2. If A, B and C are the angles of a triangle

$$D = \begin{vmatrix} e^{-2iA} & e^{iC} & e^{iB} \\ e^{iC} & e^{-2iB} & e^{iA} \\ e^{iB} & e^{iA} & e^{-2iC} \end{vmatrix} \text{ where } i = \sqrt{-1} \text{ then find the value of D.}$$

यदि A, B तथा C त्रिभुज के कोण है।

$$D = \begin{vmatrix} e^{-2iA} & e^{iC} & e^{iB} \\ e^{iC} & e^{-2iB} & e^{iA} \\ e^{iB} & e^{iA} & e^{-2iC} \end{vmatrix} \text{ जहाँ } i = \sqrt{-1} \text{ तो D का मान ज्ञात कीजिए।}$$

Ans. –4

Sol. $A + B + C = \pi$

$$D = \begin{vmatrix} e^{-i2A} & e^{iC} & e^{iB} \\ e^{iC} & e^{-i2B} & e^{iA} \\ e^{iB} & e^{iA} & e^{-i2C} \end{vmatrix}$$

$= e^{-i2A}\{e^{-i2(B+C)} - e^{i2A}\} - e^{iC}\{e^{-iC} - e^{i(A+B)}\} + e^{iB}\{e^{i(A+C)} - e^{-iB}\}$

$\Rightarrow -(1) + 2e^{i(A+B+C)} - 1$

$\Rightarrow -2 + 2e^{i\pi}$

$\Rightarrow -2-2$

$\Rightarrow -4$

3. If $x = 1+i\sqrt{3}$; $y = 1-i\sqrt{3}$ and $z = 2$, then prove that $x^p + y^p = z^p$ for every prime $p > 3$.

यदि $x = 1+i\sqrt{3}$; $y = 1-i\sqrt{3}$ तथा $z = 2$ तो सिद्ध कीजिए $x^p + y^p = z^p$ है हर एक अभाज्य $p > 3$ के लिए।

Ans.

Sol. $\left(\frac{x}{z}\right)^p + \left(\frac{y}{z}\right)^p = 1$

$(-w^2)^p + (-w)^p = 1$

$0 = 1 + w^p + w^{2p}$

p is prime so not a multiple of 3.

4. Prove that there exists no complex number z such that $|z| < \frac{1}{3}$ and $\sum_{r=1}^{n} a_r z^r = 1$ where $|a_r| < 2$.

सिद्ध कीजिए कि ऐसी कोई सम्मिश्र संख्या नहीं है ताकि $|z| < \frac{1}{3}$ और $\sum_{r=1}^{n} a_r z^r = 1$ जहाँ $|a_r| < 2$ हो।

Ans.

Sol. $1 = |\Sigma a_i z^i| \le \Sigma |a_i z^i|$

$\Rightarrow \quad 1 \le |a_1 z| + |a_2 z^2| + |a_3 z^3| + \ldots\ldots\ldots + |a_n z^n|$

$\Rightarrow \quad 1 < 2\,(|z| + |z|^2 + |z|^3 + \ldots\ldots\ldots + |z|^n)$

$\Rightarrow \quad 1 + |z| + |z|^2 + |z|^3 + \ldots\ldots\ldots\ldots + |z|^n > 3/2$

Case - I $\quad |z| < 1$

$\Rightarrow \quad 1 + |z| + |z|^2 + \ldots\ldots\ldots\ldots \infty > 3/2$

$\Rightarrow \quad \frac{1}{1-|z|} > \frac{3}{2} \quad \Rightarrow \quad 2 > 3 - 3|z|$

$\Rightarrow \quad |z| > 1/3$

Case - II $\quad |z| \ge 1$, then

Obviously $|z| < 1/3$ is not possible.

Hence, $|z| < 1/3$ and $\sum_{i=1}^{n} a_i z^i = 1$ cannot occur simultaneously for any a_i, $|a_i| < 2$.

5. Let $Z = 18 + 26i$ where $Z_0 = x_0 + iy_0$ $(x_0, y_0 \in R)$ is the cube root of Z having least positive argument. Find the value of $x_0 y_0 (x_0 + y_0)$.

यदि $Z = 18 + 26i$ जहाँ $Z_0 = x_0 + iy_0$ $(x_0, y_0 \in R)$, Z का घनमूल है जिसका न्यूनतम धनात्मक कोणांक है तो $x_0y_0(x_0 + y_0)$ का मान ज्ञात करें।

Ans. 12

Sol. $Z = 18 + 26i$

$= 10\sqrt{10}e^{i(\theta)}$ $\qquad \theta = \tan^{-1}\frac{26}{18}$

$Z = \left(10\sqrt{10}\right)^{\frac{1}{3}} e^{i\left(\frac{2K\pi+\theta}{3}\right)} = \tan^{-1}\frac{13}{9}$

$K = 0$

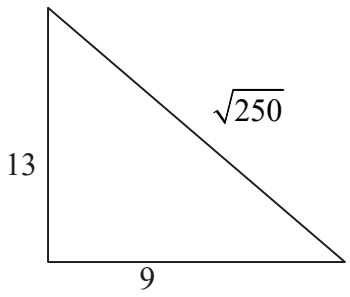
$\Rightarrow Z = \left(10\sqrt{10}\right)^{\frac{1}{3}} e^{i\frac{\theta}{3}}$

$= \left(10\sqrt{10}\right)^{\frac{1}{3}} \left\{\cos\frac{\theta}{3} + i\sin\frac{\theta}{3}\right\}$

$x_0 = \left(10\sqrt{10}\right)^{\frac{1}{3}} \cos\frac{\theta}{3}$

$y_0 = \left(10\sqrt{10}\right)^{\frac{1}{3}} \sin\frac{\theta}{3}$

Now $\tan\theta = \frac{13}{9}$



$\frac{3\tan\frac{\theta}{3} - \tan^3\frac{\theta}{3}}{1 - 3\tan^2\frac{\theta}{3}} = \frac{13}{9}$

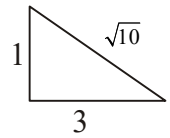
Let $\tan\frac{\theta}{3} = x$

$9(3x - x^3) = 13(1 - 3x^2)$

$27x - 9x^3 = 13 - 39x^2$

$9x^3 - 39x^2 - 27x + 13 = 0$

$\tan\frac{\theta}{3} = \frac{1}{3}$



$\Rightarrow \cos\frac{\theta}{3} = \frac{3}{\sqrt{10}}$

$\sin\frac{\theta}{3} = \frac{1}{\sqrt{10}}$

$x_0 = \left(10\sqrt{10}\right)^{\frac{1}{3}} \cdot \frac{3}{\sqrt{10}}$

$y_0 = \left(10\sqrt{10}\right)^{\frac{1}{3}} \cdot \frac{1}{\sqrt{10}}$

$x_0 y_0 (x_0 + y_0)$

$\left(10\sqrt{10}\right)^{\frac{2}{3}} \cdot \frac{3}{10}\left[\frac{4}{\sqrt{10}}\right]\left(10\sqrt{10}\right)^{\frac{1}{3}}$

$\Rightarrow \frac{10\sqrt{10}}{10\sqrt{10}} = 12$

6. If ω is the fifth root of 2 and $x = \omega + \omega^2$, prove that $x^5 = 10x^2 + 10x + 6$.

यदि ω , 2 का पांचवाँ मूल है तथा $x = \omega + \omega^2$ सिद्ध कीजिये कि $x^5 = 10x^2 + 10x + 6$.

Ans.

Sol. $\omega = 2^{\frac{1}{5}}$

$\omega^5 = 2$

$x^5 = (\omega + \omega^2)^5$

$= \omega^5 (\omega + 1)^5$

$= 2\left[{}^5C_0\omega^5 + {}^5C_1\omega^4 + {}^5C_2\omega^3 + {}^5C_3\omega^2 + {}^5C_4\omega + {}^5C_5\right]$

$= 2\left[2 + 5\omega^4 + 10\omega^3 + 10\omega^2 + 5\omega + 1\right]$

$= 2\left[5\left(\omega^4 + 2\omega^3 + \omega^2\right) + 5\left(\omega^2 + \omega\right) + 3\right]$

$= 2\,[5x^2 + 5x + 3] \qquad = 10x^2 + 10x + 6$

7. Let $z_1, z_2 \in C$ such that $z_1^2 + z_2^2 \in R$. If $z_1(z_1^2 - 3z_2^2) = 10$ and $z_2(3z_1^2 - z_2^2) = 30$, Find the value of $(z_1^2 + z_2^2)$.

यदि $z_1, z_2 \in C$ इस प्रकार है कि $z_1^2 + z_2^2 \in R$. यदि $z_1(z_1^2 - 3z_2^2) = 10$ तथा $z_2(3z_1^2 - z_2^2) = 30$ है, तो $(z_1^2 + z_2^2)$ का मान ज्ञात कीजिए।

Ans. 10

Sol. $z_1, z_2 \in C$

$z_1^2 + z_2^2 \in R$

$z_1^3 - 3z_1z_2^2 = 10 \;\&\; z_2\left(3z_1^2 - z_2^2\right) = 30$

$z_1^3 - 3z_1z_2^2 + iz_2 3z_1^2 + iz_2^3 = 10 + 30i$

$\left[z_1 + iz_2\right]^3 = 10 + 30i$ _____(A)

$\&\ \left[z_1 - iz_2\right]^3 = 10 - 30i$ _____(B)

multiply (A) & (B)

$\left\{\left(z_1 + iz_2\right)\left(z_1 - iz_2\right)\right\}^3 = 1000$

$z_1^2 + z_2^2 = 10$

8. If the equation $(z + 1)^7 + z^7 = 0$ has roots $z_1, z_2, ... z_7$, find the value of

(a) $\sum_{r=1}^{7} \mathrm{Re}(Z_r)$ and (b) $\sum_{r=1}^{7} \mathrm{Im}(Z_r)$

यदि समीकरण $(z + 1)^7 + z^7 = 0$ के मूल $z_1, z_2, ... z_7$, है तो मान ज्ञात कीजिए।

(a) $\sum_{r=1}^{7} \mathrm{Re}(Z_r)$ तथा (b) $\sum_{r=1}^{7} \mathrm{Im}(Z_r)$

Ans. (a) $-\frac{7}{2}$, (b) zero

Sol. $(z+1)^7 + z^7 = 0$

$\Rightarrow (z+1)^7 = -z^7$

$|z+1|^7 = |-z|^7$

$|z+1| = |z|$

$\Rightarrow$ z is equidistant from (0,0) & (–1,0)

So lies on $x = -\frac{1}{2}$

So all complex roots have real part $-\frac{1}{2}$

So $z_1 + z_2 + z_3 + z_4 + z_5 + z_6 + z_7 = -\frac{7}{2}$ and imaginary parts will be zero.

Thus $\sum_{r=1}^{7} \text{Im}(z_r) = 0$

9. If the expression $z^5 - 32$ can be factorised into linear and quadratic factors over real coefficients as $(z^5 - 32) = (z-2)(z^2 - pz + 4)(z^2 - qz + 4)$ then find the value of $(p^2 + 2p)$.

यदि व्यंजक $z^5 - 32$ को रेखीय तथा द्विघात गुणनखण्डों में (जिसके गुणांक वास्तविक है) में व्यक्त किया गया है $(z^5 - 32) = (z-2)(z^2 - pz + 4)(z^2 - qz + 4)$ तब $(p^2 + 2p)$ का मान ज्ञात करों।

Ans. 4

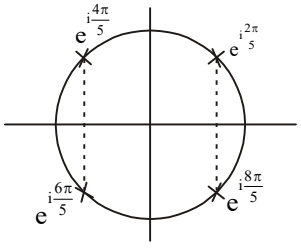
Sol. $z^5 - 32 = 0$

$z^5 = 2^5$

$= 25\ e^{i2k\pi}$

$z = 2e^{\frac{i2k\pi}{5}}$

K = 0,1,2,3,4



$\Rightarrow z^5 - 32 = (z-2)\left(z - 2e^{i\frac{2\pi}{5}}\right)\left(z - 2e^{i\frac{4\pi}{5}}\right)\left(z - 2e^{i\frac{6\pi}{5}}\right)\left(z - 2e^{i\frac{8\pi}{5}}\right)$

$$=(z-2)\left\{z-2e^{i\frac{2\pi}{5}}\right\}\left\{z-2e^{-i\frac{2\pi}{5}}\right\}\left\{z-2e^{i\frac{4\pi}{5}}\right\}\left\{z-2e^{-i\frac{4\pi}{5}}\right\}$$

$$=(z-2)\left\{z^2-2ze^{-i\frac{2\pi}{5}}-2ze^{i\frac{2\pi}{5}}+4\right\}\left\{z^2-2ze^{-i\frac{4\pi}{5}}-2ze^{i\frac{4\pi}{5}}+4\right\}$$

$$p=2\left(e^{-i\frac{2\pi}{5}}+e^{i\frac{2\pi}{5}}\right)$$

$$=2\left\{2\cos\frac{2\pi}{5}\right\}$$

$$\Rightarrow 4\cos\frac{2\pi}{5}$$

$$q=4\cos\frac{4\pi}{5}$$

$p^2 + 2p$

$$16\cos^2\frac{2\pi}{5}+2\cdot4\cos\frac{2\pi}{5}$$

$$16\cos^2\frac{2\pi}{5}+8\cos\frac{2\pi}{5}$$

$$8\left[2\cos^2\frac{2\pi}{5}+\cos\frac{2\pi}{5}\right]$$

$$8\left[2\sin^2 18^\circ+\sin 18^\circ\right]$$

$$8\left[1-\cos 36^\circ+\sin 18^\circ\right]$$

$$8\left[1+\frac{\sqrt{5}-1}{4}-\frac{\sqrt{5}+1}{4}\right]=4$$

10. Let z_i $(i=1,2,3,4)$ represent the vertices of a square all of which lie on the sides of the triangle with vertices (0,0), (2,1) and (3, 0). If z_1 and z_2 are purely real, then area of triangle formed by z_3, z_4 and origin is $\frac{m}{n}$ (where m and n are in their lowest form). Find the value of (m + n).

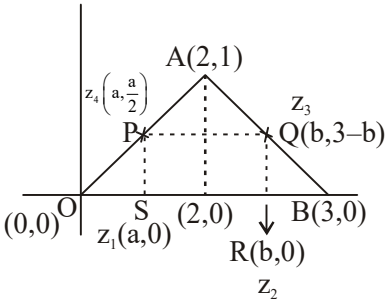
यदि z_i $(i = 1, 2, 3, 4)$ एक वर्ग के शीर्ष को व्यक्त करते है जिनमें से सभी एक त्रिभुज जिसके शीर्ष (0,0), (2,1) तथा (3, 0) की भुजाओं पर है यदि z_1 तथा z_2 पूर्णतया वास्तविक है तब z_3 , z_4 और मूलबिन्दु से बनने वाले त्रिभुज का क्षेत्रफल $\frac{m}{n}$ (जहां m तथा n न्यूनतम समकक्ष रूप में है). $(m + n)$ का मान ज्ञात करों।

Ans. 41

Sol. Equation of line OA

$$y = \frac{1}{2}x$$

$$\therefore \text{ co-ordinates of } P\left[a, \frac{a}{2}\right]$$



Equation of line AB

$$y = \frac{1}{-1}(x - 3)$$

Coordinates of Q (b, 3 – b)

Now $\frac{a}{2} = 3 - b$ _____(1)

$b - a = \frac{a}{2}$

$2b = 3a$ _____(2)

Solve (1) and (2)

$\therefore a = \frac{3}{2}, b = \frac{9}{4}$ $\Rightarrow \text{side} = \frac{3}{4}$

$$\text{Ar. of } \Delta PQO = \frac{1}{2}\left(\frac{3}{4}\right) \times \frac{3}{4}$$

$$= \frac{9}{32}$$

$$\therefore \frac{9}{32} = m + n$$

$m + n = 41$

11. (i) Let C_r's denotes the combinatorial coefficients in the expansion of $(1+x)^n$, $n \in N$. If the integers

$$a_n = C_0 + C_3 + C_6 + C_9 + ...$$
$$b_n = C_1 + C_4 + C_7 + C_{10} + ...$$
and $$c_n = C_2 + C_5 + C_8 + C_{11} + ..., \quad \text{then}$$

prove that (a) $a_n^3 + b_n^3 + c_n^3 - 3a_n b_n c_n = 2^n$, (b) $(a_n - b_n)^2 + (b_n - c_n)^2 + (c_n - a_n)^2 = 2$.

(ii) Prove the identity: $(C_0 - C_2 + C_4 - C_6 + ...)^2 + (C_1 - C_3 + C_5 - C_7 + ...)^2 = 2^n$

(i) यदि Cr's, $(1+x)^n$, $n \in N$ के प्रसार में द्विपद गुणांक है यदि पूर्णांक

$$a_n = C_0 + C_3 + C_6 + C_9 + ...$$
$$b_n = C_1 + C_4 + C_7 + C_{10} + ...$$
तथा $$c_n = C_2 + C_5 + C_8 + C_{11} + ..., \quad \text{तो}$$

सिद्ध करो (a) $a_n^3 + b_n^3 + c_n^3 - 3a_n b_n c_n = 2^n$, (b) $(a_n - b_n)^2 + (b_n - c_n)^2 + (c_n - a_n)^2 = 2$.

(ii) सर्वसमिका सिद्ध करो : $(C_0 - C_2 + C_4 - C_6 + ...)^2 + (C_1 - C_3 + C_5 - C_7 + ...)^2 = 2^n$

Ans.

Sol. $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + ... + {}^nC_n x^n$

Let $x = \omega$

$(1+\omega)^n = {}^nC_0 + {}^nC_1 \omega + {}^nC_2 \omega^2 + {}^nC_3 + ... + {}^nC_n \omega^n = (-\omega^2)^n$ _____(1)

$x = \omega^2$

$(1+\omega^2)^n = {}^nC_0 + {}^nC_1 \omega^2 + {}^nC_2 \omega^4 + {}^nC_3 + ... = (-\omega)^n$ _____(2)

Now let $x = 1$

$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + ... + {}^nC_n$ _____(3)

Add (1), (2) & (3)

$$\left(1+\omega\right)^n + \left(1+\omega^2\right)^n + 2^n = 3\left\{{}^nC_0 + {}^nC_3 + ...\right\}$$

$$\Rightarrow {}^nC_0 + {}^nC_3 + {}^nC_6 + ... = \frac{\left(-\omega^2\right)^n + \left(-\omega\right)^n + 2^n}{3} = a_n$$

Similarly, $\omega^2 \times (1) + \omega \times (2) + (3)$

$$b_n = \frac{2^n + \omega \times \left(-\omega\right)^n + \omega^2 \times \left(-\omega^2\right)^n}{3}$$

$$\text{and } c_n = \frac{2^n + \omega^2 \times \left(-\omega\right)^n + \omega \times \left(-\omega^2\right)^n}{3}$$

We know

$a_n^3 + b_n^3 + c_b^3 - 3a_n b_n c_n = (a_n + b_n + c_n)(a_n + b_n \omega + c_n \omega^2)(a_n + b_n \omega^2 + c_n \omega)$

$$\left(\frac{3\cdot 2^n}{3}\right)\times\left(\frac{3\left(-\omega^2\right)^n}{3}\right)\times\left(\frac{3\left(-\omega\right)^n}{3}\right)$$

$$=2^n\cdot\left(\omega^3\right)^n=2^n\left(1\right)^n=2^n$$

(b) $a_n^3+b_n^3+c_n^3-3a_nb_nc_n=2^n$

$$\Rightarrow\left(a_n+b_n+c_n\right)\left(a_n^2+b_n^2+c_n^2-a_nb_n-b_nc_n-c_na_n\right)$$

$$\Rightarrow\left(a_n+b_n+c_n\right)\frac{1}{2}\left\{\left(a_n-b_n\right)^2+\left(b_n-c_n\right)^2+\left(c_n-a_n\right)^2\right\}$$

Now $a^n + b^n + c^n = 2^n$

Put $x = 1$

$2^n = C_0 + C_1 + C_2 + C_3 + ...$

$$\therefore 2^n=2^n\cdot\frac{1}{2}\left\{\left(a_n-b_n\right)^2+\left(b_n-c_n\right)^2+\left(c_n-a_n\right)^2\right\}$$

$$\Rightarrow\left(a_n-b_n\right)^2+\left(b_n-c_n\right)^2+\left(c_n-a_n\right)^2=2$$

12. Let z_1, z_2, z_3, z_4 be the vertices A, B, C, D respectively of a square on the Argand diagram taken in anticlockwise direction then prove that :

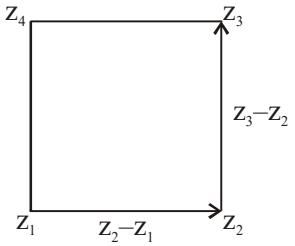
(i) $2z_2=(1+i)z_1+(1-i)z_3$ and (ii) $2z_4=(1-i)z_1+(1+i)z_3$

यदि z_1, z_2, z_3, z_4 सम्मिश्र तल में बने हुये वर्ग के शीर्षो A, B, C, D जो कि वामावर्ती दिशा में लिये गये है, को व्यक्त करते है तो सिद्ध कीजिये :

(i) $2z_2=(1+i)z_1+(1-i)z_3$ तथा (ii) $2z_4=(1-i)z_1+(1+i)z_3$

Ans.

Sol. (i)



$$z_3-z_2=\left(z_2-z_1\right)e^{\frac{i\pi}{2}}$$

$$=(z_2-z_1)i$$

$$z_3-z_2=z_2i-z_1i$$

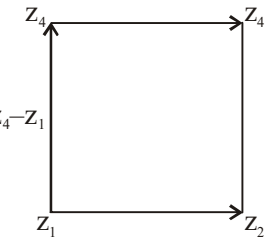
$$z_2(1+i)=z_3+z_1i$$

$$z_2 = \frac{z_3}{1+i} + \frac{z_1 i}{1+i}$$

$$z_2 = \frac{z_3(1-i)}{2} + \frac{z_1 i(1+i)}{(1+i)^2}$$

$$= \frac{z_3(1-i)}{2} + \frac{z_1 i(1+i)}{2i}$$

$$2z_2 = z_3(1-i) + z_1(1+i)$$

(ii) 

$$z_4 - z_1 = (z_2 - z_1)e^{\frac{i\pi}{2}}$$

$$z_4 = (z_2 - z_1)i + z_1$$

$$z_4 = z_2 i + z_1(1-i) \qquad ...(1)$$

Again

$$z_2 - z_1 = z_3 - z_4$$

$$z_2 = z_3 - z_4 + z_1 \qquad ...(2)$$

from (1) & (2)

$$z_4 = (z_3 - z_4 + z_1)i + z_1(1-i)$$

$$z_4(1+i) = z_3 i + z_1(1)$$

$$z_4 = \frac{z_3 i}{1+i} + \frac{z_1}{1+i}$$

$$= \frac{z_3 i(1+i)}{2i} + \frac{z_1(1-i)}{2}$$

$$2z_4 = z_3(1+i) + z_1(1-i)$$

13. Prove that :

(a) $\cos x + {}^nC_1 \cos 2x + {}^nC_2 \cos 3x + ... + {}^nC_n \cos (n + 1) x = 2^n . \cos^n \frac{x}{2} . \cos \left(\frac{n + 2}{2}\right) x$

(b) $\sin x + {}^nC_1 \sin 2x + {}^nC_2 \sin 3x + ... + {}^nC_n \sin (n + 1) x = 2^n . \cos^n \frac{x}{2} . \sin \left(\frac{n + 2}{2}\right) x$

सिद्ध कीजिये :

(a) $\cos x + {}^nC_1 \cos 2x + {}^nC_2 \cos 3x + ... + {}^nC_n \cos (n + 1) x = 2^n . \cos^n \frac{x}{2} . \cos \left(\frac{n + 2}{2}\right) x$

(b) $\sin x + {}^nC_1 \sin 2x + {}^nC_2 \sin 3x + ... + {}^nC_n \sin (n + 1) x = 2^n . \cos^n \frac{x}{2} . \sin \left(\frac{n + 2}{2}\right) x$

Ans.

Sol. $(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + ... + {}^nC_n x^n$

Put $x = e^{i\theta}$

$[1 + e^{i\theta}]^n = {}^nC_0 + {}^nC_1 e^{i\theta} + {}^nC_2 e^{i2\theta} + ... + {}^nC_n e^{in\theta}$

$e^{i\theta}(1 + e^{i\theta})^n = {}^nC_0 e^{i\theta} + {}^nC_1 e^{i2\theta} + {}^nC_2 e^{i3\theta} + ... + {}^nC_n e^{i(n + 1)\theta}$

$= \{{}^nC_0 \cos\theta + {}^nC_1 \cos2\theta + {}^nC_2 \cos3\theta + ... + {}^nC_n \cos(n+1)\theta\} + 1\{{}^nC_0 \sin\theta + {}^nC_1 \sin2\theta + ... + {}^nC_n \sin(n+1)\theta\}$

Now $e^{i\theta}\left[1+e^{i\theta}\right]^n$

$e^{i\theta}\left[\left(1+\cos\theta\right)+i\sin\theta\right]^n$

$e^{i\theta}\left[2\cos^2\frac{\theta}{2}+i2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right]^n$

$e^{i\theta}2^n\cos^n\frac{\theta}{2}\left\{\cos\frac{\theta}{2}+i\sin\frac{\theta}{2}\right\}^n$

$2^n\cos^n\frac{\theta}{2}\left(e^{\frac{i\theta}{2}}\right)^n\cdot e^{i\theta}$

$2^n\cos^n\frac{\theta}{2}e^{i\theta\left[\frac{n}{2}+1\right]}$

14. Let $f(x) = ax^3 + bx^2 + cx + d$ be a cubic polynomial with real coefficients satisfying $f(i) = 0$ and $f(1 + i) = 5$. Find the value of $a^2 + b^2 + c^2 + d^2$.

माना $f(x) = ax^3 + bx^2 + cx + d$ एक तीन घातीय बहुपद है जिसके गुणांक वास्तविक है, तो $f(i) = 0$ तथा $f(1 + i) = 5$ है। $a^2 + b^2 + c^2 + d^2$ का मान ज्ञात करें।

Ans. 26

Sol. $f(x) = ax^3 + bx^2 + cx + d \begin{cases} i \\ -i \\ \alpha \end{cases}$

$$ax^3 + bx^2 + cx + d = a[x + i][x - i][x - \alpha]$$

$$\Rightarrow f(x) = a(x + i)(x - i)(x - \alpha)$$

$$f(1+i) = a(1+i+i)(1+i-i)(1+i-\alpha) = 5$$

$$a(1+2i)(1+i-\alpha) = 5$$

$$a[1+i-\alpha+2i-2-2\alpha i] = 5$$

$$a[-1+3i-\alpha-2\alpha i] = 5$$

$$[-a-a\alpha-5]+i[3a-2\alpha a] = 0$$

$$-a-a\alpha-5 = 0$$

$$a(1+\alpha) = -5 \qquad ...(1)$$

$$a(3-2\alpha) = 0$$

$$a \neq 0, \alpha = \frac{3}{2}$$

$$a = \frac{-5 \times 2}{5}$$

$$a = -2$$

$$\Rightarrow f(x) = -2(x + i)(x - i)\left(x - \frac{3}{2}\right)$$

$= -2\left(x^2+1\right)\left(x-\frac{3}{2}\right)$

$f(x) = -2\left\{x^3 - \frac{3}{2}x^2 + x - \frac{3}{2}\right\}$

$= -2x^2 + 3x^2 - 2x + 3$

$a = -2$ $\qquad a^2 + b^2 + c^2 + d^2 = 8 + 18 = 26$

$b = 3$

$c = -2$

$d = 3$

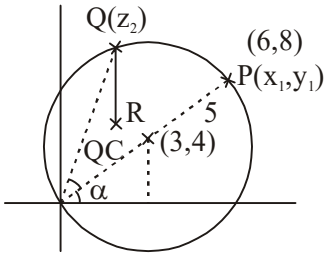
15. A particle starts to travel from a point P on the curve $C_1 : |z-3-4i| = 5$, where $|z|$ is maximum. From P, the particle moves through an angle $\tan^{-1}\frac{1}{3}$ in anticlockwise direction on $|z-3-4i| = 5$ and reaches at point Q. From Q, it comes down parallel to imaginary axis by 2 units and reaches at point R. Find the complex number corresponding to point R in the Argand plane.

एक कण बिन्दु P जो वक्र $C_1 : |z-3-4i| = 5$, पर स्थित है से प्रारम्भ होता है, जहाँ $|z|$ अधिकतम है। P से कण $|z-3-4i| = 5$ पर वामावर्त दिशा में कोण $\tan^{-1}\frac{1}{3}$ से घुमता है तथा बिन्दु Q पर पहुँचता है। Q से काल्पनिक अक्ष के सामानान्तर नीचे की ओर दो इकाई आता है तथा R पर पहुँचता है। सम्मिश्र तल में बिन्दु R के निर्देशांक ज्ञात कीजिये।

Ans. $(3+7i)$

Sol. $|z-(3+4i)| = 5$

$\theta = \tan^{-1}\frac{1}{3}$



$\tan\alpha = \frac{4}{3}$

$\tan(\theta+\alpha) = \frac{\frac{1}{3}+\frac{4}{3}}{1-\frac{4}{9}} = \frac{\frac{5}{3}}{\frac{5}{9}} = 3$

$\frac{y-0}{x-0} = 3$

$\Rightarrow y = 3x$

put in circle

$(x - 3)^2 + (y - 4)^2 = 5^2$

$(x^2 - 6x + 9) + (9x^2 - 24x + 16) = 25$

$10x^2 - 30x = 0$

$\Rightarrow x = 3, y = 9$

$\Rightarrow z_2 = 3 + 9i$

2 units down

$\Rightarrow 3 + 7i$

16. Evaluate: $\sum_{p=1}^{32} (3p + 2)\left(\sum_{q=1}^{10}\left(\sin\frac{2q\pi}{11} - i\cos\frac{2q\pi}{11}\right)\right)^p$.

मान ज्ञात कीजिये : $\sum_{p=1}^{32} (3p + 2)\left(\sum_{q=1}^{10}\left(\sin\frac{2q\pi}{11} - i\cos\frac{2q\pi}{11}\right)\right)^p$.

Ans. $48(1 - i)$

Sol. $\sum_{p=1}^{32}(3p+2)\left\{\sum_{q=1}^{10}\sin\frac{2q\pi}{11} - i\cos\frac{2q\pi}{11}\right\}^p$

Now first simplify

$\sum_{q=1}^{10}\sin\frac{2q\pi}{11} - i\cos\frac{2q\pi}{11}$

$\sum -i\left\{-\frac{1}{i}\sin\frac{2q\pi}{11} + \cos\frac{2q\pi}{11}\right\}$

$-i\sum_{q=1}^{10} e^{\frac{i2q\pi}{11}}$

Let $\frac{2\pi}{11} = \theta$

$$-i\left\{\sum_{q=1}^{10} e^{iq\theta}\right\}$$

$$-i\left\{e^{i\theta} + e^{i2\theta} + e^{i3\theta} + ... + e^{i10\theta}\right\}$$

$$-i\left\{\frac{e^{i\theta}\left(1-\left(e^{i\theta}\right)^{10}\right)}{1-e^{i\theta}}\right\}$$

$$-i\left\{\frac{e^{i\theta}\left(1-e^{i10\theta}\right)}{1-e^{i\theta}}\right\} = i$$

Now

$$\sum_{p=1}^{32}(3p+2)i^p$$

$$3\sum_{p=1}^{32}3pi^p + 2\sum_{p=1}^{32}2i^p$$

$3\ \{1i + 2i^2 + 3i^3 + ... + 32i^{32}\} + 2\ \{i + i^2 + ... + i^{32}\}$

$S = 1i + 2i^2 + 3i^3 + ... + 32i^{32}$

$Si = i^2 + 2i^3 + ... + 32i^{33}$

$S(1 - i) = 1i + i^2 + i^3 + ... + i^{32} + (-32)i$

$$S = \frac{-32i}{1-i}$$

$$\Rightarrow \frac{-32i(1+i)}{2}$$

$\Rightarrow -16(i - 1)$

$\Rightarrow 16(1 - i)$

17. Let a, b, c be distinct complex numbers such that $\frac{a}{1-b} = \frac{b}{1-c} = \frac{c}{1-a} = k$. Find the value of k.

माना a, b, c तीन भिन्न सम्मिश्र संख्यायें इस प्रकार है कि $\frac{a}{1-b} = \frac{b}{1-c} = \frac{c}{1-a} = k$ है। k का मान ज्ञात कीजिये।

Ans. $-\omega$ or $-\omega^2$

Sol. $\frac{a}{1-b} = \frac{b}{1-c} = \frac{c}{1-a} = k$

$a = k - bk \qquad b = k - kc \qquad c = k - ak$

$\Rightarrow \quad a = k - (k^2 - k^2c)$

$a = k - (k^2 - (k^3 - ak^3)]$

$a = k - k^2 + k^3 - ak^3$

$a(k^3 + 1) = k[k^2 - k + 1]$

$(ak + a - k)(k^2 - k + 1) = 0$

$k = \frac{-a}{a-1} \qquad$ or $\qquad k^2 - k + 1 = 0$

$k = -\omega, -\omega^2$ and $k = -\frac{a}{a-1}$ is not possible

Hence Proved.

Hindi $\frac{a}{1-b} = \frac{b}{1-c} = \frac{c}{1-a} = k$

$a = k - bk \qquad b = k - kc \qquad c = k - ak$

$\Rightarrow \quad a = k - (k^2 - k^2c)$

$a = k - (k^2 - (k^3 - ak^3)]$

$a = k - k^2 + k^3 - ak^3$

$a(k^3 + 1) = k[k^2 - k + 1]$

$(ak + a - k)(k^2 - k + 1) = 0$

$k = \frac{-a}{1-a} \qquad$ या $\qquad k^2 - k + 1 = 0$

$k = -\omega, -\omega^2$ तथा $k = -\frac{a}{a-1}$ असंभव है।

इति सिद्धम्

18. Let α, β be fixed complex numbers and z is a variable complex number such that, $|z-\alpha|^2 + |z-\beta|^2 = k$. Find out the limits for 'k' such that the locus of z is a circle. Find also the centre and radius of the circle.

माना α, β नियत सम्मिश्र संख्यायें है तथा z एक चर सम्मिश्र संख्या है। $|z-\alpha|^2 + |z-\beta|^2 = k$.

'k' के सीमान्त मान ज्ञात करें जबकि z का बिन्दुपथ एक वृत है। साथ ही वृत की त्रिज्या एवं केन्द्र भी ज्ञात कीजिये।

Ans. $k > \frac{1}{2}|\alpha - \beta|^2$

Sol. $|z-\alpha|^2 + |z-\beta|^2 = k$

$(z-\alpha)(\bar{z}-\bar{\alpha}) + (z-\beta)(\bar{z}-\bar{\beta}) = k$

$2z\bar{z} - (\bar{\alpha}+\bar{\beta})z - (\alpha+\beta)\bar{z} + |\alpha|^2 + |\beta|^2 - k = 0$

$\Rightarrow z\bar{z} - \left(\frac{\bar{\alpha}+\bar{\beta}}{2}\right)z - \left(\frac{\alpha+\beta}{2}\right)\bar{z} + \frac{|\alpha|^2 + |\beta|^2 - k}{2} = 0$

center : - $\left(\frac{\alpha+\beta}{2}\right)$ $\qquad \pi = \sqrt{\left|\frac{\alpha+\beta}{2}\right|^2 - \left(\frac{|\alpha|^2 + |\beta|^2 - k}{2}\right)}$

$$\pi = \sqrt{\frac{2k - \left(|\alpha|^2 + |\beta|^2 - \alpha\bar{\beta} - \beta\bar{\alpha}\right)}{4}}$$

$$\pi = \sqrt{\frac{2k - |\alpha-\beta|^2}{4}}$$

Hence $2k > |\alpha - \beta|2$

$\Rightarrow k > \frac{1}{2}|\alpha-\beta|^2$

19. If the area of the polygon whose vertices are the solutions (in the complex plane) of the equation $x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$, can be expressed in the simplest form as $\frac{a\sqrt{b}+c}{d}$, find the value of $(a + b + c + d)$.

यदि समीकरण $x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$ के हलों को सम्मिश्र तल में निरूपित करने पर बने बहुभुज का क्षेत्रफल सरलतम रूप में $\frac{a\sqrt{b}+c}{d}$ से व्यक्त किया जा सकता है, तो $(a + b + c + d)$ का मान बताइयें।

Ans. 8

Sol. $x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$

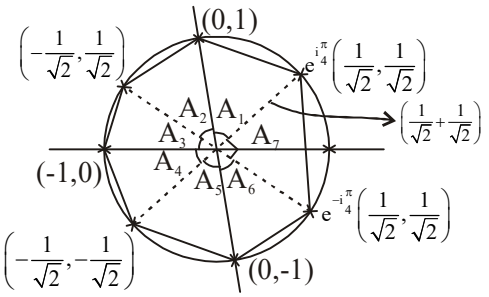
$$\frac{1\left(1-x^8\right)}{1-x} = 0; x \neq 1$$

$1 - x^8 = 0$

$x^8 = 1$

$= e^{i2k\pi}$

$x = e^{\frac{i2k\pi}{8}}$



$A_1 + A_2 + A_3 + A_4 + A_5 + A_6 = 6A_1$

$= 6 \cdot \frac{1}{2} \cdot 1 \cdot 1 \cdot \sin\frac{\pi}{4}$

$= \frac{3}{\sqrt{2}}$

$A_7 = \frac{1}{2} \times 1 \times 1 \times \sin\frac{\pi}{2}$

$= \frac{1}{2}$

$\text{Total area} = \frac{3}{\sqrt{2}} + \frac{1}{2}$

$\Rightarrow \frac{3\sqrt{2}+1}{2}$

$a = 3, b = 2, c = 1, d = 2$

$a + b + c + d = 8$

20. If a and b are positive integer such that $N = (a + ib)^3 - 107i$ is a positive integer. Find N.

यदि a तथा b धनात्मक पूर्णांक है इस प्रकार है कि $N = (a + ib)^3 - 107i$ एक धनात्मक पूर्णांक है। N ज्ञात करें।

Ans. 198

Sol. $N = (a+ib)^3 - 107i$

$a^3 - b^3 i + 3abi(a+ib) - 107i$

$a^3 - b^3 i + 3a^2 bi - 3ab^2 - 107i$

$= (a^3 - 3ab^2) + i(3a^2 b - b^3 - 107)$

Now for the integer

$3a^2b - b^3 - 107 = 0$

$3a^2b - b^3 = 107$

$(3a^2 - b^2)\cdot b = 107$

$= 107 \times 1$

$3a^2 - b^2 = 107$ & $b = 1$

$3a^2 = 108$

$a^2 = 36$

$a = 6, b = 1$

$\therefore N = 6^3 - 3\cdot 6\cdot 1^2$

$= 216 - 18$

$= 198$

21. If the biquadratic $x^4 + ax^3 + bx^2 + cx + d = 0$ (a, b, c, d $\in$ R) has 4 non real roots, two with sum $3 + 4i$ and the other two with product $13 + i$. Find the value of 'b'.

यदि एक चार घातीय समीकरण $x^4 + ax^3 + bx^2 + cx + d = 0$ (a, b, c, d $\in$ R) के 4 अवास्तविक मूल है जिनमें दो का योग $3 + 4i$ है तथा अन्य दो का गुणन $13 + i$ है। 'b' का मान ज्ञात करें।

Ans. 51

Sol. Let roots be $\alpha, \beta, \bar{\alpha}, \bar{\beta}$

$\alpha + \beta = 3 + 4i \qquad \bar{\alpha}\bar{\beta} = 13 + i$

$\Rightarrow \bar{\alpha} + \bar{\beta} = 3 - 4i \qquad \Rightarrow \alpha\beta = 13 - i$

$b = \alpha\beta + \alpha\bar{\alpha} + \alpha\bar{\beta} + \bar{\alpha}\beta + \beta\bar{\beta} + \bar{\alpha}\bar{\beta}$

$= \left(\alpha\beta + \bar{\alpha}\bar{\beta}\right) + \left(\alpha\bar{\alpha} + \alpha\bar{\beta} + \bar{\alpha}\beta + \beta\bar{\beta}\right)$

$\left(13 - i + 13 + i\right) + \left(\alpha + \beta\right)\left(\bar{\alpha} + \bar{\beta}\right)$

$26 + (3 + 4i)(3 - 4i)$

$26 + 9 + 16 = 51$

22. If ω is the imaginary cube root of unity, then find the number of pairs of integers (a, b) such that $|a\omega + b| = 1$.

यदि ω इकाई का काल्पनिक घनमूल है, तो क्रमागत पूर्णांक युग्मों (a, b) की संख्या बताइयें जिनके लिये $|a\omega + b| = 1$ है।

Ans. 0006

Sol. We have $|a\omega + b| = 1$

$\Rightarrow |a\omega + b|^2 = 1 \Rightarrow (a\omega + b)(a\bar{\omega} + b) = 1 \Rightarrow a^2 + ab(\omega + \bar{\omega}) + b^2 = 1 \Rightarrow a^2 - ab + b^2 = 1$

$\Rightarrow \quad (a-b)^2 + ab = 1 \quad ....(1) \quad (\text{As } 1 + \omega + \omega^2 = 0)$

when $(a-b)^2 = 0$ and $ab = 1$ then $(1, 1); (-1, -1)$

when $(a-b)^2 = 1$ and $ab = 0$ then $(0, 1); (1, 0); (0, -1); (-1, 0)$

Hence $(0, 1); (1, 0); (0, -1); (-1, 0); (1, 1); (-1, -1)$ i.e. 6 ordered pairs **Ans.**

Paragraph for Question Nos. 23 to 25

Let A, B, C be three sets of complex numbers as defined below.

माना A, B, C सम्मिश्र संख्याओं के तीन समुच्चय है जो निम्न प्रकार से परिभाषित है।

$A = \{z : \text{Im}\, z \geq 1\} \qquad B = \{z : |z - 2 - i| = 3\} \qquad C = \{z : \text{Re}((1-i)z) = \sqrt{2}\}$

23. The number of elements in the set $A \cap B \cap C$ is : **[JEE-2008]**

समुच्चय $A \cap B \cap C$ में सदस्यों की संख्या निम्न है : **[JEE-2008]**

(A) 0 (B) 1 (C) 2 (D) ∞

Ans. B

Sol. $\text{Re}\left[(1-i)(x+iy)\right] = \sqrt{2}$

$x + y = \sqrt{2}$

For R

$(x-2)^2 + (y-1)^2 = 9$

$x = 0$

$(y-1)^2 = 5$

$y = 1 \pm \sqrt{5}$

$A \cap B \cap C = \{\text{Point S}\}$

Only $\perp$ point

24. Let z be any point in $A \cap B \cap C$. Then, $|z + 1 - i|^2 + |z - 5 - i|^2$ lies between s : **[JEE-2008]**

(A) 25 and 29 (B) 30 and 34 (C) 35 and 39 (D) 40 and 44

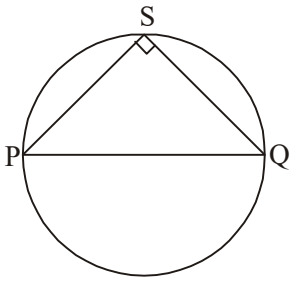
माना z, $A \cap B \cap C$ का कोई एक बिन्दु है। तब $|z + 1 - i|^2 + |z - 5 - i|^2$ निम्न के बीच स्थित है : **[JEE-2008]**

(A) 25 तथा 29 (B) 30 तथा 34 (C) 35 तथा 39 (D) 40 तथा 44

Ans. C

Sol. Z = Point S

= POI of $x + y = \sqrt{2}$ & $(x - 2)^2 + (y - 1)^2 = 9$



$|z + 1 - i|^2 + |z - 5 - i|^2 = PS^2 + QS^2$

$= (PQ)^2$

$= 6^2$

$= 36$

25. Let z be any point in $A \cap B \cap C$ and let w be any point satisfying $|w - 2 - i| < 3$. Then, $|z| - |w| + 3$ lies between: **[JEE-2008]**

(A) – 6 and 3 (B) –3 and 6 (C) – 6 and 6 (D) – 3 and 9

माना z, $A \cap B \cap C$ में कोई एक बिन्दु है तथा w, $|w - 2 - i| < 3$ को संतुष्ट करने वाला कोई बिन्दु है। तब, $|z| - |w| + 3$ निम्न के बीच स्थित है: **[JEE-2008]**

(A) – 6 तथा 3 (B) –3 तथा 6 (C) – 6 तथा 6 (D) – 3 तथा 9

Ans. BCD

Sol. $|w - 2 - i| < 3 \Rightarrow$ Points inside circle $B \equiv w$

$|z| - |w| + 3$

$|z - w| < 6 \Rightarrow |z| - |w| \in (-6, 6)$

$|z| - |w| + 3 \in (-3, 9)$

Paragraph for Question Nos. 26 to 28

Let $(1 + x)^n = C_0 + C_1x + C_2x^2 + ...+ C_nx^n$. For sum of series $C_0 + C_1 + C_2 + ...$, put x = 1. For sum of series $C_0 + C_2 + C_4 + C_6 + ...$, or $C_1 + C_3 + C_5 + ...$ add or substract equations obtained by putting x = 1 and x = – 1.

For sum of series $C_0 + C_3 + C_6 + ...$ or $C_1 + C_4 + C_7 + ...$ or $C_2 + C_5 + C_8 + ...$ we subsitute x = 1, $x = \omega$, $x = \omega^2$ and add or manupulate results.

Similarly, if suffixes differe by 'p' then we substitute p^{th} roots of unity and add.

माना $(1 + x)^n = C_0 + C_1x + C_2x^2 + ...+ C_nx^n$। श्रेणी के योग $C_0 + C_1 + C_2 + ...$ के लिए, x = 1 रखें। श्रेणी के योग $C_0 + C_2 + C_4 + C_6 + ...$, या $C_1 + C_3 + C_5 + ...$ के लिए उन समीकरणों को जोड़े या घटायें जो कि x = 1 तथा x = – 1 रखने पर प्राप्त होती है।

श्रेणी के योग $C_0 + C_3 + C_6 + ...$ या $C_1 + C_4 + C_7 + ...$ या $C_2 + C_5 + C_8 + ...$ के लिए हम x = 1, $x = \omega$, $x = \omega^2$ रखते है तथा प्राप्त समीकरणों का योग अथवा कोई अन्य कार्यवाही करते है।

समरूपतः, यदि प्रत्ययों के मध्य 'p' का फर्क हो तो हम इकाई के pवें मूलों को रखते है तथा योग करते है।

26. $C_0 + C_3 + C_6 + C_9 + \ldots\ldots = $

(A) $\frac{1}{3}\left[2^n - 2\cos\frac{n\pi}{3}\right]$ (B) $\frac{1}{3}\left[2^n + 2\cos\frac{n\pi}{3}\right]$

(C) $\frac{1}{3}\left[2^n - 2\sin\frac{n\pi}{3}\right]$ (D) $\frac{1}{3}\left[2^n + 2\sin\frac{n\pi}{3}\right]$

Ans. B

Sol. $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 \ldots\ldots\ldots$

$2^n = C_0 + C_1 + C_2 + C_3 \ldots\ldots\ldots$ .....(1)

$(1+\omega)^n = C_0 + C_1\omega + C_2\omega^2 + C_3\,\omega^3\ldots\ldots\ldots$ .....(2)

$(1+\omega^2)^n = C_0 + C_1\omega^2 + C_2\omega^4 + C_3\,\omega^6\ldots\ldots\ldots$ .....(3)

on adding (1) + (2) + (3) जोड़ने पर

$2^n + (1+\omega)^n + (1+\omega^2)^n = 3[C_0 + C_3 + C_6 \ldots\ldots]$

$$\frac{2^n + (-\omega^2)^n + (-\omega)^n}{3} = C_0 + C_3 + C_6\ldots\ldots$$

$$C_0 + C_3 + C_6 \ldots\ldots\ldots = \frac{2^n + e^{i\frac{n\pi}{3}} + e^{-i\frac{n\pi}{3}}}{3}$$

$$= \frac{1}{3}\left[2^n + 2\cos\frac{n\pi}{3}\right] \qquad \text{(B)}$$

27. $C_1 + C_5 + C_9 + \ldots. = $

(A) $\frac{1}{4}\left[2^n - 2^{n/2}2\cos\frac{n\pi}{4}\right]$ (B) $\frac{1}{4}\left[2^n + 2^{n/2}2\cos\frac{n\pi}{4}\right]$

(C) $\frac{1}{4}\left[2^n - 2^{n/2}2\sin\frac{n\pi}{4}\right]$ (D) $\frac{1}{4}\left[2^n + 2^{n/2}2\sin\frac{n\pi}{4}\right]$

Ans. D

Sol. $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + C_5x^5 + C_6x^6 \ldots\ldots\ldots$

put x = 1 रखने पर $2^n = C_0 + C_1 + C_2 + C_3 + C_4 + C_5 + C_6 + C_7 + C_8$ ..... (1)

put x = – 1 रखने पर $0 = C_0 - C_1 + C_2 - C_3 + C_4 - C_5 + C_6 - C_7 + C_8$ ..... (2)

put x = i रखने पर $(1+i)^n = C_0 + iC_1 - C_2 - iC_3 + C_4 + iC_5 - C_6 - iC_7 + C_8$ ..... (3)

put x = – i रखने पर $(1-i)^n = C_0 - iC_1 - C_2 + iC_3 + C_4 - iC_5 - C_6 + iC_7 + C_8$ ..... (4)

Now अब (1) – (2) – i(3) + i(4)

$4(C_1 + C_5 + C_9\ldots\ldots) = 2^n - 0 - i(1+i)^n + i(1-i)^n$

$= 2^n + (-i)\,[(1+i)^n - (1-i)^n]$

$= 2^n + (-i)\,[(1+i)^n - (1-i)^n]$

$$= 2^n - i2^{n/2}\left[e^{i\frac{n\pi}{4}} - e^{-i\frac{n\pi}{4}}\right]$$

$$= 2^n - i\,2^{n/2}\,.\,2i\sin\left(\frac{n\pi}{4}\right)$$

$$\therefore \qquad C_1 + C_5 + C_9 \ldots\ldots = \frac{1}{4}\left[2^n + 2^{n/2}\,.2\sin\frac{n\pi}{4}\right]$$

28. $C_2 + C_6 + C_{10} + \ldots\ldots =$

(A) $\frac{1}{4}\left[2^n - 2^{n/2}\,2.\cos\frac{n\pi}{4}\right]$ (B) $\frac{1}{4}\left[2^n + 2^{n/2}\,2.\cos\frac{n\pi}{4}\right]$

(C) $\frac{1}{4}\left[2^n - 2^{n/2}\,2.\sin\frac{n\pi}{4}\right]$ (D) $\frac{1}{4}\left[2^n + 2^{n/2}\,2.\sin\frac{n\pi}{4}\right]$

Ans. A

Sol. $(1 + x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + C_5x^5 + C_6x^6 + \ldots\ldots$

Put $x = 1, i, -1, -i$

$$2^n = C_0 + C_1 + C_2 + C_3 + C_4 + C_5 + C_6 + \ldots\ldots \qquad \ldots(1)$$

$$(1+i)^n = C_0 + C_1i - C_2 - C_3i + C_4 + C_5i - C_6 - C_7i + \ldots\ldots \qquad \ldots(2)$$

$$0^n = C_0 - C_1 + C_2 - C_3 + C_4 - C_5 + C_6 - C_7 \ldots\ldots\ldots\ldots \qquad \ldots(3)$$

$$(1-i)^n = C_0 - C_1i - C_2 + C_3i + C_4 - C_5i - C_6 + C_7i \ldots\ldots\ldots\ldots \qquad \ldots(4)$$

(1) – (2) + (3) – (4)

$$2^n - (1+i)^n - (1-i)^n = 4(C_2 + C_6 + C_{10} + \ldots\ldots)$$

$$2^n - 2^{\frac{n}{2}}\left[e^{\frac{in\pi}{4}} + e^{-\frac{in\pi}{4}}\right] = 4(C_2 + C_6 + C_{10} + \ldots\ldots)$$

$$2^n - 2^{\frac{n}{2}} \cdot 2\cos\frac{n\pi}{4} = 4(C_2 + C_6 + C_{10} + \ldots\ldots)$$

$$C_2 + C_6 + C_{10} + \ldots\ldots = \frac{1}{4}\left[2^n - 2^{\frac{n}{2}} \cdot 2\cos\frac{n\pi}{4}\right]$$

Paragraph for Question Nos. 29 to 30

29. If $w(\neq 1)$ is a cube root of unity and $(1+w)^7 = A + Bw$, then A and B are respectively the numbers :

यदि $w(\neq 1)$ इकाई का घनमूल है तथा $(1+w)^7 = A + Bw$, तो A तथा B क्रमशः है :

(A) 0 , 1 (B) 1 , 1 (C) 1 , 0 (D) −1 , 1

Ans. B

Sol. $(1+w)^7 = A + Bw$

$1 + w = -w^2$

$-w^{14} = A + Bw$

$-w^2 = A + Bw$

$1 + w = A + Bw$

$A = 1, B = 1$

30. If $(w \neq 1)$ is a cube root of unity then $\begin{vmatrix} 1 & 1+i+w^2 & w^2 \\ 1-i & -1 & w^2-1 \\ -i & -i+w-1 & -1 \end{vmatrix} =$

यदि $(w \neq 1)$ इकाई का घनमूल है, तो $\begin{vmatrix} 1 & 1+i+w^2 & w^2 \\ 1-i & -1 & w^2-1 \\ -i & -i+w-1 & -1 \end{vmatrix} =$

(A) 0 (B) 1 (C) i (D) w

Ans. A

Sol. $\begin{vmatrix} 1 & 1+i+w^2 & w^2 \\ 1-i & -1 & w^2-1 \\ -i & -i+w-1 & -1 \end{vmatrix}$

$C_3 \to C_3 + C_1$

$\begin{vmatrix} 1 & i-w & w^2 \\ 1-i & -1 & w^2-1 \\ 1-i & -1 & w^2-1 \end{vmatrix} = 0$

MATCH THE COLUMN

31. Match the statements given in Column I with the values given in Column II : **[JEE-2011]**

Column I	**Column II**
(A) If $\vec{a} = \hat{j} + \sqrt{3}\hat{k}, \vec{b} = -\hat{j} + \sqrt{3}\hat{k}$ and $\vec{c} = 2\sqrt{3}\hat{k}$ form a triangle, then the internal angle of the triangle between $\vec{a}$ and $\vec{b}$ is :	(p) $\frac{\pi}{6}$
(B) If $\int_a^b (f(x) - 3x)\,dx = a^2 - b^2$, then the value of $f\left(\frac{\pi}{6}\right)$ is	(q) $\frac{2\pi}{3}$
(C) The value of $\frac{\pi^2}{\ln 3}\int_{7/6}^{5/6} \sec(\pi x)\,dx$ is	(r) $\frac{\pi}{3}$
(D) The maximum value of $\left\lvert Arg\left(\frac{1}{1-z}\right)\right\rvert$ for $\lvert z\rvert = 1, z \neq 1$ is given by	(s) π
	(t) $\frac{\pi}{2}$

Column I में दिये गये कथनों को Column II में दिये गये मानों के साथ सूमेलित कीजिये : **[JEE-2011]**

Column I	**Column II**
(A) यदि $\vec{a} = \hat{j} + \sqrt{3}\hat{k}, \vec{b} = -\hat{j} + \sqrt{3}\hat{k}$ तथा $\vec{c} = 2\sqrt{3}\hat{k}$ एक त्रिभुज बनाते है, तो $\vec{a}$ तथा $\vec{b}$ के मध्य आन्तरिक कोण होंगा :	(p) $\frac{\pi}{6}$
(B) यदि $\int_a^b (f(x) - 3x)\,dx = a^2 - b^2$, है तो $f\left(\frac{\pi}{6}\right)$ का मान होगा	(q) $\frac{2\pi}{3}$
(C) $\frac{\pi^2}{\ln 3}\int_{7/6}^{5/6} \sec(\pi x)\,dx$ का मान होगा	(r) $\frac{\pi}{3}$
(D) $\left\lvert Arg\left(\frac{1}{1-z}\right)\right\rvert$, जहां $\lvert z\rvert = 1, z \neq 1$, का अधिकतम मान है	(s) π
	(t) $\frac{\pi}{2}$

Ans. (A) q (B) p or pqrs and t; (C) s (D) t

Sol. (A) $\vec{a} - \vec{b} = -1 + 3 = 2$

$|\vec{a}| = 2, |\vec{b}| = 2$

$\cos\theta = \frac{2}{2\times 2} = \frac{1}{2}$

$Q = \frac{\pi}{3}, \frac{2\pi}{3}$ but its $\frac{2\pi}{3}$ as its opposite to side of maximum length.

(B) $\int_a^b \left(f(x) - 3x\right) dx = a^2 - b^2$

$$\int_a^b f(x)\,dx = \frac{3}{2}\left(b^2 - a^2\right) + a^2 - b^2 = \frac{-a^2 + b^2}{2}$$

$\Rightarrow f(x) = x$

(C) $$\frac{\pi^2}{\ell n 3}\left(\frac{\ell n\left|\sec \pi x + \tan \pi x\right|_{7/6}^{5/6}}{\pi}\right) = \frac{\pi}{\ell n 3}\left(\ell n\left|\sec\frac{5\pi}{6} + \tan\frac{5\pi}{6}\right| - \ell n\left|\sec\frac{7\pi}{6} + \tan\frac{7\pi}{6}\right|\right) = \pi$$

(D) Let $u = \frac{1}{1-z} \Rightarrow z = 1 - \frac{1}{u}$

$$|z| = 1 \Rightarrow \left|1 - \frac{1}{u}\right| = 1$$

$\Rightarrow |u - 1| = |u|$

$\Rightarrow$ locus of u is perpendicular bisector of line segment joining 0 and 1.

$\Rightarrow$ maximum arg u approaches $\frac{\pi}{2}$ but will not attain.

32. Match the statements given in Column I with the intervals/union of intervals given in column II : **[JEE-2011]**

Column I	**Column II**
(A) The set $\left\{\text{Re}\left(\frac{2iz}{1-z^2}\right); z \text{ is a complex}, \lvert z\rvert = 1, z \neq \pm 1\right\}$ is	(p) $(-\infty, -1) \cup (1, \infty)$
(B) The domain of the function $f(x) = \sin^{-1}\left(\frac{8(3)^{x-2}}{1-3^{2(x-1)}}\right)$ is	(q) $(-\infty, 0) \cup (0, \infty)$
(C) If $f(\theta) = \begin{vmatrix} 1 & \tan\theta & 1 \\ -\tan\theta & 1 & \tan\theta \\ -1 & -\tan\theta & 1 \end{vmatrix}$ then the set $\left\{f(\theta) : 0 \le \theta < \frac{\pi}{2}\right\}$ is	(r) $[2, \infty)$
(D) If $f(x) = x^{3/2}(3x - 10), x \ge 0$ then f(x) is increasing in	(s) $(-\infty, -1] \cup [1, \infty)$
	(t) $(-\infty, 0] \cup [2, \infty)$

Column I में दिये गये कथनों को Column II में दिये गये अन्तरालों के साथ सूमेलित कीजिये : **[JEE-2011]**

Column I	**Column II**
(A) समुच्चय $\left\{\text{Re}\left(\frac{2iz}{1-z^2}\right); z \text{ d l fteJ l ka; k, } \lvert z\rvert=1, z\neq\pm1\right\}$ है	(p) $(-\infty,-1)\cup(1,\infty)$
(B) फलन $f(x)=\sin^{-1}\left(\frac{8(3)^{r-2}}{1-3^{2(x-1)}}\right)$ का प्रान्त होगा	(q) $(-\infty,0)\cup(0,\infty)$
(C) यदि $f(\theta)=\begin{vmatrix}1 & \tan\theta & 1\\ -\tan\theta & 1 & \tan\theta\\ -1 & -\tan\theta & 1\end{vmatrix}$ है तो समुच्चय $\left\{f(\theta): 0\le\theta<\frac{\pi}{2}\right\}$ है	(r) $[2,\infty)$
(D) यदि $f(x)=x^{3/2}(3x-10),\ x\ge0$ है तो f(x) वर्धमान होगा	(s) $(-\infty,-1]\cup[1,\infty)$
	(t) $(-\infty,0]\cup[2,\infty)$

Ans. $A\to s;\ B\to t;\ C\to r;\ D\to r$

Sol. (A) $z=\dfrac{2i(x+iy)}{1-(x+iy)^2}=\dfrac{2i(x+iy)}{1-(x^2-y^2+2ixy)}$

Using $1-x^2=y^2$

$$Z=\frac{2ix-2y}{2y^2-2ixy}=-\frac{1}{y}$$

$$\because -1\le y\le1\Rightarrow -\frac{1}{y}\le-1 \text{ or } -\frac{1}{y}\ge1$$

(B) For domain

$$-1\le\frac{8\cdot3^{x-2}}{1-3^{2(x-1)}}\le1$$

$$\Rightarrow -1\le\frac{3^x-3^{x-2}}{1-3^{2x-2}}\le1$$

Case–I : $\dfrac{3^x-3^{x-2}}{1-3^{2x-2}}-1\le0$

$$\Rightarrow\frac{(3^x-1)(3^{x-2}-1)}{(3^{2x-2}-1)}\ge0$$

$\Rightarrow x \in (-\infty, 0] \cup (1, \infty)$

Case–II : $\dfrac{3^x - 3^{x-2}}{1 - 3^{2x} - 2} + 1 \geq 0$

$\Rightarrow \dfrac{\left(3^{x-2} - 1\right)\left(3^x + 1\right)}{\left(3^x \cdot 3^{x-2} - 1\right)} \geq 0$

$\Rightarrow x \in (-\infty, 1) \cup [2, \infty)$

So, $x \in (-\infty, 0] \cup [2, \infty)$

(C) $R_1 \to R_1 + R_3$

$$f(\theta) = \begin{vmatrix} 0 & 0 & 2 \\ -\tan\theta & 1 & \tan\theta \\ -1 & -\tan\theta & 1 \end{vmatrix}$$

$= 2\left(\tan^2\theta + 1\right) = 2\sec^2\theta \qquad \Rightarrow 2\sec^2\theta \in [2, \infty)$

(D) $f'(x) = \dfrac{3}{2}(x)^{\frac{1}{2}}(3x - 10) + (x)^{\frac{3}{2}} \times 3 = \dfrac{15}{2}(x)^{\frac{1}{2}}(x - 2)$

Increasing, when $x \geq 2$